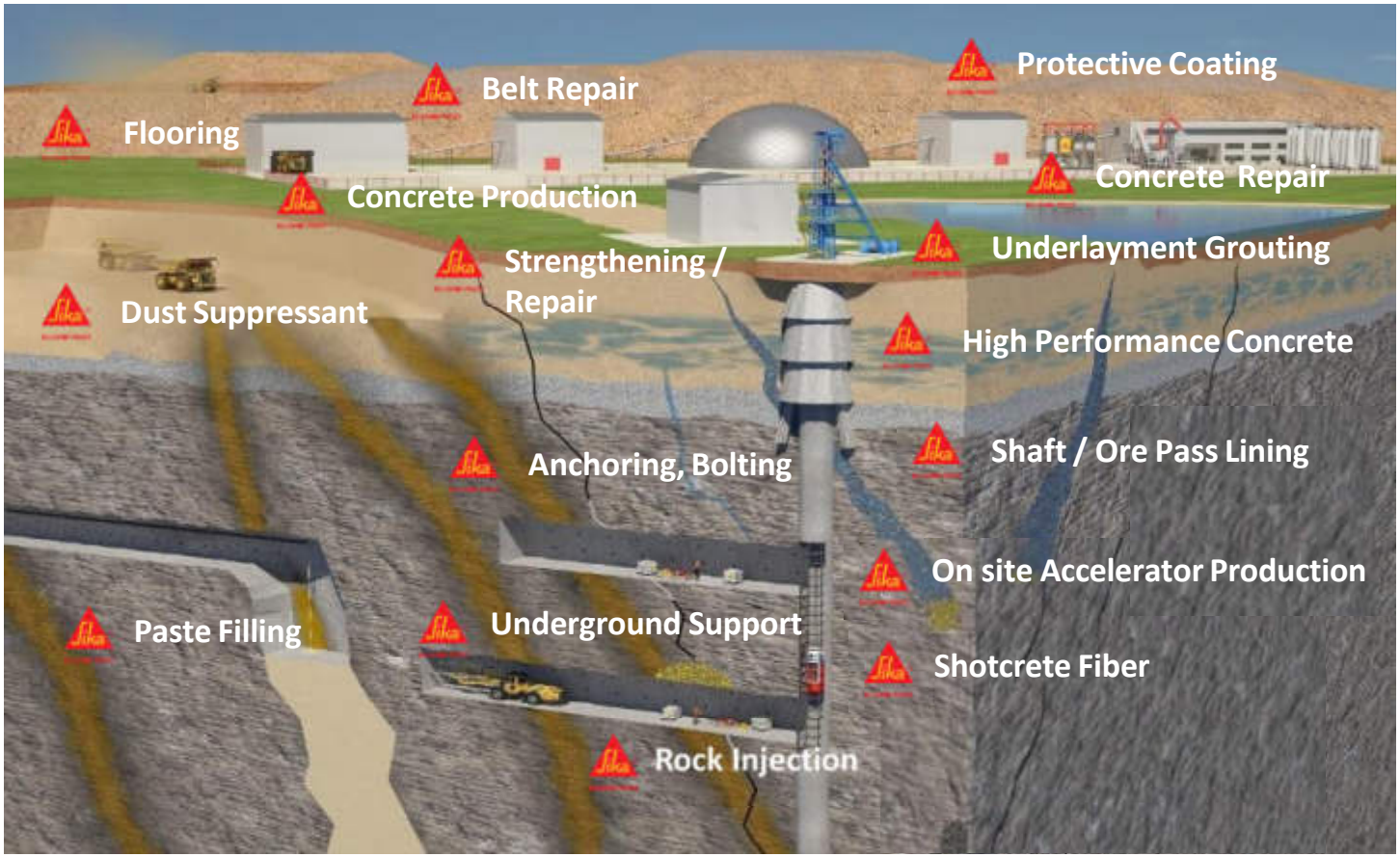


SIKA IN THE MINING INDUSTRY

2026_ABDALLAH IJANI

BUILDING TRUST

TODAYS MINES



HOW TO OPERATE INTERACTIVE PPTX

- **CLICK ON ”SIKA SYMBOL” NEXT TO ”TOPIC OF INTEREST** WHICH WILL OPEN DEDICATED PRESENTATION.
- **END OF SPECIFIC PPTX, CLICK ON ”BACK TO PRESENTATION”** WHICH RETURNS TO ”TODAYS MINES” MAIN PAGE” TO START OVER.

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CONCRETE PRODUCTION

AN INTRODUCTION TO SIKA CONCRETE ADMIXTURES AND SURFACE PRODUCTS

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THE BASICS OF CONCRETE

Concrete is produced from mixing proportions of:

- Aggregates
- Sands
- Cement
- Cement
- Water
- Admixtures

The aggregates and sands act as bulking material

The cement and water act as paste



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SOLUTIONS GROUPS



Cement Producer



Concrete Producer



Precaster



Contractor



Gypsum Board Manufacturer



Mortar Producer



Mining Company

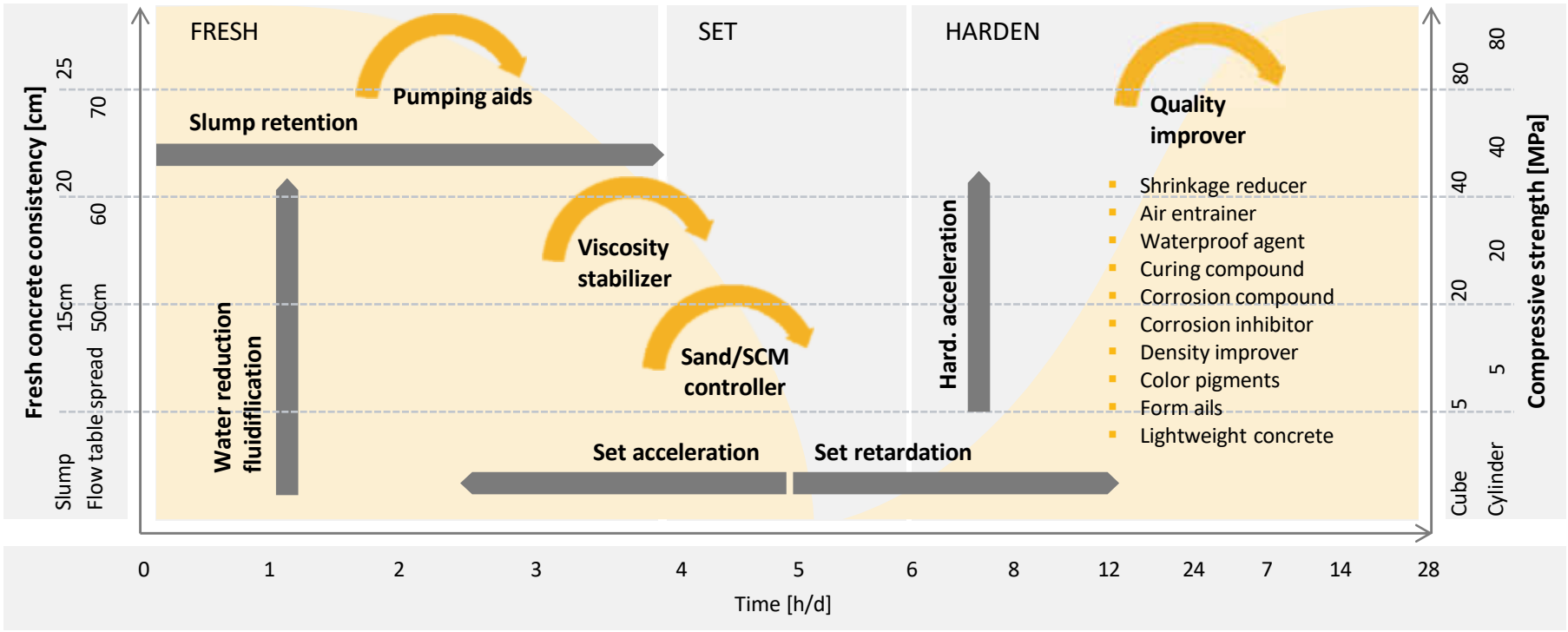


Distribution

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PRODUCTS GROUPS



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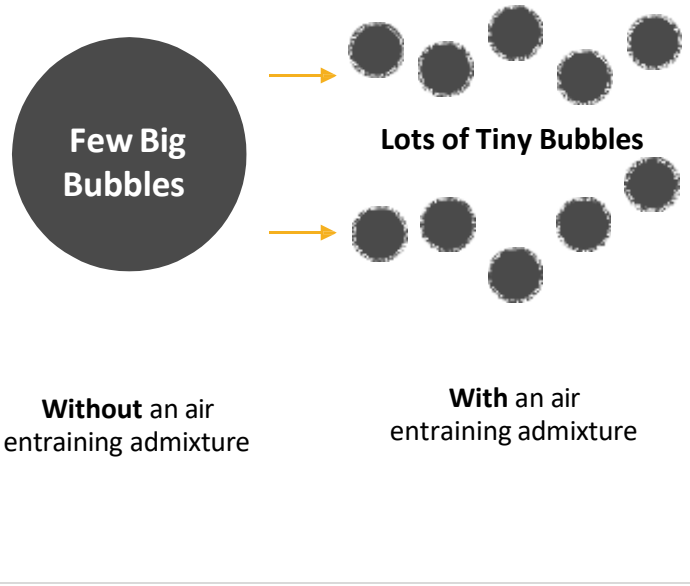
BETTER WORKABILITY WITH
AIR ENTRAINING ADMIXTURES

All concrete generates some air on its own

Air-entrainers produce an entrained air void system which create many tiny air bubbles to be trapped in the wet concrete, improving workability and frost resistance.

AIR ENTRAINING ADMIXTURES:

- **Sika® AER**



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SLOWING DOWN THE SET TIME
OF CONCRETE USING “RETARDERS”

- Slows down the setting process
- Ensures that the concrete is still placeable when the truck arrives on site
- Particularly useful in hot or windy weather

RETARDING ADMIXTURES:

- **Sika® Retarder**
- **SikaTard® -930**



SPEEDING UP THE SET TIME OF CONCRETE
USING “ACCELERATORS”

Speed up the concrete setting process

Ensures that the concrete sets fast enough

Particularly useful in cold weather

ACCELERATING ADMIXTURES:

- **SikaSet®**
- **Sika Rapid®1**



Note! *If using a set accelerating admixture, it must be added to every batch of concrete to ensure a consistent colour*

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MAKING CONCRETE EASIER TO USE
WITH ‘PLASTICISERS’ (WATER REDUCERS)

Improve workability (slump) without adding more water

More water means weaker concrete!



WATER REDUCING ADMIXTURES:

- **Sikaplast®**

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MAKING IT FLOW EASILY
WITH ‘SUPERPLASTICISERS’

Decreases the amount of water needed in the mix

Increases flow of the mix to make placement easier

SUPERPLASTICISER ADMIXTURES:

- **Sika® ViscoCrete®**



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KEEPING IT FLOW EASIER AND LONGER
WITH ‘SUPERPLASTICISERS’

Workability enhancing admixture for concrete based on Sika® PCE technology

Define the workability time based on project requirements without a negative effect on early strength development.

SUPERPLASTICISER ADMIXTURES:

- **Sika® ViscoFlow®**

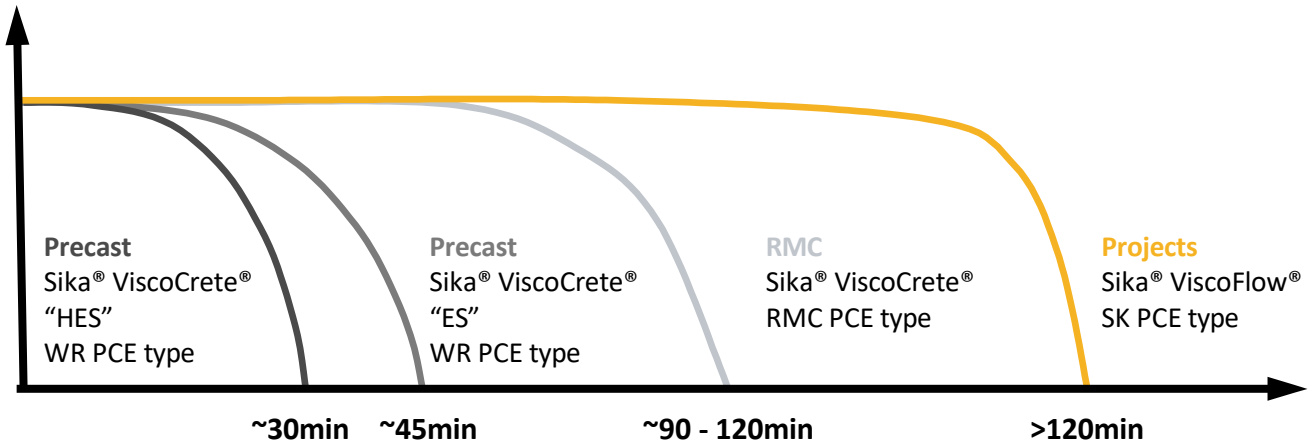


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SLUMP LIFE (WORKABILITY)

Slump for different timeframes of workability



Note:

ViscoCrete® & ViscoFlow® arebased on 3 main polycarboxilates base polymers:

- WR = Water reducer
- RMC = Slump retention
- SK = Slump keeper

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EASIER FLOW THROUGH A CONCRETE PUMP
WITH “SIKA PUMP”

Pumping aids reduce pressure at the pump making it easier to pump concrete

Especially useful and helpful when the sand used presents lack of fines (particle < 150 µm)

PUMPING AIDADMIXTURE:

- **SikaPump®**



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WATERTIGHT CONCRETE

Concrete is made less permeable by adding **Sika-1+** or **Sika WT-200 P** (crystalline self-healing)

Helps make below ground concrete more watertight

Suitable for both positive and negative water pressure

WATERTIGHTNESS ADMIXTURES:

- **Sika® -1+**
- **Sika® WT-200 P**



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IMPROVED DURABILITY
USING SIKAFIBRE

MICRO FIBER

- Non structural fibers
- Improved abrasion resistance
- Reduced plastic shrinkage cracking
- Improves fire protection in tunnels



IMPROVED DURABILITY ADDITIVE:

- **SikaFiber®**





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IMPROVED STRUCTURAL DUCTILITY
USING SIKAFIBRE

MICRO FIBER

- Structural fibers
- Improves shotcrete (sprayed concrete) structural ductility.
- Higher build in vertical and overhead application



IMPROVED DURABILITY ADDITIVE:

- **SikaFiber®**



More information is given in Sika Shotcrete (Sprayed Concrete) Solutions

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IMPROVED STRUCTURAL DUCTILITY
USING SIKA CONTROL 40 (NZ)

- Helps control shrinkage as the newly placed concrete cures
- Reduces shrinkage cracking which can be common as new concrete cures

SHRINKAGE CONTROL ADDITIVE:

- **Sika® Control-xyz**



An example of plastic shrinkage cracking

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IMPROVING CONCRETE SURFACE FINISH

STABILIZER-4 R

Stabilizer-4 R is a VMA “Viscosity Modifier”

Helps prevent high-flow concrete from segregating

MIX STABILITY ADMIXTURE:

- **Sika® Stabilizer-4 R**



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IMPROVING DURABILITY OF CONCRETE
WITH SIKA FUME

Added to concrete it will:

- Reduce permeability
- Increase all mechanical strengths
- Improve sulphate and chemical resistance
- Increase impact damage

IMPROVED DURABILITY ADDITIVE:

- **SikaFume®** *powder*
- **SikaCrete® L** *slurry*





- Added to concrete it will also:
  - Increase abrasion resistance

- Added to shotcrete it will:
  - Reduce rebound

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SURFACE PRODUCTS

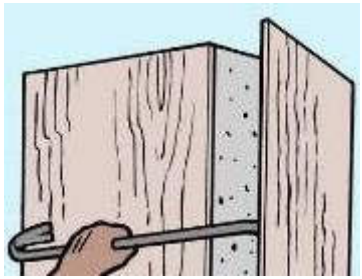
- PRODUCTS USED ON THE SURFACE OF FRESHLY PLACED CONCRETE

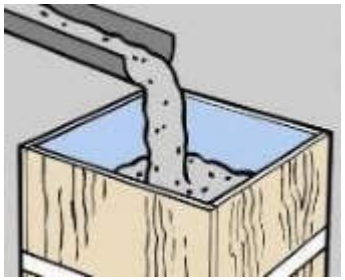
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EASY REMOVAL OF MOULDS AND FORMWORK

WITH A 'RELEASE AGENT'

- Easier formwork removal and improved concrete finish
- Products based on:
  - Mineral oils
  - Solvents
  - Water based emulsions
- Not only the mould release technology, but also the type of formwork and even curing is important
- When applying, "Less is Best"







FORM RELEASE AGENT:

- **Sika® Formoil**
- **Sika® Separol®**



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REDUCING RISK OF CRACKING IN NEW CONCRETE
WITH 'CURING AGENTS / CURING MEMBRANES'

Prevents cracking due to excessive evaporation from the concrete

CURING MEMBRANES:

- **Antisol A (acrylic)**
  Can be removed and overcoated (refer PDS)
- **Antisol E (wax)**
  Difficult to remove and cannot be overcoated
- **Antisol S (silicate)**
  Reacto-penetrative curing compound can be overcoated



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MAKE DECORATIVE OR NON-SLIP SURFACES
WITH A 'SURFACE RETARDER' (RUGASOL)

Produce an exposed aggregate surface on fresh wet concrete

SURFACE RETARDERS:

- **Rugasol® C -** sprayed on horizontal surfaces
  Rugasol® C Extra - sprayed on horizontal surfaces over 20MPa
- **Rugasol® MH or F -** applied to formwork for vertical surfaces
- **Rugasol® G –** Gel type.
  Undiluted acts as type MH or F.
  Diluted act as type C





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ANTI-DUST COATING

FOR NEW OR OLD CONCRETE

- Clear surface hardener
- Dust-proofer
- Sealer for hardened concrete
- Curing aid for fresh concrete

ANTI-DUSTING PRODUCTS:

- **Sikafloor® CureHard®-24**
- **Purigo®-5 S**



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BACK TO PRESENTATION

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SIKAFLOOR® SYSTEMS - GENERAL OVERVIEW

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SIKA IN FLOORING & COATING

INTRODUCTION- MAIN APPLICATIONS



- Storage, logistics and sales areas



- Production and processing area



- Cleanroom Floors &
- **E**lectro**S**tatic **D**ischarge protection



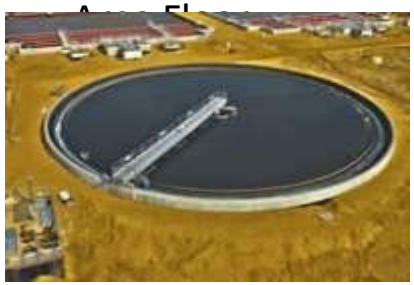
- Car Park Floors



- Commercial, public and residential floors



- Walls and ceilings



- Tank linings & water treatment facilities



Other Industrial Sectors i.e., Mining

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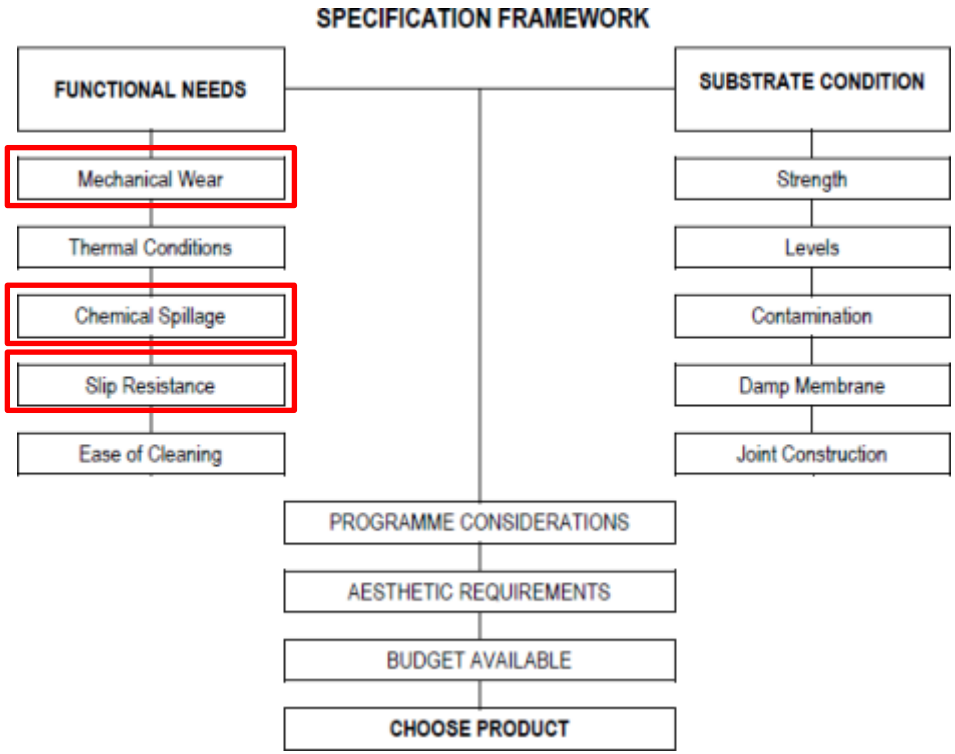
BUILDING TRUST

28

SIKA IN FLOORING & COATING

DESIGN & SELECTION OF RIGHT SYSTEM

- FLOOR DESIGN CONSIDERATIONS / FLOOR SYSTEM SELECTION … **As per FeRFA**



- FeRFA: **R**esin **F**looring **A**ssociation (From UK)

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SIKA IN FLOORING & COATING

DESIGN & SELECTION OF RIGHT SYSTEM

- REQUIREMENTS THAT A FLOOR MAY HAVE TO RESIST



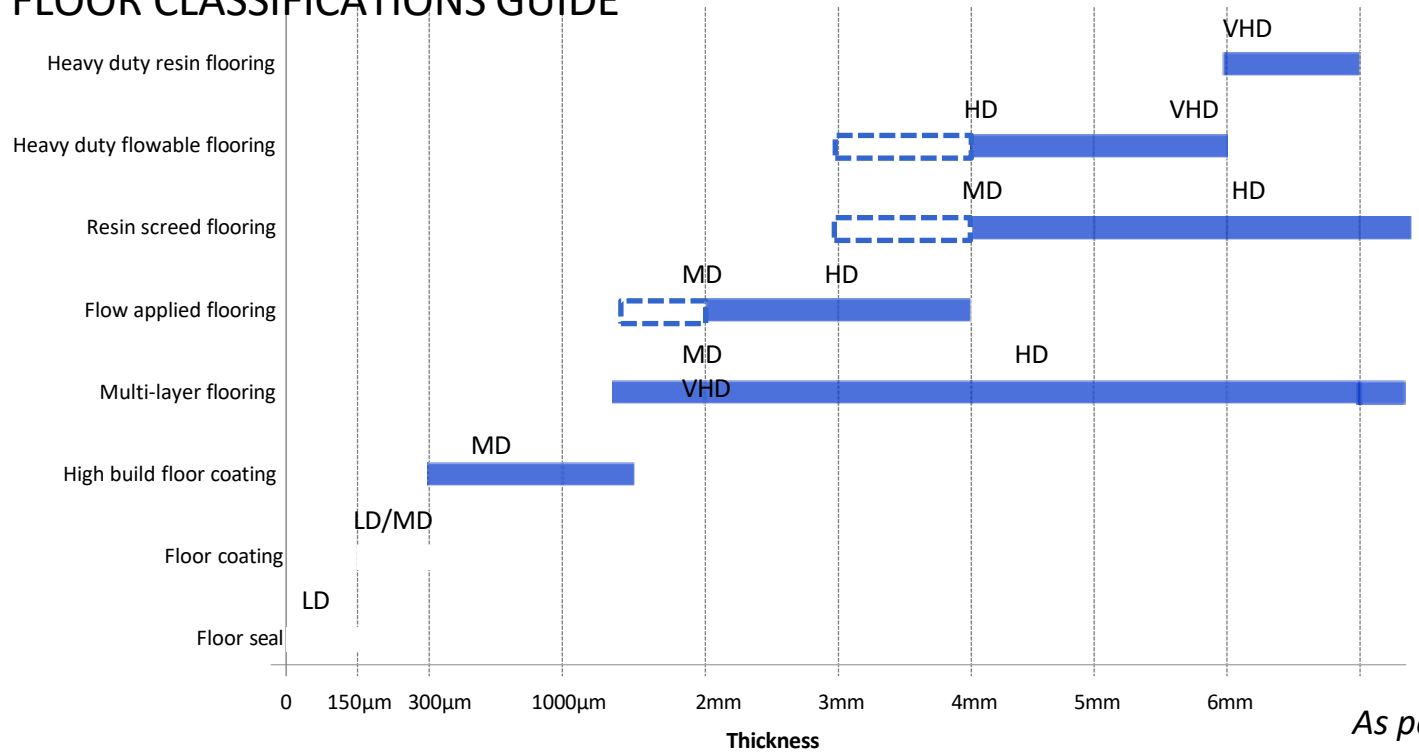

FLOORING & WALL COATING – PROJECT CHECK LIST

1) FLOOR REQUIREMENTS

SIKA IN FLOORING & COATING

DESIGN & SELECTION – TRAFFIC CONSIDERATION

RESIN FLOOR CLASSIFICATIONS GUIDE



As per FeRFA

LD (Light duty)	e.g. light foot traffic, occasional rubber-tyred vehicles
MD (Medium duty)	e.g. regular foot traffic, frequent fork lift truck traffic, occasional had plastic-wheeled trolleys
HD (Heavy duty)	e.g. constant fork lift truck traffic, hard plastic wheeled trolleys, some impact
VHD (very heavy duty)	e.g. severe heavily-loaded traffic and impact

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SIKA IN FLOORING & COATING

MAIN TECHNOLOGIES TYPES

- **Cementitious** Dry Shake Floor Hardeners, Self levelling (Underlayment & Overlayment) & Liquid Type Sealers & Hardeners
- **Epoxy** (EP) Based Coatings, Self levelling, Broadcasted and Screed Systems
- **Hybrid Cementitious (EP or PU Cementitious based)** Modified Based Coatings, Self levelling, Broadcasted and Screed Systems
- **Polyurethane** (PUR) Based Coatings, Self-levelling, broadcasted Systems
- **Vinyl Ester** (VE) Based & **Metacrylate** (PMMA) Fast Cure Coating and Lining Systems

Decorative finish





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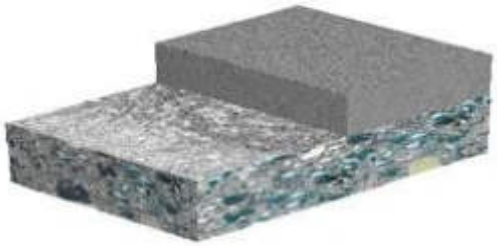
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SIKA IN FLOORING & COATING

SELECTED FLOORING SYSTEMS / BUILD UP – CEMENTITIOUS



- Dry Shake & Liquid Floor Hardener



- Mortar Screed (Dry Mix)



- Mortar Screed incl. Self levelling Mixtures (Underlayment & Overlayment products)

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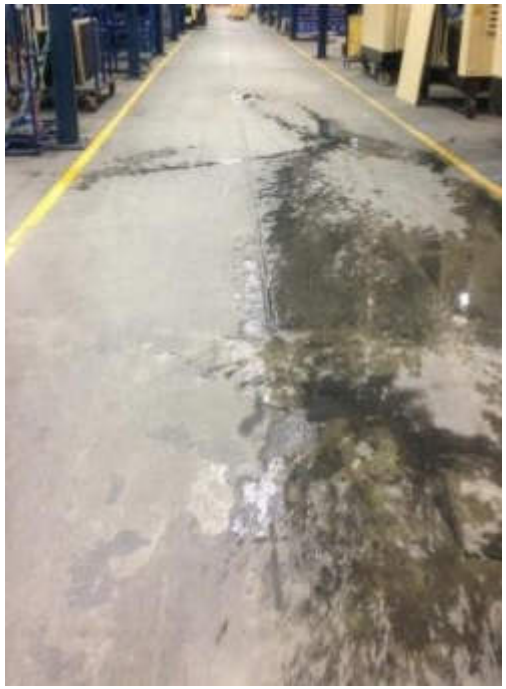
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SIKA IN FLOORING & COATING

SELECTED FLOORING SYSTEMS / BUILD UP – CEMENTITIOUS

Typical main requirements for heavy duty floors





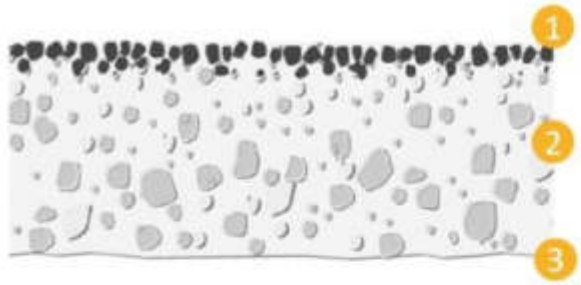


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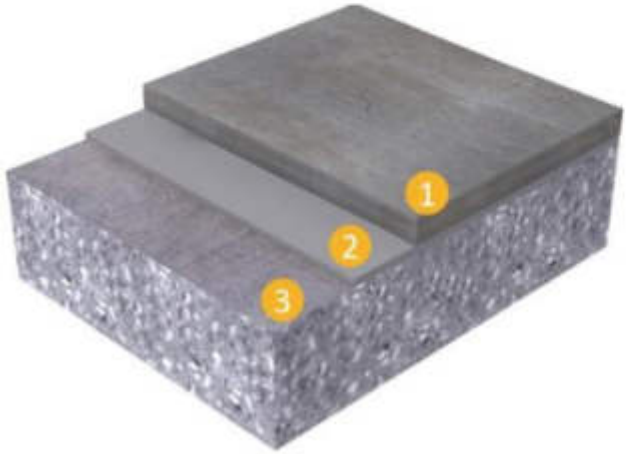
SIKA IN FLOORING & COATING

SELECTED FLOORING SYSTEMS / BUILD UP – CEMENTITIOUS



New Concrete

1. Sikafloor® -2 SynTop-2 SCD
2. Newly poured Concrete
3. Ground



Old Concrete – Refurbishment

1. Sikafloor® -2 Syntop-2 SCD
2. Sikafloor® SCD Primer or Sikadur ® -32 TH
3. Concrete Floor

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SIKA IN FLOORING & COATING

SELECTED FLOORING SYSTEMS / BUILD UP – CEMENTITIOUS

FEATURE	Sikafloor-2 SynTop 2 SCD	Sikafloor-2 SynTop 1 SCD	Sikafloor-2 SynTop 1 SCD HF
Water Content	10%	~8.6%	14%-15%
Aggregate Type	Corundum + Metallic	Alloy	Natural + Alloy
1d Compressive Strength	C50	C40	C20
28d Compressive Strength*	C90	C80	C60
Ease of Application (as Topping)	0	0	++
Dosage kg/m²/mm	2.6	2.4	2.0
Application thickness mm	10-50mm	10-50mm	10-20mm

- Compressive Strength as per JC/T906, ~ 68 MPa (ASTM C109), (as noted in pds)
- > 80 MPa (Sika Quality Control lab Minimum Requirement)(~ASTM C109)

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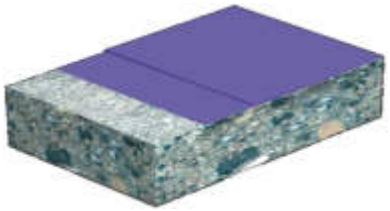
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SIKA IN FLOORING & COATING

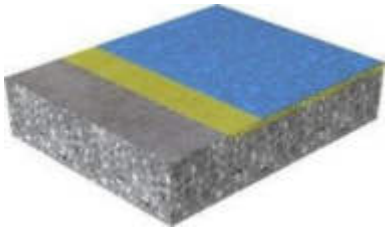
SELECTED FLOORING SYSTEMS / BUILD UP - EPOXY MAINLY



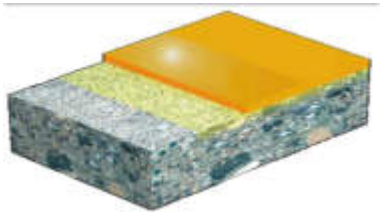
- Seal Coat



- High Build Coat



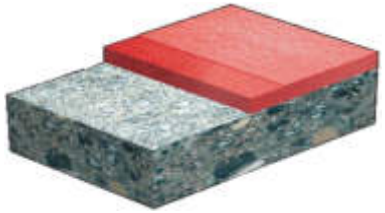
- Textured Coat



- Self Levelling ("SL") Layer



- Broadcast Layer



- Screed



- ESD/ECF Systems

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37

SIKA IN FLOORING & COATING

OUR FLOORING PRODUCTS NAMES

Sikafloor® MultiDur **E**S-24 ECF

Code system: 2 characters

Chemical base of main component

E = Epoxy
P = Polyurethane
A = Acrylic
H = Hybrid
M = One Component (mono)
W = Water based coating system
R = PMMA (rapid)

Sikafloor® MultiDur ES-**24** ECF

Code system: 2 numbers

Numbers

00 – 19	Thin layer system
20 – 29	Medium thickness system (self-levelling coating)
30 – 39	High chemical resistance
40 – 49	NN
50 – 59	Systems with water proofing membrane
60 – 69	Systems with acoustic isolation
70 – 79	NN
80 – 89	NN
90 – 99	NN

Sikafloor® MultiDur ES-24 **S** ECF

Code system: 2 characters

Chemical base of main component

E = Epoxy
P = Polyurethane
A = Acrylic
H = Hybrid
M = One Component (mono)
W = Water based coating system
R = PMMA (rapid)

Surface finish

S = Smooth
B = Broadcast with aggregate
T = Textured
M = Mortar

Sikafloor® Multidur ES-24 **ECF**

Suffix: Indication of special characteristics

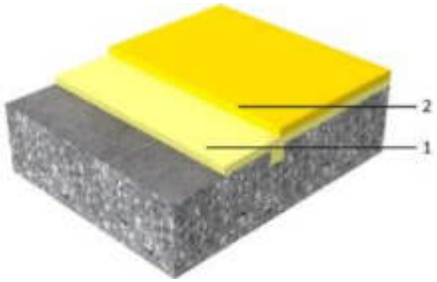
Examples

ECF	Electrostatic Conductive Floor
ESD	Electro Static Discharge
Flake/Quartz/...	for type of decorative filler
Gloss	Glossy finish
EQ	Environmental Quality, higher sustainability performance
HD	Heavy Duty
V	Vertical application
UV	Non yellowing
ECC	Based on Sikafloor EpoCem
Hygienic	Hygienic wall coatings
S	Sika FloorJoint system for standard applications
PD	Sika FloorJoint system for Park Deck
XS	Sika FloorJoint small

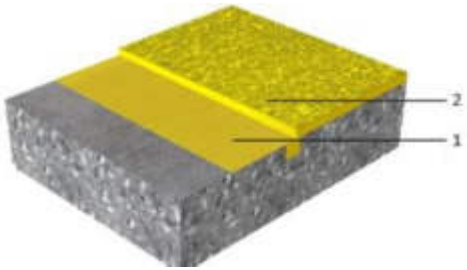
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SIKA IN FLOORING & COATING

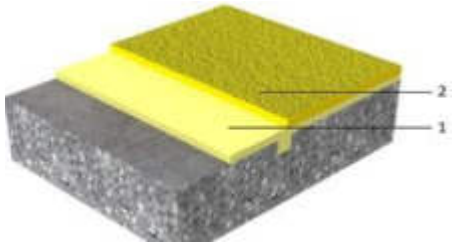
SELECTED FLOORING SYSTEMS / BUILD UP – PU CEMENT MAINLY



- PU Cement Self levelling



- PU Cement Mortar (Thick)



- PU Cement Broadcast system

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SIKA FLOORING & COATING

TECHNOLOGY SUMMARY – FOR INDICATION

- FLOORING TECHNOLOGIES ARE NOT ALL SUITED TO SAME REQUIREMENTS

Characteristics *(For Indication)*	Cement	Epoxy	PU	PU Cement	PMMA	TILES
Mechanical Resistance	0/+	+	0/+	+	+	0/+**
Impact Resistance	-/0	-/0	0/+	+	0/+	-/0
Chemical Resistance	-	+*	+*	+*	0/+*	0/+
Thermal Shock Resistance	-/0	-/0***	-/0***	0/+***	0**	-/0**
Flexibility/Crack Bridging	-	-	0/+	-	+**	-
Hygienic Level	-	+	+	+	+	-(joint)/+
Low Odor (of fresh mixture)	+	0/+	0/+	0/+	-	+
Slip Resistance of smooth finish	-/0	-	0	0/+	0	-
Aesthetic Incl. Color Variety	-	+	0/+**	-/0	0/+**	+
Curing Speed & Application . by low temp.	-	-/0	-/0	0/+	+	0
Cleanability	-	0/+***	0/+***	0/+***	0/+***	-/0***

+: Good/Main feature **0**: Average/Sufficient for most application **-**: Low/Not main feature

* See Chemical Charts ** Depends on Type or Formula ***Depends on Build-Up and/or texture

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40

SIKA IN FLOORING & COATING
SELECTED FLOORING PRODUCT/SYSTEM

- Dry Shake Type Floor Hardener



- Economical & Tough Floor in Warehouse mainly.
- Several Colors Available
- Different Grades of Dry Shake : Standard Type (Quartz Sand): Sikafloor 3 QuartzTop
  Synthetic Type: Sikafloor 2 SynTop-1/-2 type
  Metallic Type: Sikafloor 1 MetalTop type

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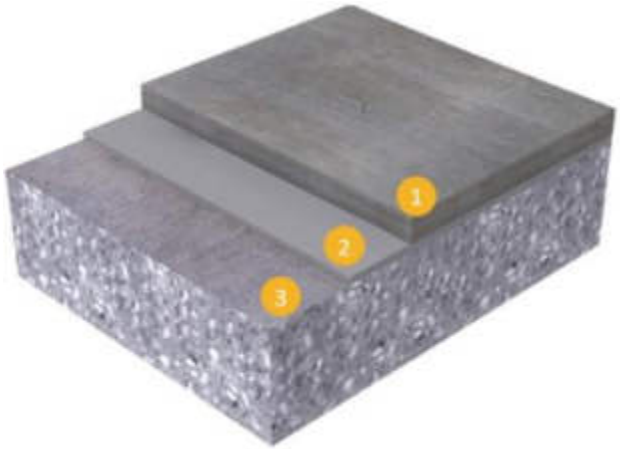
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SIKA IN FLOORING & COATING

SELECTED FLOORING PRODUCT/SYSTEM

- Heavy Duty Screed Cementitious Screed





Old Concrete – Refurbishment
1. Sikafloor® -2Syntop-1 SCD (HF)

Or Sikafloor® -2 Syntop-2 SCD
2. Sikafloor® SCD Primer or Sikadur 732
3. Concrete Floor

New Concrete
1. Sikafloor® -2Syntop-1 SCD (HF)
   or Sikafloor® -2Syntop-2 SCD
2. Newly poured Concrete
3. Ground

- Tough Floor with good impact resistance
- Synthetic Aggregates (i.e. Corundum type) Sikafloor 2 SynTop-1/-2 SCD

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SIKA IN FLOORING & COATING

SELECTED FLOORING PRODUCT/SYSTEM – EVER USED IN MINING

- PU Cementitious System (typically 4 – 6 mm thick)



Sikafloor 21 PurCem (LP) : Body Coat
Or Sika UCRETE MF
Sikafloor 21 PurCem (LP) : Scratch coat
Or Sika UCRETE MF

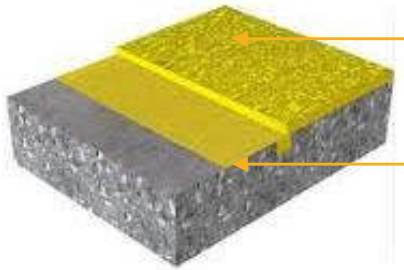
- HACCP Attestation of Sikafloor Purcem / Sika Ucrete
- Typical Slip Resistance. R10 SRV Dry 70, SRV Wet 60 (Rapra 4S Slider)
- CSM – Biological Resistance - Riboflavin test

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SIKA IN FLOORING & COATING
SELECTED FLOORING PRODUCT/SYSTEM

- PU Cementitious System (typically 7 to 9 mm thick)



Sikafloor 20 PurCem (LP) : Body Coat
Or Sika Ucrete UD 200

Sikafloor 21 PurCem (LP) : Scratch coat
Or Sika Ucrete MF

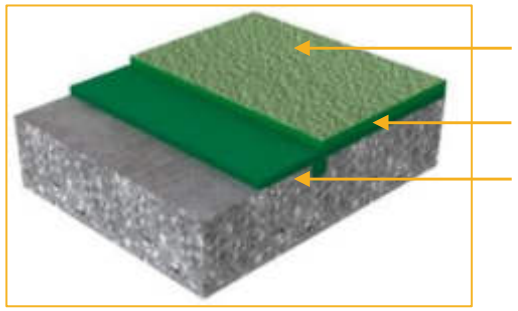
- HACCP Attestation of Sikafloor Purcem / Sika Ucrete
- High Impact Resistance
- High Temperature resistance and thermal shocks
- Typical Slip Resistance. R11, SRV Dry 70, SRV Wet 65 (Rapra 4S Slider)
- CSM – Biological Resistance - Riboflavin test

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SIKA IN FLOORING & COATING
SELECTED FLOORING PRODUCT/SYSTEM

- PU Cementitious System (typically 5 to 6 mm thick)
- Or Sika Ucrete DP System



Sikafloor 31 PurCem (LP) : Topcoat
Sikafloor 21 PurCem (LP) : Body coat with Sand Broadcast
Sikafloor 21 PurCem (LP) : Scratch coat

- HACCP Attestation of Sikafloor PurCem and Sika Ucrete
- Typical Slip Resistance. R12 to R13

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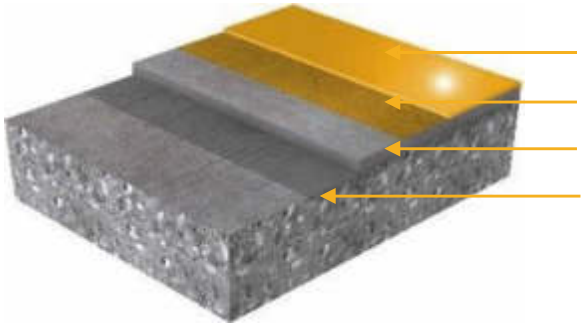
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45

SIKA IN FLOORING & COATING
SELECTED FLOORING PRODUCT/SYSTEM



- Epoxy Cement Sikafloor 81 EpoCem System (Min. 2 mm thick needed)



- Temporary Moisture based on Epoxy-Cement Hybrid combination. (3 components)
- Becomes a permanent barrier solution when sealed with an impervious epoxy system
- To be used only in case of high moisture in substrate and/or of rising moisture prior to Impervious systems (mostly epoxy)
- 35 Years of successful projects Worldwide

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SIKA IN FLOORING & COATING
SELECTED FLOORING PRODUCT/SYSTEM



- **Sikafloor 81 EpoCem** System: Saving Time Solution

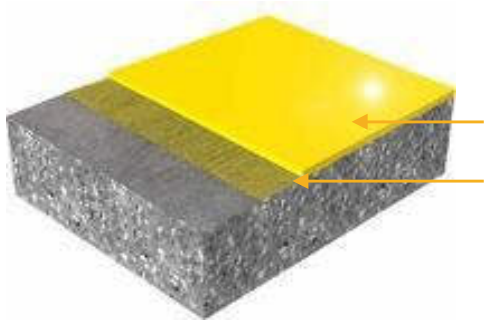


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47

SIKA IN FLOORING & COATING
SELECTED FLOORING PRODUCT/SYSTEM

- **Epoxy Self Leveling (1.5 – 2mm thick) : Sikafloor 263 SL HC** System

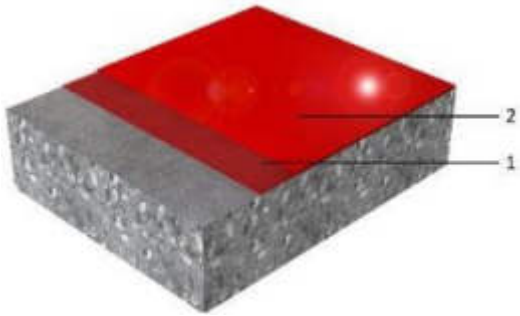


Sikafloor 263 SL HC : Self levelling pigmented Epoxy
Sikafloor 161: Epoxy Primer

  - Slip Resistance: Low as smooth system
  - CSM – Particle Emission
    - OutGassing
    - Biological Resistance
    - Riboflavin test
  - LEED v4 Attestation. VOC Content & VOC Emission

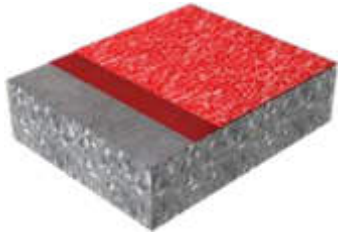
SIKA IN FLOORING & COATING
SELECTED FLOORING PRODUCT/SYSTEM

- **Epoxy Roller Coat (less than 0.6 mm thick typically) : Sikafloor 264 HC** System



Sikafloor 264 : Pigmented Epoxy Topcoat(s)
Sikafloor 161: Epoxy Primer

- Slip Resistance: Low as smooth system Up to Medium High as Textured systems
- CSM – Particle Emission
  OutGassing
  Biological Resistance
  Riboflavin test
- LEED v4 Attestation. VOC Content & VOC Emission



- Textured & thicker Systems Also available

SIKACOR® SYSTEMS SOLUTIONS

What are SikaCor® ?

-Special cements, laminate systems and coatings to protect surfaces from highest mechanical, chemical and thermal stress – based on synthetic resins

Where are SikaCor® used?

- Everywhere where it is necessary to protect against aggressive & high concentrated acids, alkaline, solvents, oxidising agents and other chemicals at high temperatures







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SIKACOR® VEL TH LAMINATE

- Formulated vinyl-ester resin with *extended pot-life*
- Ready to apply 2 K system [minimize mixing failure]
- Grey top coat color available

- Chemical resistance: to solvents, acids (organic & in-organic), alkaline,
- oxidizing agents [**Group 1, 2, 3,4, 4a, 4b, 5, 5a, 6, 6b, 7, 7a, 8, 9, 9a, 10, 11, 12, 13,14, 15 and 15a**]

- Examples: Nitric Acid 30%, HF 20%,
- Sodium Hypochloride 12%
- Nitric Acid 30%, HF 20%
- Chlorine Dioxide
- Sulfuric Acid 80%
- Methyl-Di-Ethanol-Amine 100%
- Methyl Ethyl Ketone MEK 100% (spillage only)
- Phenol 2% (spillage only)]



- Temperature resistance: **up to 100 C**
- Mechanical resistance: medium to high
- Abrasion resistance: medium to high
- Crack-bridging: hair cracks up to 0,2 mm (0,6 mm)

- Color : brown-gray / optional grey

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SIKACOR® VEL LAMINATE

- Formulated vinyl-ester resin
- Ready to apply 2 K system [minimize mixing failure]
- Grey top coat color available

- Chemical resistance: To solvents, acids (organic & in-organic), alkaline,
- oxidizing agents [**Group 1, 2, 3,4, 4a, 4b, 5, 5a, 6, 6b, 7, 7a, 8, 9, 9a, 10, 11, 12, 13,14, 15 and 15a**]

- Examples: Nitric Acid 65%, HF 50%,
- Sodium Hypochloride 12%
- Nitric Acid 65% + HF 3%
- Chlorine Dioxide
- Sulfuric Acid 90% (spillages up to 96%)
- Methylene Chloride (spillage only)
- Methyl Ethyl Ketone MEK (spillage only)
- Phenol (spillage only)

- Temperature resistance: up to 100 C
- Mechanical resistance: medium to high
- Abrasion resistance: medium to high
- Crack-bridging: hair cracks up to 0,2 mm (0,6 mm)

- Color : brown-gray / optional dark-gray


Source: Sika Services

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SIKACOR® SYSTEMS SOLUTIONS

	SikaCor VEL	SikaCor VEL TH	SikaCor ET
Chemical Resistance	++	(++)	+
Nitric Acid	65%	30%	30%
Sulfuric Acid	90%	80%	75%
HCl Acid	37%	37%	37%
NaOH	50%	50%	50%
Pot Life (30^0C)	~ 10 min.	~ 10 min.	~ 12-15 min.
Color	Brownish Transparent	Brownish Transparent	Brownish Transparent

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SIKA IN FLOORING & COATING

ADDITIONAL DESIGN CONSIDERATION – CARE & MAINTENANCE





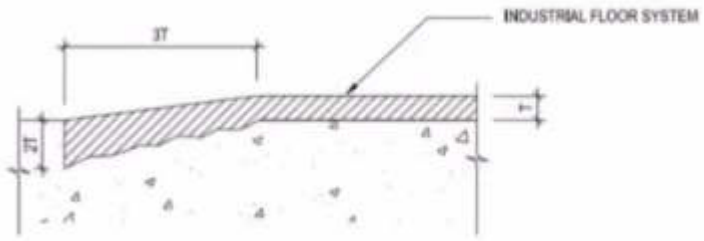
- **CLEANING**
  - Manual
  - Mechanical
  - Chemicals
  - High pressure
  - Steam
- **MAINTENANCE GUIDELINE**
  - Frequency
  - Type

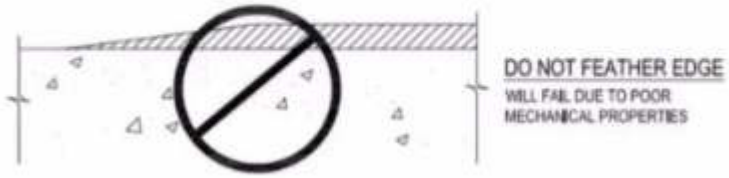
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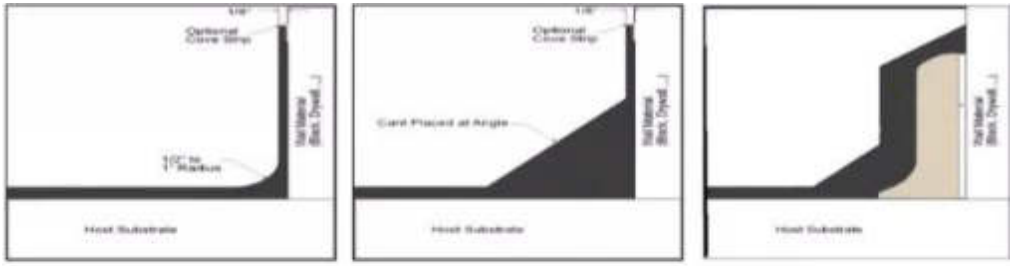
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SIKA IN FLOORING & COATINGS

ADDITIONAL DESIGN CONSIDERATION – TERMINATION & DETAILING







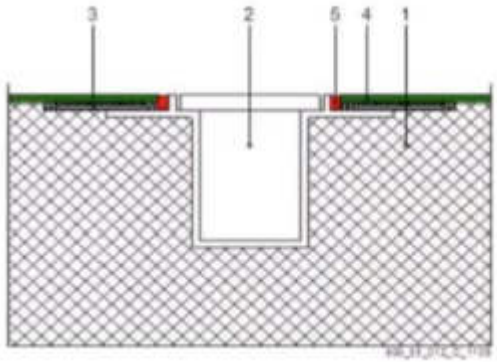






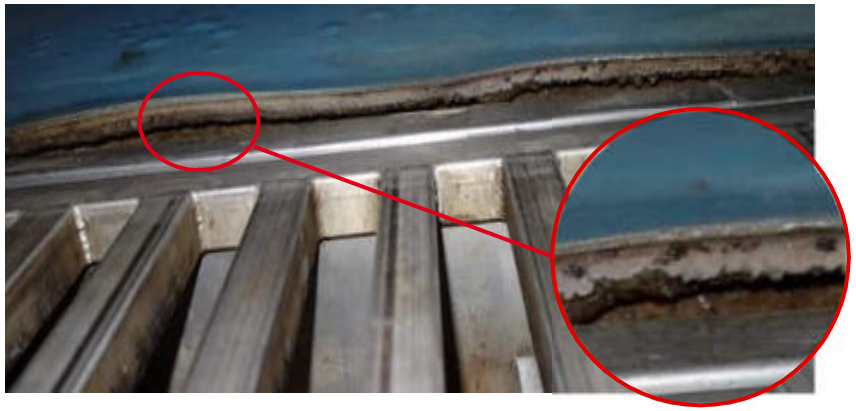
SIKA IN FLOORING & COATING

ADDITIONAL DESIGN CONSIDERATION – TERMINATION & DETAILING



1. Existing substrate prepared in accordance to the most recent issue of local Product Data Sheet for the product concerned.
2. Drainage Channel.
3. Waterproof connection if required.
4. Selected Flooring System.
5. Flexible joint sealant.



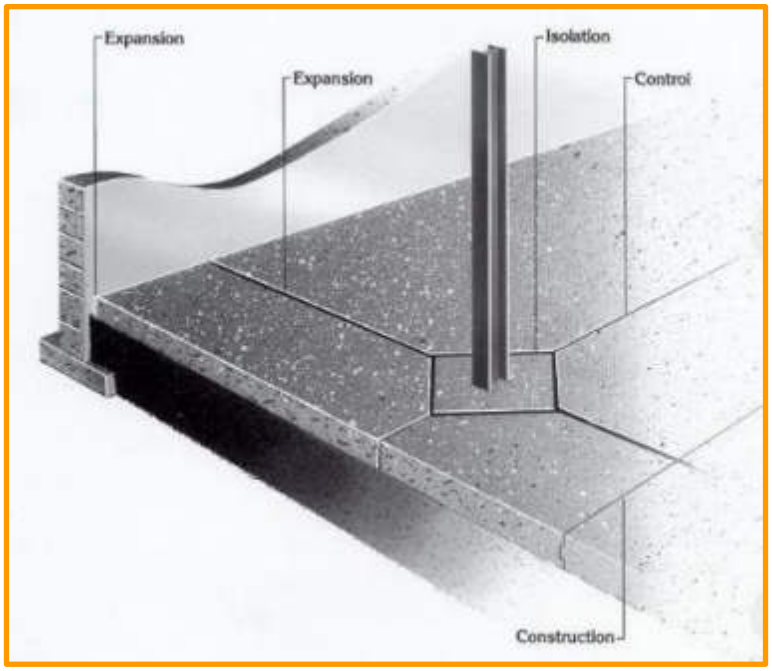




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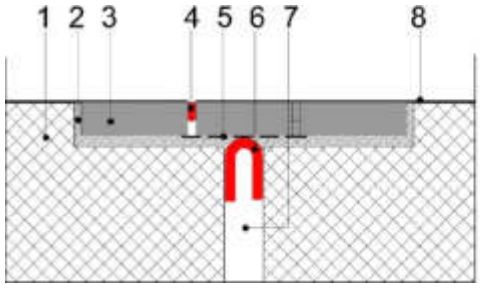
SIKA IN FLOORING & COATING

ADDITIONAL DESIGN CONSIDERATION – TERMINATION & DETAILING









1. Existing substrate prepared in accordance to the most recent issue of local Product Data Sheet for the product concerned.
2. Epoxy Adhesive
3. FloorJoint
4. Flexible Sealant
5. Masking tape. The tape is an essential measure in order to allow a movement between the two plates of the joint panel.
6. Screed edge stripping or backer rod
7. Maximum width of joint is 3.0 cm
8. Flooring system

- Sika floorJoint Profiles

57

SIKAFLOOR® REFERENCE IN MINING

Country	Client/ Owner	Project	Customer	Year	Product	Quantities m2
Indonesia	Freeport Indonesia	Warehouse	Freeport Indonesia	2009-2011	Sikafloor®-3 SynTop 2 SCD	10,000
Indonesia	Freeport Indonesia	Institutional area	Freeport Indonesia	2010	Sikafloor®-261 ID	1,500
Indonesia	Freeport Indonesia	Warehouse MP 72	Anggun Permai	2013	Sikafloor®-21 PurCem FG	1,880
Indonesia	Freeport Indonesia	Bus Terminal MP 72	Anggun Permai	2013-2014	Sikafloor®-21 PurCem FG	6,000
Indonesia	Freeport Indonesia	Melati Mass Kitchen	Anggun Permai	2014	Sikafloor®-21/20 PurCem FG	1,400
Indonesia	Freeport Indonesia	OB 4 Building	Anggun Permai	2014	Sikafloor®-21 PurCem FG	6,000
Indonesia	Freeport Indonesia	Underground	Anggun Permai	2015	Sikafloor®-263 SL FP HC	6,500
Indonesia	Freeport Indonesia	Underground	Anggun Permai Tekindo	2016	Sikafloor®-263 SL FP HC	8,000
Indonesia	Freeport Indonesia	Superlock MP 72	Anggun Permai Tekindo	2016	Sikafloor®-21 PurCem LP, 20 PurCem LP	3,500
Indonesia	Freeport Indonesia	Underground	Anggun Permai Tekindo	2022	Sikafloor®-263 SL FP HC	8,000
Indonesia	PT Freeport Indonesia	Underground	Anggun Permai Tekindo	2022	Sikafloor®-20 PurCem LP, 21 PurCem LP, 31 PurCem	800

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BACK TO PRESENTATION

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SIKA COATINGS FOR MINING

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60

SIKA COATINGS FOR MINING

- Sika has a wide range of coatings for different applications. In mining, coatings are applied to different structures to increase the chemical resistant for ore processing and containment structures and as well as for shielding them against mechanical abrasion
- Products are applied using spraying equipment or through manual application
- The goal is to increase the life time of the structures and to reduce downtime
- For Sika, the coating business is usually coupled with high margins
- Specialized applicators are used by Sika to apply the products

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COMMON MINING INFRASTRACTURE THAT NEED COATINGS

- Floatation cells
- Thickening units
- CIL leach tanks
- Ore Bins
- Coke Bins
- Concentrate Bins
- Fuel storage tanks
- Retentian basins
- Overflow basins
- Secure containment structures for Acid storage tanks

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REQUIREMENTS FOR TANK AND BUND LININGS

- prevent the concrete/steel from chemical attack of the stored liquid
- prevent the stored liquid from leaching substances out off the concrete/steel structure
- long term chemical and abrasion resistance to the stored liquid
- possibly crack bridging properties to prevent leakages even if the substrate fails
- easy to clean surface to shorten time for cleaning regimes
- possibly thermal resistance in case of elevated temperature of the stored liquid



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STANDARDS FOR TANK AND BUND LININGS

Standards:

- no unified standards available
- required testing depends on the region or country
- many companies have own specifications
- it is difficult to have and to provide the requested test reports
- project related adjustments are often necessary

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PROTECTIVE COATINGS

Lining of Containment Structures



Sika offers a variety of sprayed membranes for sealing containment structures in mining

Corrosion Protection of Steel Structures



Sika also developed highly effective products for the corrosion protection of steel works for every environment.

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65

LINING OF CONTAINMENT STRUCTURES



In this picture; BHP copper concentrate storage facility, Porto Coloso, Antofagasta



Sika's coating products protect specific mining infrastructure on a large scale such as process water retention basins, thickeners, leach tanks, flotation cells and tailings processing and storage facilities.

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SOME EXAMPLES



Mill-slurry thickeners for dewatering of the waste material coming from the Kiruna Mill complex, Kiruna Mine, LKAB, Sweden

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SOME EXAMPLES

Repair of drains around thickener

Sikalastic®-841 ST 3224 kg

SikaQuick®-506 1000 kg

6 000 m^3 water / h





Mill-slurry thickeners for dewatering of the waste material coming from the Kiruna Mill complex, Kiruna Mine, LKAB, Sweden

SOME EXAMPLES



- Mill-slurry thickeners for dewatering of the waste material coming from the El Teniente Mill complex, El Teniente Mine, Codelco, Chile

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SOME EXAMPLES



Waste slurry thickener, Grasberg Mine, Freeport McMoran, Indonesia

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70

SOME EXAMPLES





CIL tank lining at ….Mine processing facility, Ghana, Secondary containment bund lining around CIL leach tanks.

SOME EXAMPLES



Kansanshi Mine construction, First Quantum, Zambia. Large thickeners in the background.

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SOME EXAMPLES



Copper concentrate storage facility for concentrate from Escondida Mine, BHP Billiton, Puerto Coloso, Chile

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SOME EXAMPLES

- Sikalastic®-841 ST
- 3.000 m^2





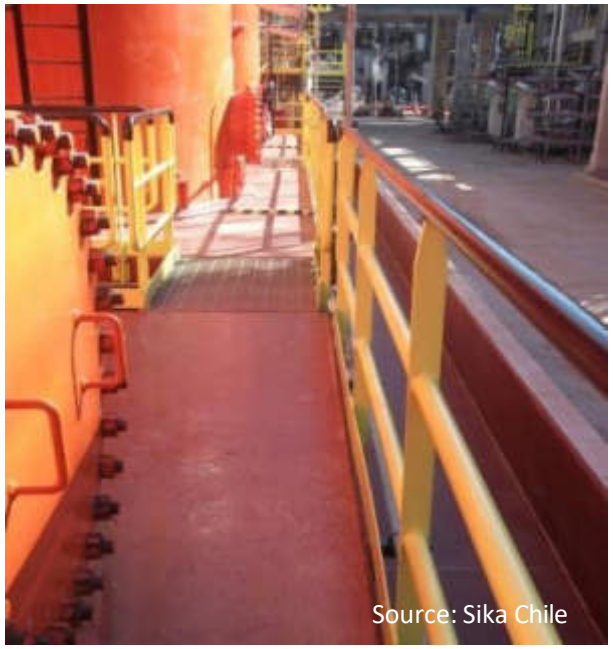
Copper concentrate storage facility (wet conc.), Mina Mantos Blancos, Anglo American, Antofagasta, Chile

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SOME EXAMPLES

- Sikaguard Antiacido
- approx. 3000 m²





Secondary containment Bund for hydrocarbon storage tanks, RPC, Concón/ Enap, Chile

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TECHNOLOGIES POSISITIONING VERSUS CHEMICAL RESISTANCE REQUIREMENTS

Product base	Chemical resistance	Product Name
Flexible PUA	Moderate to Good	Sikalastic®-8800 Sikalastic®-841 ST
Tough PUA	Moderate	Sikalastic®-844XT
Epoxy	Moderate to Good	Sikagard®-63N, Sikagard®-62,
Vinylester	Excellent	SikaCor® VEL

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PRODUCT CHARACTERISTICS SIKALASTIC®-8800

Metrics	Sikalastic®-8800		
Chemical resistance	Not resistant to biogenic sulphuric acid	Tensile strength	> 10 N/mm²
Technology	Flexible PUA, pure polyurea	Abrasion resitance	~400mg (H22/1000 g/1000 cy)
UV Resitance	NA	Application temperature	-20°C / +50°C
Mode of application	Sprayed	Sevice temperature	-30°C / + 100°C dry
Gel time	~11 seconds at + 20 °C	Crack Bridging	Class A5 Static Class B4.2 Dynamic
Waiting time overcoating	1–2 minutes at +20 °C	Consumption	~1.05 kg /m² / mm
Curing time	24 hours at +20 °C	Min thickness sytem	> 2mm
Solide content	>99%		
Shear strength	NA	Certification	Kiwa Polymer institute GmbH BBA, CE Marking EN 1504-2
Shore D	> 50	Sustainability	LEED v4 MRc2, V2009 IEQc 4.2

PRODUCT CHARACTERISTICS SIKALASTIC®-841 ST

Metrics	Sikalastic®-841 ST		
Chemical resistance	Not resistant to biogenic sulphuric acid	Tensile strength	> 15 N/mm²
Technology	Flexible PUA, pure polyurea	Abrasion resitance	~230mg (H22/1000 g/1000 cy)
UV Resitance	NA	Application temperature	+1°C / +50°C
Mode of application	Sprayed	Sevice temperature	-30°C / + 100°C dry
Gel time	~11 seconds at + 20 °C	Crack Bridging	Class A5 Static Class B4.2 Dynamic
Waiting time overcoating	1–2 minutes at +20 °C	Consumption	~1.05 kg /m² / mm
Curing time	24 hours at +20 °C	Min thickness sytem	> 2mm
Solide content	>99%		
Shear strength	NA	Certification	Kiwa Polymer institute GmbH BBA, ETA 033
Shore D	45 - 50	Sustainability	Conforms LEED EQ Credit 4.2
Elongation at break	~360%		

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78

PRODUCT CHARACTERISTICS SIKAGARD-7000 CR (FOMERLY MASTERSEAL 7000 CR)

Metrics	Sikalastic®-841 ST		
Chemical resistance	Resistant sulphuric acid (50%)	Tensile strength	> 20 N/mm²
Technology	Flexible tough Xolutec	Abrasion resitance Taber	194 (required < 3000)
UV Resitance	NA	Application temperature	+1°C / +35°C
Mode of application	Hand applied	Sevice temperature	-20°C / + 80°C dry Up to + 60°C wet
Workability time (5 kg/kit)	~15 min at + 30 °C	Crack Bridging	Class A2 (+70°C) Static Class B3.1 (+23°C) Dynamic
Waiting time overcoating	4 hours at +20 °C	Consumption	~2 x 1.10-1.20 kg /m²
Curing time	7 days at +20 °C	Min thickness sytem	1.0 – 1.1 mm DFT
Solide content	NA		
Shear strength	NA	Certification	CE Surface protection product / coating EN 1504-2: Principles 1.3 / 2.2 / 5.1 / 6.1 / 8.2
Shore D	80		

PRODUCT CHARACTERISTICS SIKALASTIC®-844 XT

Metrics	Sikalastic®-844 XT / 7000 CR		
Chemical resistance	Resistant to biogenic sulphuric acid	Tensile strength	> 10 N/mm²
Technology	Tough PUA, pure polyurea	Mechanical strength	~100mg (H22 17/1000 gr /1000 cy)
UV Resitance	Yes	Application temperature	-15°C / +50°C
Mode of application	Sprayed	Sevice temperature	-60°C / + 100°C dry
Gel time	5 – 15 seconds +20 °C	Crack Bridging	Class A4 Static Class B3.1 Dynamic
Tack Free	60 - 120 seconds	Consumption	~1.08 kg /m² / mm
Curing time	24 hours	Min thickness sytem	> 2mm
Solide content	>99%	Certification	Kiwa Polymer institute GmbH KB-Hoch (reaction to fire)
Shore D	45-50	Sustainability	LEED v4 MRc2, V2009 IEQc 4.2
Elongation at break	50-100%		

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PRODUCT CHARACTERISTICS SIKAGARD®-63 N

Metrics	Sikagard®-63 N		
Chemical resistance	Moderate to Good	Tensile adhesion strength	> 1.5 N/mm² (to concrete) > 15 N/mm² (to steel) > 10 N/mm² (to aluminium)
Technology	Epoxy	Application temperature	+10 °C min. / +30 °C max.
UV Resitance	NA	Sevice temperature	Permanent + 40°C dry heat Max 3 days + 60°C dry heat
Mode of application	Airless Sprayed, Brush, Roller	Consumption	~0.15 kg /m² / layer DFT = 0.1 mm
Waiting time overcoating	5 hours at +20 °C, 4 hours at +30 °C	Min thickness sytem	Min 3 layers
Curing time	9 days at +20 °C 5 days at +30 °C	Certification	CE Marking DoP EN 1504-2
Solide content	100%	Sustainability	LEED v4 MRc4, V2009 IEQc 4.2
Pot life	30 min at +10 °C 20 min at +20 °C 10 min at +30 °C		

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PRODUCT CHARACTERISTICS SIKAGARD®-62

Metrics	Sikagard®-62		
Chemical resistance	Medium to moderate	Tensile adhesion strength	> 1.5 N/mm² (to concrete)
Technology	Epoxy	Application temperature	+8°C min. / +40°C max.
Mode of application	Airless Sprayed, Brush, Roller	Sevice temperature	Permanent + 50°C dry heat Max 7 days + 80°C dry heat
Shore D	~80	Consumption	~0.30 kg /m² / layer DFT = 0.2 mm
Waiting time overcoating	10 hours at +20 °C, 6 hours at +30 °C	Min thickness sytem	Min 3 layers
Curing time	10 days at +20 °C 5 days at +30 °C	Mechanical strength	~560 mg (H22/1000 gr /1000 cy)
Solide content	100%	Certification	CE Marking DoP EN 1504-2
Pot life	30 min at +10 °C 20 min at +20 °C 10 min at +30 °C	Sustainability	LEED v2009 IEQc 4.2

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PRODUCT CHARACTERISTICS SIKACOR®-VEL

Metrics	SikaCor®-VEL		
Chemical resistance	Excellent	Pot life	30 min at +15 °C
Technology	Conductive vinylester laminate system	Application temperature	+5°C min. / +30°C max.
Mode of application	Troweling, laminating, rolling = undiluted	Sevice temperature	Dry heat up to approx. + 100°C
Tensile strength at break	~73 N/mm² (horizontal layer)	Chemial resistance	Additional building inspectorate approval for the following materials: - hydrochloric acid ≤ 37 % - sulfuric acid ≤ 70 % - nitric acid ≤ 65 % - aqueous sodium hypochlorite (12 % active chlorine) - hydrogen peroxide ≤ 30 % - chromic acid ≤ 50 %
Crack bridging ability	Up to max 0.2 mm		
Pot life	30 min at +15 °C		

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TANK & BUND LINING

Sikalastic®-8800

- Pure Polyurea, spray applied, membrane
- SL-8800 applicable with Polyurea spray equipment
- Polyurea for universal use
- UV-resistant but not UV-stable
- SL-8800 for tank lining jobs in open spaced areas with aggressive water, i.e. sewage plants, thickeners
- SL-8800 as abrasion resistant coating in multi-purpose use



Source: Sika Services

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84

TANK & BUND LINING

Sikalastic®-844 XT

- Pure Polyurea, spray applied, membrane
- SL-844 XT applicable with Polyurea spray equipment
- Polyurea with excellent chemcial resistance
- UV-resistant but not UV-stable
- SL-844 XT for bund lining
- SL-844 XT for tank lining jobs in confined areas with aggressive water and atmosphere, i.e. digester tanks, leach tanks


Source: Sika Services

TANK & BUND LINING

Sikalastic®-821 ST

- Very fast reactivity and curing time
- Almost immediate return-to-service time
- Applicable in temperatures from +1°C to +50°C
- Performs in constant dry temperatures from -30°C to +100°C
- Excellent crack bridging properties
- Good chemical resistance
- Excellent abrasion resistance
- UV light exposure may lead to yellowing and chalking

- Not resistant to biogenic sulphuric acid



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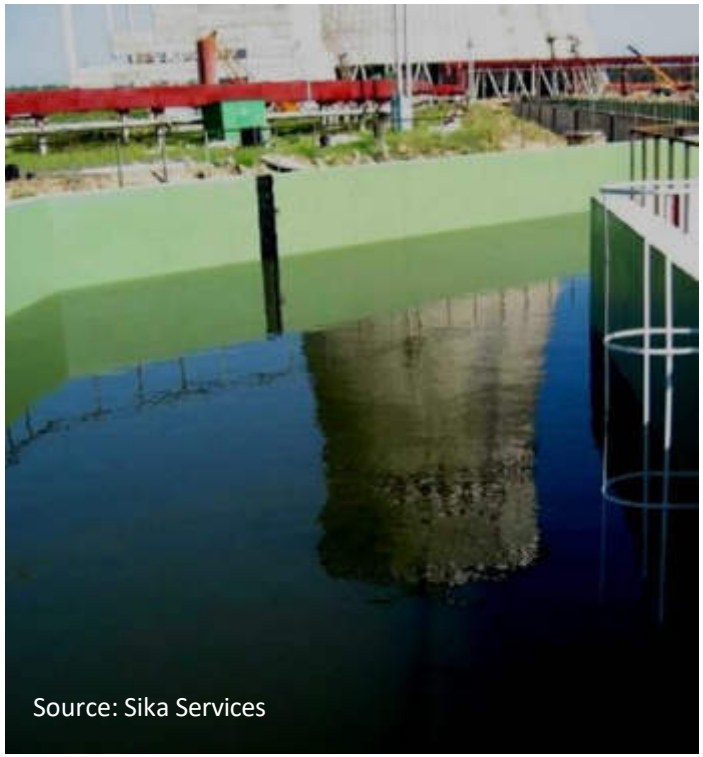
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86

TANK & BUND LINING

Sikagard®-62/ 63 N

- Use in sewage plants, fermenters, process water and waste water installations
- Epoxy resin based coating systems
- solvent free
- 3 layer system, each layer approx. 150µm
- Can be reinforced with fabric
- good chemical resistance, not resistant to BSA (biogenic sulphuric acid)



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87

TANK & BUND LINING

SikaCor® VEL

- Vinyester resin based coating system
- Use in harsh chemical environment
- glass fabric reinforced
- excellent chemical resistance
- chemical resistant to concentrated acids and oxidising agent
- crack bridging due to reinforcement

Sikaguard Anti Acido

- Epoxy Novolac combination provided from Sika Chile



Source: Sika Services

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88

EQUIPMENT FOR PU/PUA

Application of the sprayed PU/PUA membranes are straight forward using dedicated spraying/pumping equipment and specialised contractors available in most countries







BACK TO PRESENTATION

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90



UNDERLAYNMENT GROUTING

CEMENTITIOUS, EPOXY & PU READY TO USE GROUT

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91

UNDERLAYMENT GROUTING





Heavy machinery such as mills, crushers, hoists, drives and electricity generators are key components in large scale mining operations.

Grouts play an important role in transferring vibrations and heavy loads to the foundations.

Expanding, non-shrinking grouts, fill tight voids with highly effective bearing areas, fast and with high strengths.

Sika offers a whole range of grouting solutions including epoxy, cementitious, and polyurethane materials.

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92

UNDERLAYMENT GROUTING

Description	Epoxy Based	PU based	Cement Based
Modules	3 component (No components to measure)	2 component (No components to measure)	1 component (must adjust water for flow & strength)
Type Loading	dynamic / vibration (low E-Modulus 10 – 20 GPa)	vibration absorbing, load-bearing, flexible grout	static loads (high E-Modulus > 25 – 40 GPa)
Working Temp.	max. 60°C	min. -40°C max. 80°C	> 100 to ~150°C (continuous with no cycles)
Shrinkage	non-shrink	non-shrink	shrinkage compensated
Type of Curing	none required	none required	must be protected
Tensile Strength	high	3 N/mm2	low ~10% of CS
Bond Strength	High to steel and concrete	Good	Good
Curing	Fast (24 – 36 hours)	Fast (4 h – 23°C – Shore A 45) Final Shore A 70	Slow (7 – 28 days)
Performance	Impervious to most liquids, chemicals lubricants and freeze thawcycles	Can be affected by Mineral oils, diesel fuel	Can be effected by all
Sika Production	centralised	centralised	local

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GROUTING MACHINE BASES



- Filling small voids under machinery bases.
- Machinery and equipment which have precise tolerances for alignment or require uniform support cannot be placed directly on finished concrete surfaces.
- Both the concrete surface and the machine base have irregularities which result in alignment difficulties and bearing load concentrations.

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BASE PLATES





- As the concrete surface and the machine base have irregularities, machine bases or soleplates are aligned and leveled by shimming or other means and the resulting space between the machine base and the foundation filled with a load transfer material.

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BRIDGE BEARINGS





- All cementitious systems will shrink and only through the introduction of various admixtures that create expansion and, thereby, compensate for the expected drying shrinkage within the cementitious matrix. Therefore , the term “shrinkage -compensating” is more accurate than “non-shrink”.

DOWELING BARS



Anchor reinforcing bars for continuity (EN 1504-6)

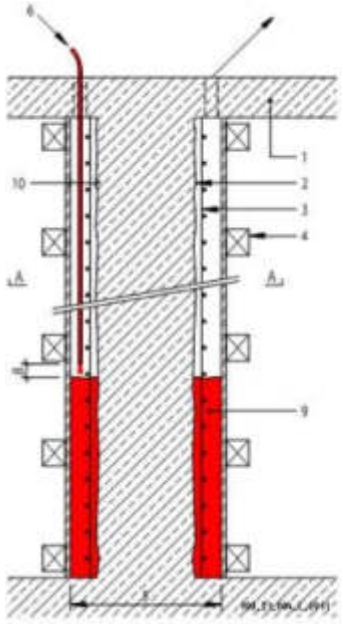
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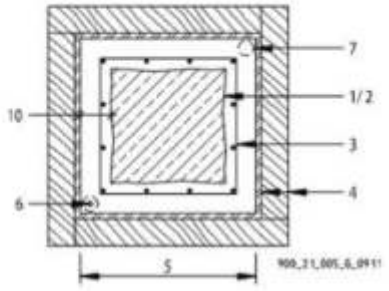
PERFORMANCE RELEALTED ANCHORING, CHEMICAL ANCHORS USING RESIN GROUT IN TENSION IS CLASSIFIED BY SIKA AS A "DEMANDING *APPLICATION*

VOIDS FILLING



- Another typical use for a cementitious grout is to fill the voids between precast concrete segments.
- Example : Grouting step irons in pre-cast concrete access chamber rings.





Section A – A

1	Host concrete structure
2	Prepared concrete substrate
3	Cleaned and prepared steel reinforcement bars
4	Formwork
5	Extent of refurbished concrete
6	Opening made in existing structure for material application
7	Opening made in existing structure for air release
8	Maintain application ~ 2 cm above material level
9	Pouring mortar suitable for methods 3.2, 4.4, 7.1 and 7.2 to European Standard EN 1504-9 e.g. SikaGrout®-318
10	Application thickness

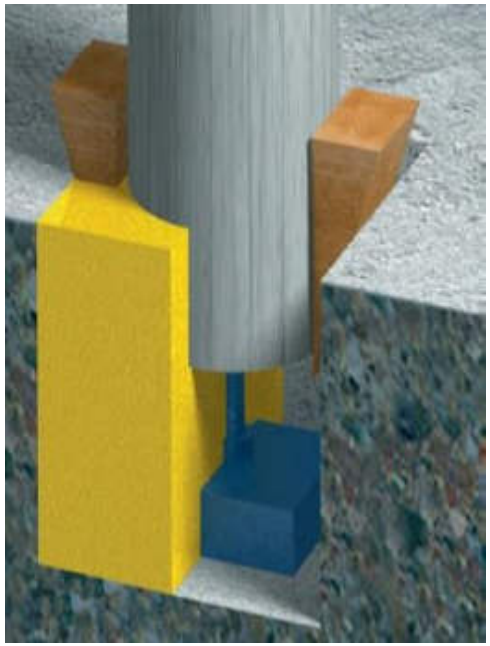
- Concrete repair by pouring Cementitious grout or.. .. ..
- Pouring Extended grout.

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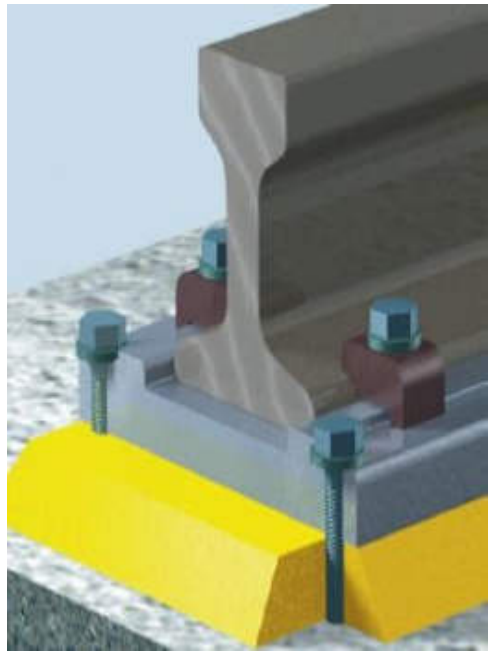
PRECISION GROUTING



Heavy Equipment / Machinery Bases



Pre-cast Column Bases



Rail Track Bedding *(Rigid system)*

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99

ANCHORING AND FIXING



Anchor Bolts / Bars



Bearing Plates and Pads



Rail / Post Fixing

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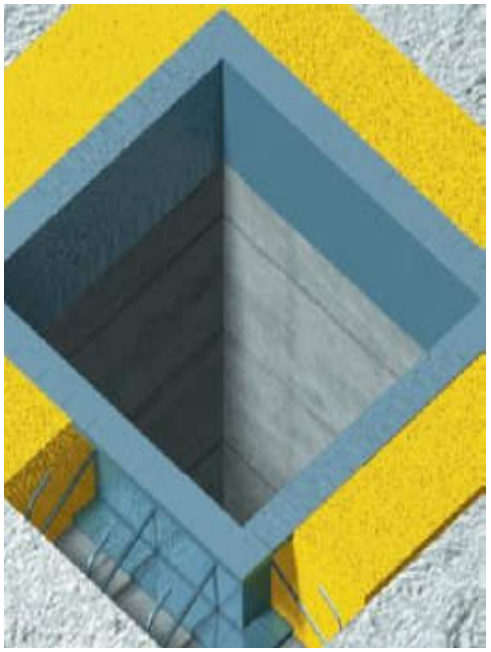
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CONCRETE RESTORATION /STRENGTHENING



Increasing Thickness



Steel Frame Bedding



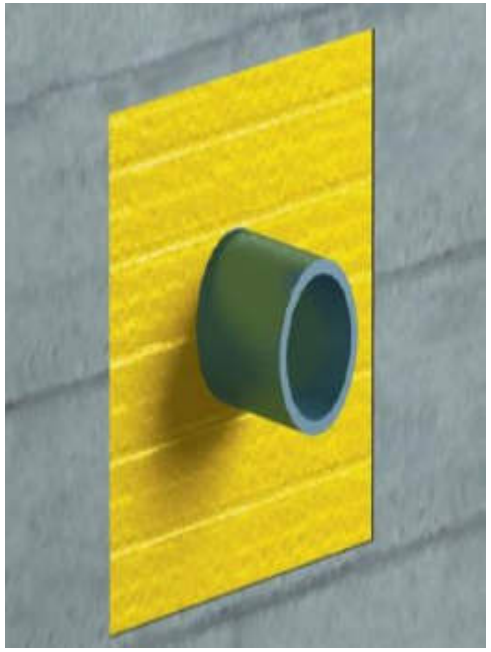
Replacement of Steel Joints

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101

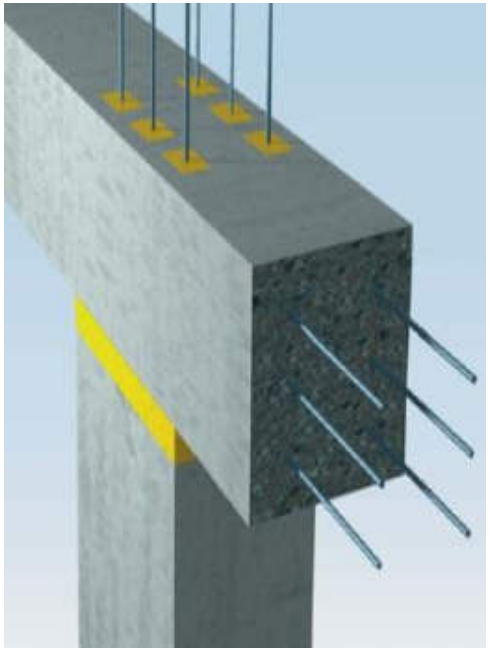
FILLING



Sealing around Penetrations



Filling of Voids



Filling of cavities or joints

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102

SPECIAL PERFORMANCES OF SIKAGROUT SERIE



FATIGUE/DYNAMIC LOAD

- Ability to resist a minimum and maximum stress load at a given frequency and number of cycles

Identifying names:
3000 series



VERY HIGH FINAL STRENGT
Generally considered

- 80 - >100 MPa after 28 or 91 days

Identifying names:
Ultra, VHS, HP



HIGH EARLY STRENGTH
Generally:

- 1 d Compressive strength 30 - >40 Mpa.

1 or 2 hours > 10 MPa

Identifying names:
HES, HP



RAPID SETTING

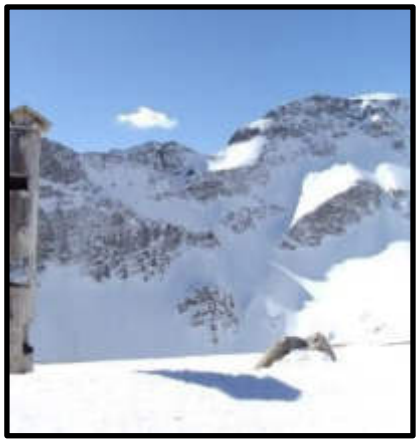
- Short pot life, initial set < 30 – 45 minutes normal strength development
- May also minimum compressive strength 4 - 8 hours > 10 MPa.

Identifying names:
RAPID, R

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103

SPECIAL PERFORMANCES



COLD WEATHER

- Accelerated grouts to aid hydration of cement in low temperatures, typically ambient conditions of less than +5°C.

Identifying names:
Arctic



SLOW SETTING/SUMMER

- In very warm climates typically > +35°C, retarders to delay initial set or prolong pot life time

Identifying names:
S, LP (long potlife)



VERY HIGH TEMPERATURE

- A grout developed to resist temperatures e.g. > +100°C with no significant loss in strength

Identifying names:
TR (temperature resistant)



- UNDER WATER

- Grouts able to push out water in formwork repairs and have adequate adhesion

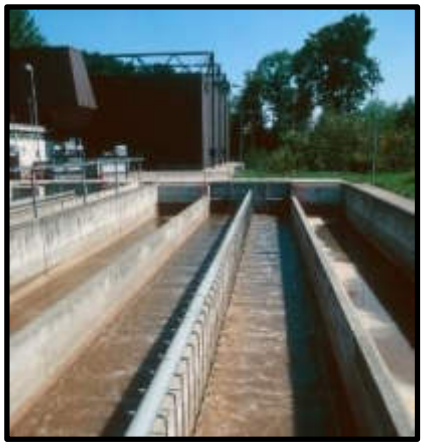
Identifying names:
UW

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104

TECHNICAL PERFORMANCES



CHEMICAL RESISTANCE

- Adequately tested and demonstrate resistance to certain chemicals e.g. hydrocarbons, oils

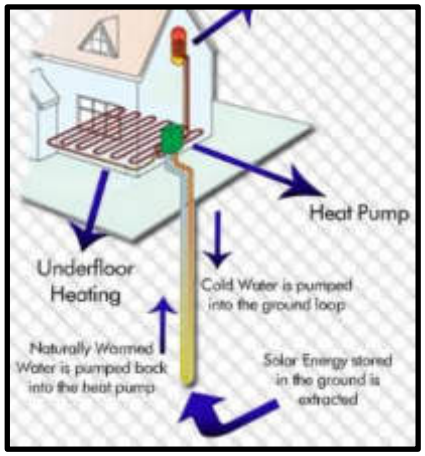
Identifying name:
HP



SULPHATE RESISTANCE

- Formulated to withstand sulphate attack on the hardened mortar

Identifying name:
SR



THERMAL CONDUCTIVITY

- Approximately twice the level of thermal conductivity compared to conventional grout.

Identifying name:
Sika Geotherm



ROCK BOLT ANCHORING

- High early strength pre/post pumpable grout free from chlorides to provide protection from corrosion

Identifying name:
Sika Rock, Rokkon

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SPECIAL PERFORMANCES – TARGET VALUES

Table 4 – Performance value guide

Item N°	Minimum Performance Characteristic	Example Reference Substrate	Example Test Methods	Performance		
				Gold	Silver	Bronze
N1	Compressive strength [a, b]	none	EN 12190	≥ 80 MPa	≥50 - 70 MPa	≥ 35 - 45 MPa
			ASTM C 1107	≥ 11600 PSI	≥7500 –11600 PSI	≥ 5000 – 7000 PSI
N2	Flexural strength [a, b]	none	EN 12190 / ASTM C 348	≥ 9 MPa	≥ 6 MPa	≥ 3 MPa
N3	Tensile splitting strength [a, b]	none	EN 12390-6 / ASTM 496	≥ 3.5 MPa/500 PSI	3.5 MPa/500 PSI	No requirement
N4	Segregation and bleeding [b]	none	EN 480-4 / ASTM C 940	None		
N5	Shrinkage [a, b]	none	EN 12617-4	≤ 600µm	≤ 1000µm	≤ 1500µm
N6	Swelling [c]	none	CRD C-621	≤ 0.30 ‰ after 28/91 days		
N7a	Flow charactersitics [d]	none	EN 13395-1	Typically flow characteristics vary subject to grain size e.g. d_{max} 3 mm the flow > 300 mm and d_{max} 1 mm > 320 mm		
N7b	Flow characteristics [d]	none	EN 13395-2	F2[f]	F2[f]	F1[f]
N7c	Flow Characteristics	none	ASTM C1437 / ASTM C 939	Plastic flow 100-125%; Flowable 126-145% Fluid having efflux time 10-30 seconds		

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106

SPECIAL PERFORMANCES – TARGET VALUES

Table 4 – Performance value guide

Item N°	Minimum Performance Characteristic	Example Reference Substrate	Example Test Methods	Performance		
				Gold	Silver	Bronze
N8	Effective bearing area	none	ASTM C 1339	> 85 %	70 – 85 %	> 70 %
N9	Elastic modulus [e]	none	EN 13412 / ASTM C 469	≥ 30 GPa	≥25 - 30 GPa	≥ 18 – 25 GPa
N10	Adhesive bond [a, b]	MC(0,40)	EN 1542 / ASTM C 1583	≥ 2.0 MPa	≥1.20 – 2.0 MPa	≥ 1.00 MPa
N11	Creep in compression		EN 13584 / ASTM C 1181	Declared value		No requirement

Legend

a - Performance value refers to 28 day test result, procedure according to Standard

b - The water: powder mixing ratio or tested consistency e.g. plastic, flowing or fluid shall clearly be identified in the PDS

c - Storage under water in accordance with Standard

d - Using defined water ratio measured < 5 minutes after mixing in accordance with Standard

e - Static E-Modulus in compression

f – To German Richtlinie Standard. Refer to appendix C.

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107

HOW TO SPECIFY A "GROUT"
QUESTIONS TO ASK

Cementitious Grouts are use for many different applications, they have different performances and can be sold through directly or distribution channels

The project, service temperatures
Grouting thickness
Early strength requirements
Final strength requirements

} Determine the performance requirements

Construction /application method
Maximum grain size
Ambient temperatures during application
Open/working time/pot life

} Determine the application method

Special requirements e.g. vibration/dynamic loading

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108

CORPORATE PRODUCTS
ENGINEERED GROUTS

SikaGrout®-212

General Performance, low shrinkage, expanding pouring mortar

SikaGrout®-311 /-314 /-318
3 to 10 / 10 to 40 / 25 to 80 High Performance, low shrinkage, expanding pouring mortar; 1d ~30MPa, 28d ~80MPa, E-Modulus ~30 GPA

SikaGrout®-334
High performance expanding low shrinkage pouring mortar wit reduced carbon footprint.
10 – 125 mm 1d ~30MPa, 28d ~80MPa, E-Modulus ~38 GPA

SikaGrout®-3000 series

Precision grouts for the wind energy

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CORPORATE PRODUCTS – SPECIAL APPLICATION

POST TENSIONED SYSTEMS

SikaGrout®-300 PT

High performance, zero bleed, sand free, cementitious grout for post-tensioned structures

SikaGrout®-300 PP

“PowerPack” additive to produce cementitious grout



GROUTING PRODUCTS

ALTERNATIVE – RIGID EPOXY - **SIKADUR®-42 (READY TO USE)**

Sikadur®-42 Epoxy Grouts are used for:

HIGH STRENGTH GROUTING AND FIXING OF:

- Starter bars
- Anchors
- Fasteners
- Tie rods
- Crash barrier posts
- Fence and railing posts

UNDER GROUTING AND BEDDING OF:

- Precision seating of base plates
- Machine bases, base-plates for light and heavy machinery including heavy impact and vibrating
- machinery, static engines, compressors, pumps, presses, etc.
- Bridge bearings
- Mechanical joints (i.e. road/bridge/deck types, etc.)

SLEEPER LESS, DIRECT RAIL FIXING:

Crane tracks

Light rail and permanent way in tunnels

Light rail and permanent way over bridges



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GROUTING PRODUCTS

ALTERNATIVE – RIGID EPOXY - **SIKADUR®-42 (READY TO USE)**

SIKADUR-42 HE

Sikadur-42 HE is a three-component, high performance, high precision, solvent-free, moisture tolerant, epoxy resin based grouting system. For use at ambient and substrate temperatures between +5°C and +30°C.

SIKADUR-42 LE

Sikadur-42 LE is a three-component, high performance, high precision, low exothermic, solvent-free, moisture tolerant, epoxy resin based grouting system.

For use at ambient temperatures between +20°C to +45°C and on substrates from +15°C to +45ºC.

SIKADUR-42 MP NORMAL AND SLOW

Sikadur-42 MP Normal is a three-component, multi-purpose, solvent-free, moisture tolerant, epoxy resin based grouting system.

For use at substrate and ambient temperatures between +10°C and +30°C.

Sikadur-42 MP Slow is a three-component, multi-purpose, solvent-free, moisture tolerant, epoxy resin based grouting system. For use at substrate and ambient temperatures between +20°C and +40°C.

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112

GROUTING PRODUCTS

ALTERNATIVE – RIGID EPOXY - **SIKADUR®-42 (READY TO USE)**

SIKAFLOW®-648 (FORMERLY MASTERFLOW®-648)

Precision epoxy resin grout, consisting of 3 components – resin, hardener and specially blended inert aggregates. On mixing, the components yield a high flow, high strength grout. Upto 300 mm thickness

Possibility to produce standard and high flow of the grout.

RE-GROUT SOLEPLATE & SUBSTRATE CRUSHER#6 & SAG MILL #3



SIKAFLOW-648 (FORMERLY MASTERFLOW 648 CP) AT FREEPORT
MOTOR FAN BASE PLATE GROUTING, PORT SITE









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114

SIKADUR-996

TWO – COMPONENT 100% SOLIDS, HIGH STRENGTH EPOXY SYSTEM





WHERE TO USE

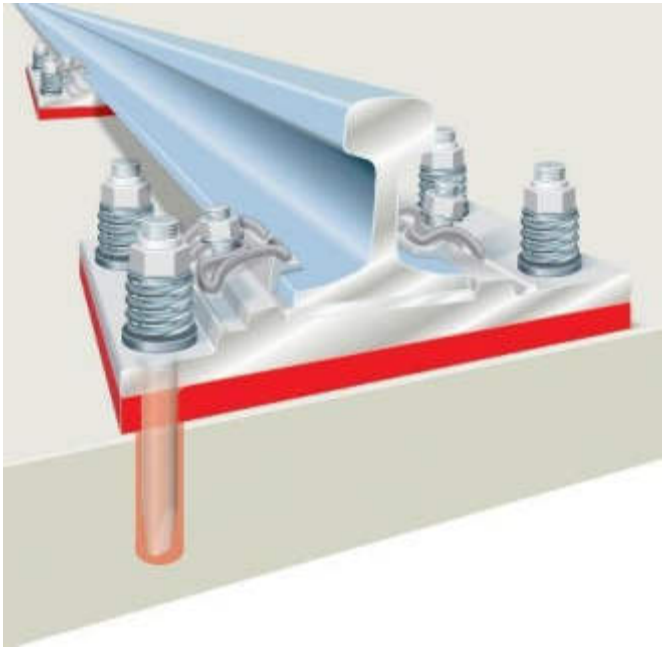
- Useful as a support system in crushing and grinding equipment.
- As a backup system in gyratory or cone crushers wear plates.
- Mills shielding and steel plates reinforcement
- As an injection or gravity filling system on joints or cracks of the above-mentioned applications.



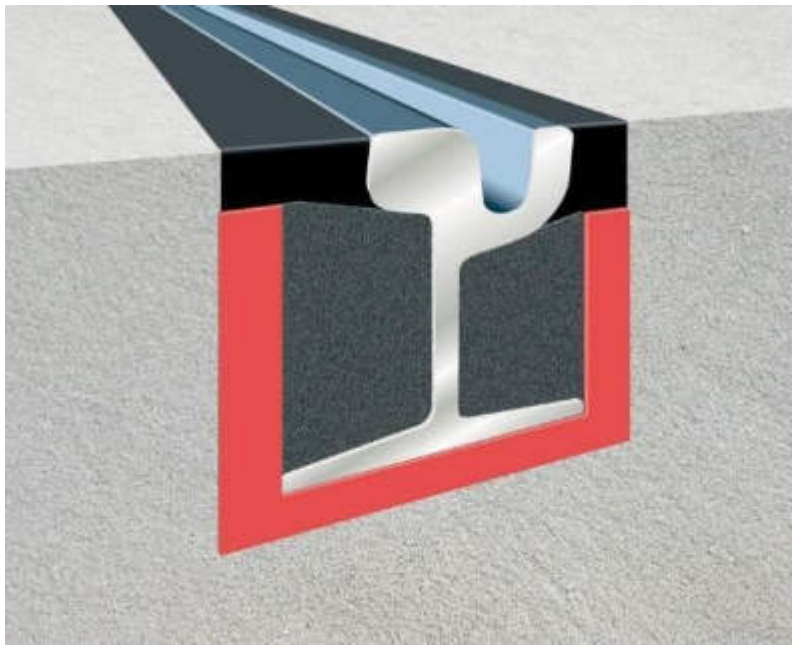
GROUTING PRODUCTS

ALTERNATIVE – FLEXIBLE PU - **ICOSIT® KC RANGE (READY TO USE)**

DISCRETE FIXATION WITH STEEL BASEPLATE



EMBEDDED (FLOATING) RAIL DESIGN



116

GROUTING PRODUCTS

ALTERNATIVE – FLEXIBLE PU - **ICOSIT® KC RANGE (READY TO USE)**

ICOSIT KC 330 AND KC 340 GRADES

2-pack Polyurethane-based grouts properties:

- flexible, viscoelastic (damping, compressible)
- load-bearing (outstanding mechanical resistance)
- efficent electric insulation
- excellent adhesion on various substrates (shrink compensating)
- Icosit KC 340 is moisture-insensitive

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117

GROUTING PRODUCTS

ALTERNATIVE – FLEXIBLE PU - **ICOSIT® KC RANGE (READY TO USE)**

EMBEDDED (FLOATING) RAIL DESIGN



Working pit in maintenance workshop



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SIKABOND® R&B-100 FOR CONVEYOR BELT REPAIR

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120

INNOVATION FOR CONVEYOR BELTS

Over the past years, Sika developed exciting new solutions for the repair of rubber conveyor belts in the mining and materials processing industry using state of the art materials and bonding technology

Main objective of the products is to reduce the down-time of conveyors during emergency-repair stops and maintenance windows

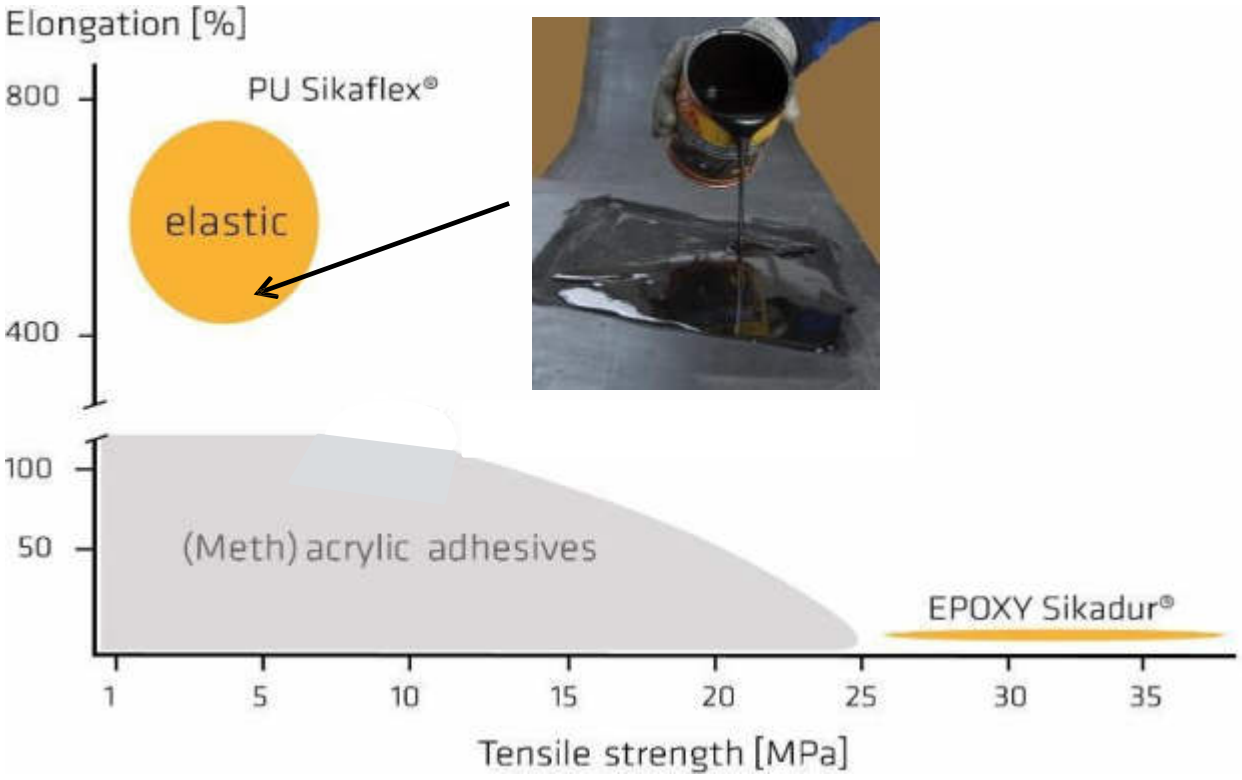
The products are fast curing and easy and safe to apply with minimal equipment needs (no power, no vulcanizing presses, limited solvents)

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121

SIKA BONDING TECHNOLOGIES



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122

SIKABOND® R&B-100, CONVEYOR BELT REPAIR KIT

- Easy application, even under conditions where access is limited
- Rapid curing and fast commissioning at ambient temperature
- Good adhesion to mechanically prepared rubber substrates
- High elasticity and tensile strength
- High chemical and mechanical resistance
- Self-leveling properties
- Safe handling and use



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REPAIRING

TEXTILE CARCASS PROCEDURE



Mark area to be prepared



- Remove top cover



- Grind edges (45-deg slope) with a tungsten sander at < 5,000 rpm



- Brush edges with a steel brush

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124

REPAIRING

TEXTILE CARCASS PROCEDURE



- Clean prepared surface with a soft brush or cold blower (use clean air).



- Apply SikaBond© R&B-100 Primer on cleaned surface in a thin film.



- Clean and degrease the surface using SikaBond © R&B-100 Cleaner.



- Add Comp-B into can containing Comp-A and mix thoroughly for 1 minute.

REPAIRING

TEXTILE CARCASS PROCEDURE



- Pour mixed SikaBond© R&B-100 on prepared repair area



- If necessary, distribute still liquid product evenly over area to be repaired, using provided spatula.



- Leave repaired area for curing.

SIKABOND R&B-100, IRON ORE PORT, AUSTRALIA





SIKABOND® R&B-100, QUARRY OPERATION, GERMANY





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128

SIKABOND R&B-100, MOBILE CRUSHER, GERMANY





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129

SIKABOND R&B-100, RMC PLANT GERMANY







BACK TO PRESENTATION

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CONCRETE STRUCTURES STRENGTHENING (REPAIR)

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CONCRETE STRUCTURES - STRENGTHENING

REASONS OF STRENGTHENING A CONCRETE STRUCTURE

STRUCTURAL STRENGTHENING =
Increasing the load bearing capacity to restore structural safety and serviceability requirements

STRUCTURAL IMPROVEMENTS
- Load increase
- Vibration / deflection reduction
- Impact / seismic / blast resistance
- Design errors / code changes
- Construction errors



STRUCTURAL MODIFICATIONS
- Column / wall removals
- Slab openings
- Floor additions



STRUCTURAL RECOVERY
- Corrosion
- Mechanical damage
- Fire / explosion
- Impact
- Earthquake



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CONCRETE STRUCTURES - STRENGTHENING

HOW TO STRENGTHEN

- Increase concrete cross section
- Add steel or concrete members
- Add bonded steel plates
- Add bonded CFRP plates
- Add bonded fabrics (CFRP)
- Add near surface mounted strips (NSMS)
- Add external post-tensioned reinforcement
- Add shells (GFRP jackets with EP / CEM grout)

FRP =
Fibre **R**einforced **P**olymer

CFRP = **C**arbon FRP
GFRP = **G**lass FRP

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CONCRETE STRUCTURES - STRENGTHENING

HOW TO STRENGTHEN

- Increase concrete cross section



HIGH PERFORMANCE GROUT

Anchor additional rebars. Apply bonding agents. Install shuttering. Pour or pump high performance grout



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CONCRETE STRUCTURES - STRENGTHENING

HOW TO STRENGTHEN

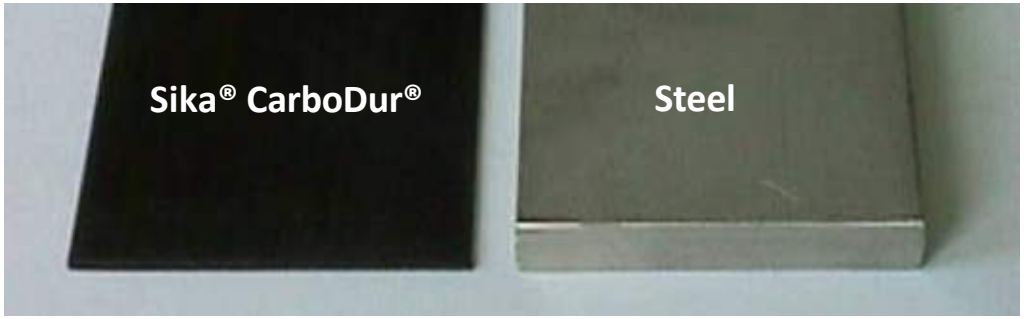
- Add bonded steel plates



STEEL PLATES

CONCRETE STRUCTURES - STRENGTHENING

HOW TO STRENGTHEN



CFRP VS. STEEL PLATES

- High material cost
- Fast installation
- Low installed cost
- Non-corrosive
- Lightweight
- No fabrication required
- Cross-overs possible
- Low maintenance

- Low material cost
- Slow installation
- High installed cost
- Corrosive
- Heavy
- Fabrication required
- Cross-over difficult
- High maintenance

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137

CONCRETE STRUCTURES - STRENGTHENING

HOW TO STRENGTHEN



CFRP FABRICS



CONCRETE STRUCTURES - STRENGTHENING

HOW TO STRENGTHEN



NSM STRIPS

CONCRETE STRUCTURES - STRENGTHENING

HOW TO STRENGTHEN

CARBON



- Virtually inert
- Very low creep
- Extreme resistance to elevated temperature
- Extreme resistance to stress rupture and corrosion
- Good fatigue resistance
- Very limited ductility

GLASS



- Good temperature resistance
- Low creep
- Inexpensive
- Limited chemical resistance (acid and bases)
- Subject to stress corrosion

ARAMID



- Very light
- Good impact / explosion performance
- Expensive
- Absorbs water
- High creep

STATE OF THE ART

Carbon fibres represent the state of the art for structural strengthening applications due to their outstanding perfor-mance and impressive track record.

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140

CONCRETE STRUCTURES - STRENGTHENING

PRODUCTS SYSTEM OVERVIEW

LAMINATES	FABRICS
Sika® CarboDur® S	**SikaWrap® C**





	Dry application	Wet application
Sikadur®-30	**Sikadur®-330**	**Sikadur®-300**

SPECIALTIES

Sika® CarboDur NSM

- near surface mounted rods.

Sika® CarboStress®

- post-tensioned tendons.

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CONCRETE STRUCTURES - STRENGTHENING

SIKA CARBODUR AND SIKAWRAP SYSTEM
SIKADUR-30

- Primer, putty and adhesive for Sika CarboDur
- Thixotropic, 2 component epoxy resin based
- High creep resistance
- Very good adhesion to various substrates
- Non sag performance also for over-head applications
- Density 1.65 kg/l

- Packaging in sets of 30 kg or 6 kg
- 24 months shelf life



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142

CONCRETE STRUCTURES - STRENGTHENING

SIKA CARBODUR AND SIKAWRAP SYSTEM

Sika CarboDur Type S

- Pultruded carbon fibre reinforced laminates
- Elastic tensile modulus 165'000 MPa
- Tensile strength > 2'800 MPa
- Elongation at break > 1.7 %

- Roll length 100 m
- Wide range of cross sections (thickness 1.2/1.4 mm, width 50, 60, 80, 90, 100, 120 and 150 mm)

→**Used for FLEXURAL STRENGTHENING**

- **NSM**
- Sika CarboDur Type S is also available in 20 mm width or in circular shape (8, 10, 12, 16 mm) for use as near Near Surface Mounted (NSM) type reinforcement.



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143

CONCRETE STRUCTURES - STRENGTHENING

SIKA CARBODUR AND SIKAWRAP SYSTEM - SIKA CARBODUR TYPE S – INSTALLATION



- Surface Preparation



- Surface Testing



- Application Sikadur-30



- Application SD-30 to Sika CarboDur plate



- Transfer the Sika CarboDur plates



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144

CONCRETE STRUCTURES - STRENGTHENING

SIKA CARBODUR AND SIKAWRAP SYSTEM
SIKAWRAP TYPE C

- Woven carbon fibre fabric with unidirectional fibres
- Elastic tensile modulus > 230’000 MPa
- Tensile strength > 3’900 MPa
- Elongation at break > 1.43 %

- Roll width 500 mm
- Roll length 100 m (50 m if > 450 g/m2)
- Wide range of fibre area densities (155, 200, 230, 300, 450, 530, 610 g/m2)



→**Used for FLEXURAL and SHEAR STRENGTHENING**
→**Used for CONFINEMENT**

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145

CONCRETE STRUCTURES - STRENGTHENING

SIKA CARBODUR AND SIKAWRAP SYSTEM
SIKADUR-330 (DRY APPLICATION METHOD)

- Primer, putty, impregnating adhesive for dry applied SikaWrap® System
- **Thixotropic**, 2 component epoxy resin based
- Density 1.31 kg/l
- Tensile strength > 30 MPa (@ 7 days)
- Elongation at break > 0.9 % (@ 7 days)

- Packaging in sets of 30 kg or 6 kg
- 24 months shelf life



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146

CONCRETE STRUCTURES - STRENGTHENING

SIKA CARBODUR AND SIKAWRAP SYSTEM - SIKADUR-330 (DRY APPLICATION METHOD)



- **SEALING / SATURATION** of the prepared concrete substrate with Sikadur-330



- **PLACEMENT OF THE DRY FABRIC** into the base layer of Sikadur-330



- **LAMINATION OF FABRIC** with special roller till the Sikdur-330 squeezes through the fabric

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BACK TO PRESENTATION

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148





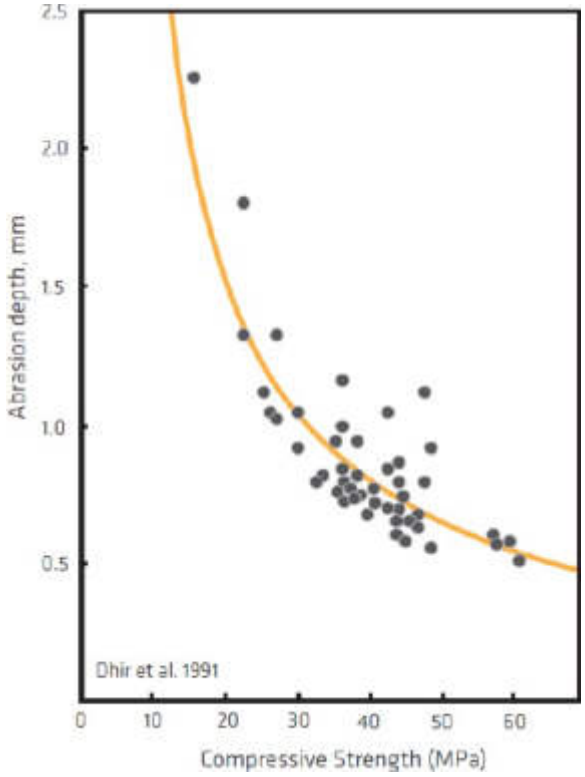
SHAFT – ORE PASS LINING

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SHAFT – ORE PASS LINING

GENERAL REMARKS ON ABRASION RESITANT LINERS

- Abrasion resistance strongly dependent on the compressive strength of the concrete and, also the hardness of the used aggregates.
- Impact resistance and flexural strength of the liner can be greatly improved by the combination of macrosynthethic macro fibers and micro steel fibers
- Liner system dependent on the overall rock quality of the host rock

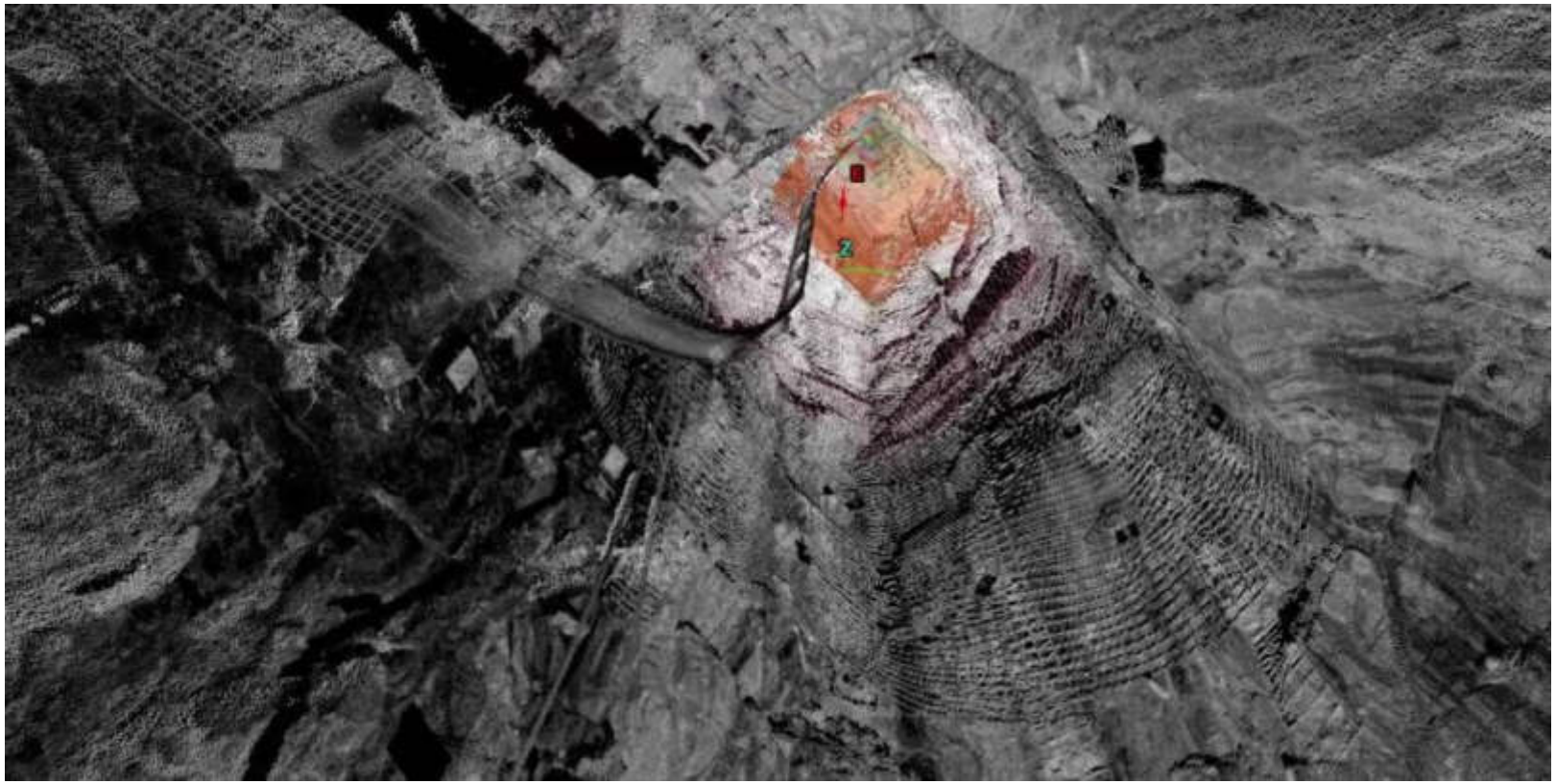


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150

SHAFT – ORE PASS LINING



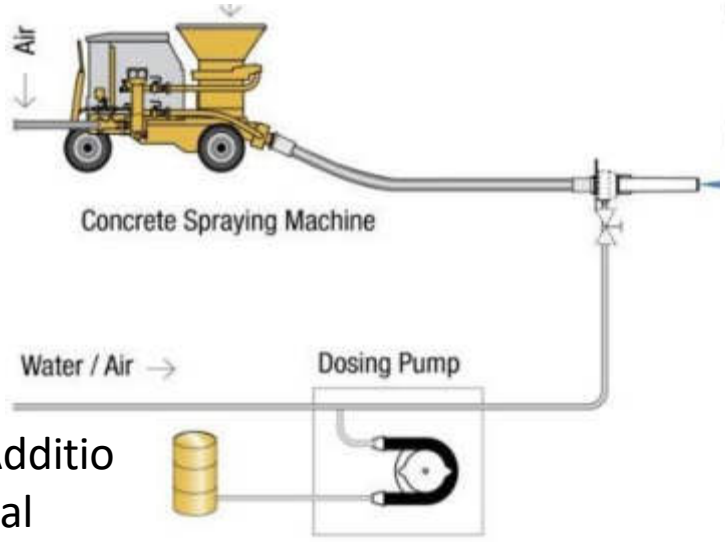
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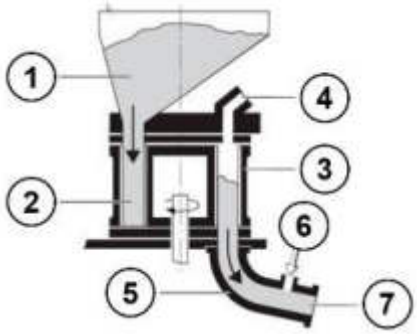
151

DRY SPRAYING = 1 SPRAYING PROCESS = 1 PUMPING SYSTEM

THIN STREAM FLOW (DRY)

RTU dewatering gunite





1. The dry mix slides from the hopper (1) into an empty chamber (2) of the rotor (3)

2. The filled chamber is rotated to the outlet position, where the mix is blown by the top air (4) into the blow-off chamber (5)

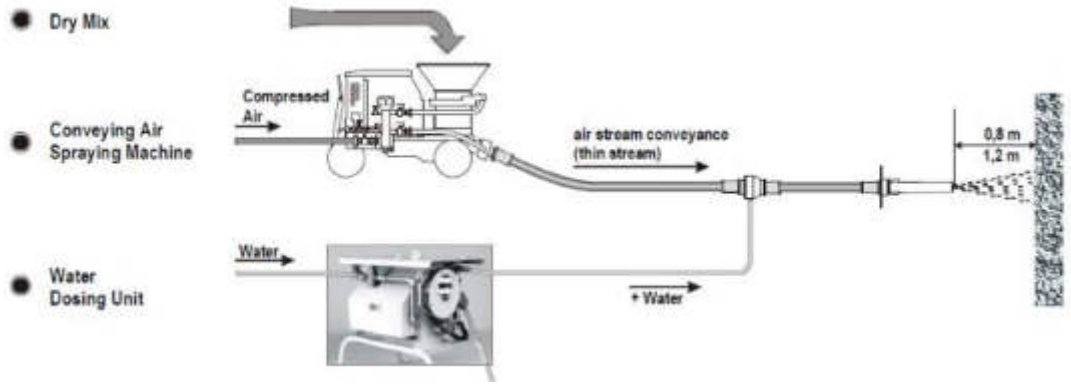
3. From there, supported by the bottom air (6), the mix reaches the conveyor line

Additional addition of liquid Acceler

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SET UP THIN STREAM FLOW (DRY) FOR GUNITE SPRAYING



Hose Ø mm	Air consumption Nm³/min. ***	
	Dry spraying	
	60 m	120 m
32	3	4

1. Dewatering sprayed gunite required low out-put as usually the maximal aggregate size of the gunite mortar will be < 4 mm
2. This lead to implement an out-put of max 1 m3/h.
3. The low out-put spraying with generally a small rotor (e.g 2 liter volume) will require to convey the

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PREBAGGED READY TO USE MATERIALS FOR SPRAYED GUNITE MORTAR

DRY SPRAYED PRODUCTS



Sikacrete® STS 1010 (Masterroc® STS 1010)
Ready to use cement based single component shotcrete mortar modified with microsilica and alkali free accelerators. The formulation is designed to match the requirements for a dry process shotcrete applied mortar. It gives high early strength, reduced rebound and maximum application thickness.



Sikalastic® TSL 865 (MasterRoc TSL 865) is a single-component polymer powder for spray-application onto soil and rock for reinforcement or weathering protection. Through good bond to most substrates, high elasticity and high tensile strength, the ground stability and stand-up-time is improved.



KING RS-D2
Rapid hardening, pre-packaged shotcrete material for dry-process applications powered by Rapid Set® technology, containing blended aggregates and other carefully selected components. It has greatly enhanced shooting characteristics providing greatly reduced setting times and very rapid strength development.

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154

SHAFT – ORE PASS LINING

PREBAGGED READY TO USE MATERIALS FOR SPRAYED GUNITE MORTAR

DRY SPRAYED PRODUCTS



RS ARMOURGUARD

Specially formulated, rapid-setting, steel fiber-reinforced, shotcrete mix designed to provide superior impact and abrasion resistant characteristics.
This product is a pre-blended, pre-packaged, dry-process shotcrete material **powered by Rapid Set® technology**, containing steel fiber, blended aggregates and other carefully selected components.
It is designed to produce superior impact and abrasion resistant characteristics, greatly reduced setting times and very rapid.



ARMOURGUARD

Specially formulated, high-performance, steel fiber-reinforced shotcrete mix designed to provide superior impact and abrasion resistant characteristics. This product is a pre-blended, pre-packaged, dry-process shotcrete material containing high early Portland cement, silica fume, steel fiber, blended aggregates, set time accelerators and other carefully selected components

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155

SHAFT – ORE PASS LINING

CHEMICAL SEAL WATERPROOFING RING

INJECTION PRODUCTS



SikaInject® MP 304 (Masterroc® MP 304)
Low viscosity, fast-reacting and highly flexible acrylic resin with adjustable reaction speed for permanent water sealing, joint repair and layer curtaining of concrete and masonry



SikaInject® MP 307 (Masterroc® MP 307)
Low-viscosity, fast-reacting acrylic resin for permanent water sealing and layer curtaining of concrete and masonry

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156

SHAFT – ORE PASS LINING - CURRENT BOLTING SYSTEMS ON THE MARKET

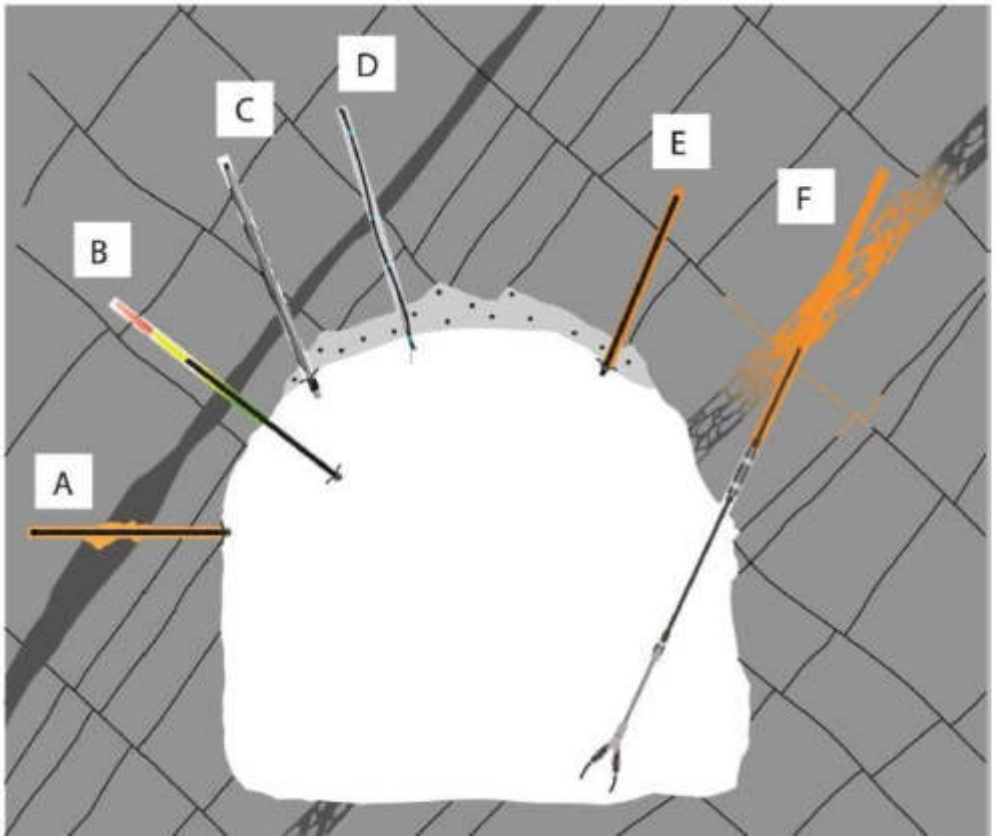


Fig.1

Current bolting systems used in mines:

A: Self drilling bolts, fully resin grouted or cementitious grouted

B: Polyester cartridges

C: Swellex bolts

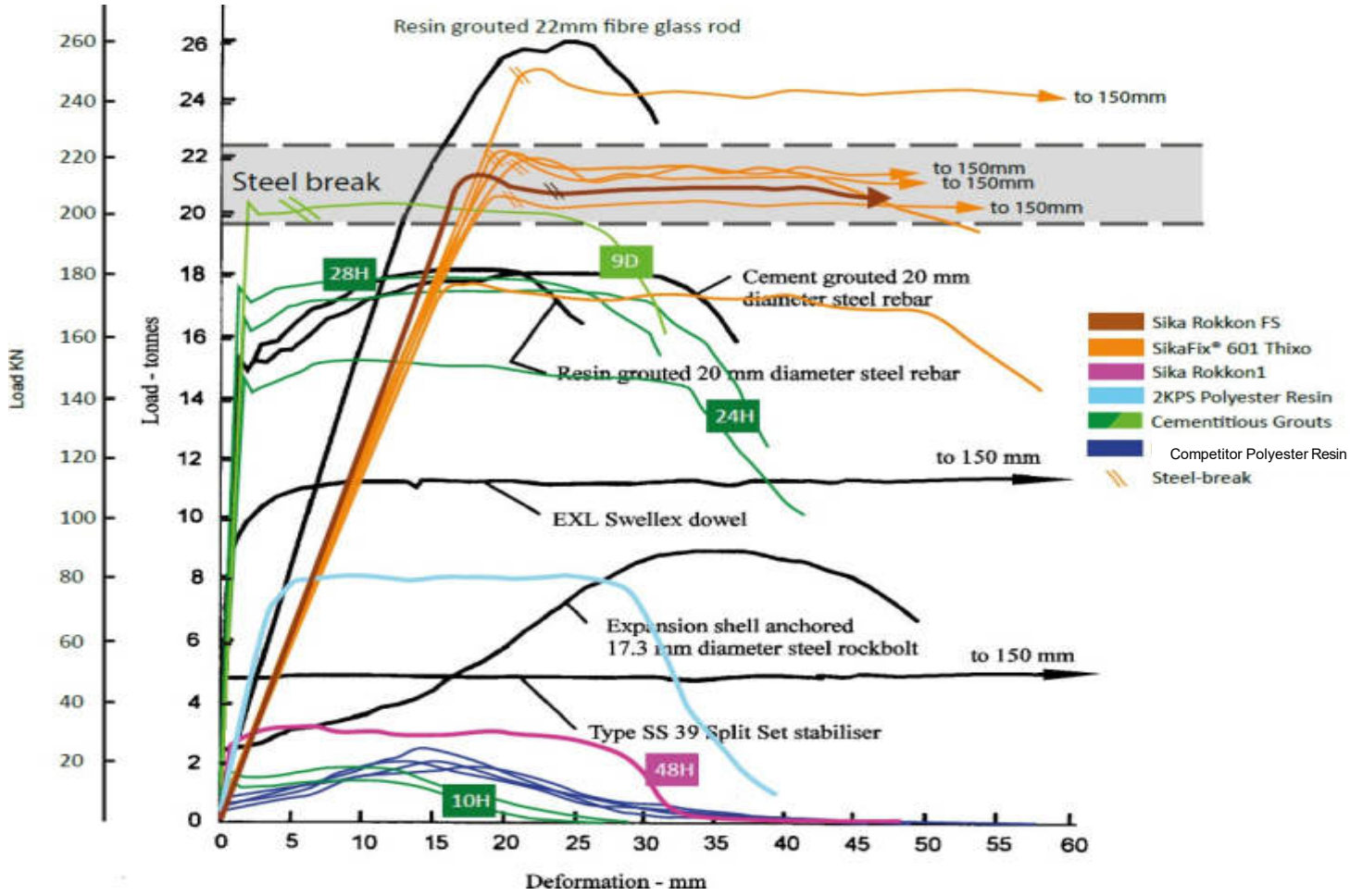
D: Fully grouted cable-bolts using a cementitious grout

E: Chemical resin grouted bolt (silicate, PU etc.) using similar technologies as for chemical rock injections **(F)**

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158

SHAFT – ORE PASS LINING - SIKA PRODUCT FOR CHEMICAL ANCHOR

CEMENTITIOUS BASE

SikaInject® MG 10 NA (MasterRoc® MG 10 NA)
Thixotropic, high yield, and shrinkage compensated grout used for placement of cable and rock bolts in tunnels and underground mines.
Pre-grouted application

SikaInject® MG 57 (MasterRoc® MG 57)
Thixotropic, high yield, shrinkage compensated grout that exhibits rapid strength gain designed for rock reinforcement for placement of cable and rock bolts in civil tunnels and underground mines.
Pre-grouted application

Sika® Rokkon- C / Rokkon®-1
Cementitious fast setting, shrinkage compensating mortar supply in sausage for the grouting of rock bolts, soils nails.
Pre-grouted application

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SHAFT – ORE PASS LINING - SIKA PRODUCT FOR CHEMICAL ANCHOR

RESIN BASE

ROKKON® R (FS)
Polyester very fast setting resin, supply in sausage for the grouting of rock bolts, soils nails.
Pre-grouted application

SikaInject®-601 THIXO 2 (SikaFix®-601 THIXO 2)
Two-component, CFC-free, halogen-free and solvent-free silicate resin with a fast reaction time for solidification and stabilization of soils.

SikaInject®-680 (formerly MasterRoc® RBA 380)
Fast reacting, two-component, instantly-thickening, solvent free polyurea silicate anchoring resin specifically designed for rock bolting. Its thixotropic feature facilitates overhead applications without additional sealing.

SITE MIXES

SikaRapid®-700 L
Liquid hardening accelerator for grout, mortar and concrete. Suitable for Cable and bolt grout
Post-grouted application

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160

CONCRETE LINING - SHAFT







BACK TO PRESENTATION

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162



SHOTCRETE / SPRAYED CONCRETE

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WORKING MECHANISM SHOTCRETE

Shotcrete…

A) …fills joints and evens out irregularities of the substrate
B) …its flexural strength prevents blocks from falling out
C) …seals, that prevents water from washing out fines and small rocks
D) …supports the rock mass by arching effect
E) …builds a beam/panel effect with the rock bolts

Source: Sprayed Concrete for Rock Support, no. 7, Norwegian Concrete Association

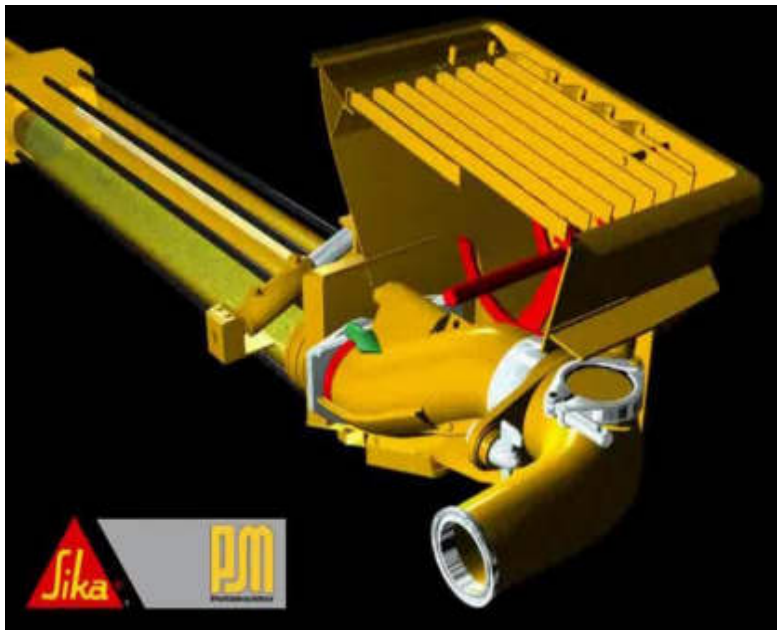
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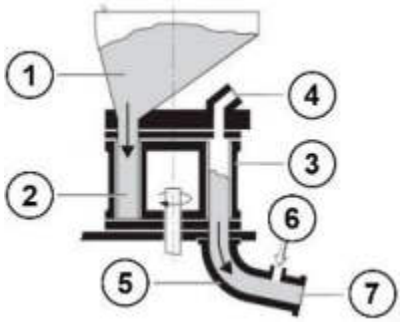
164

2 PROCESSES = 2 DIFFERENT PUMPING (SPRAYING)SYSTEMS

DENSE STREAM FLOW (WET)



THIN STREAM FLOW (DRY)





1. The dry mix slides from the hopper (1) into an empty chamber (2) of the rotor (3)
2. The filled chamber is rotated to the outlet position, where the mix is blown by the top air (4) into the blow-off chamber (5)

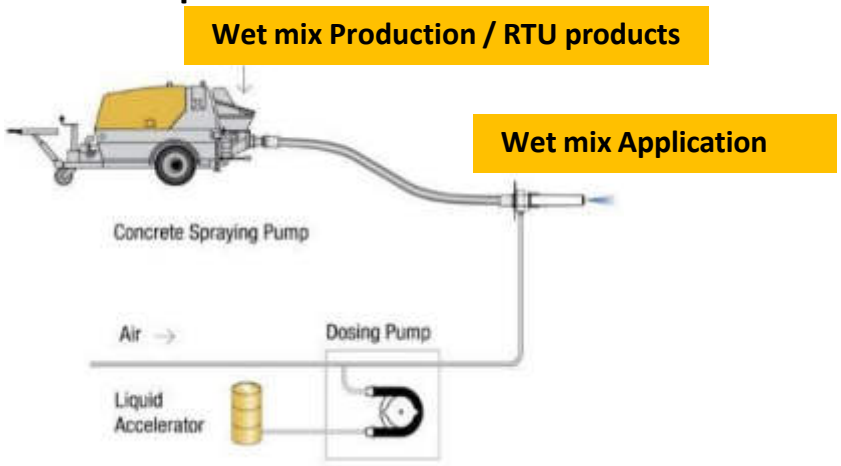
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165

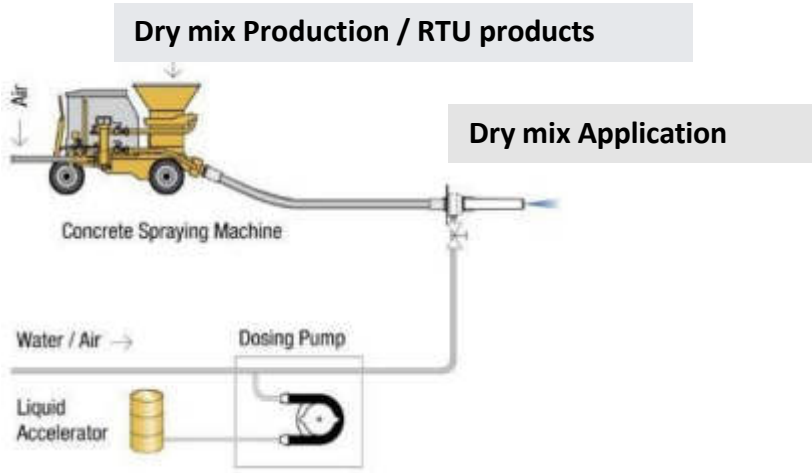
TWO SHOTCRETE PRODUCTION & APPLICATION PROCESSES

Dense-flow process



- Ready mixed concrete
- **“on-demand” wet mix (prebagged)**
- Conveyed like pump concrete
- Adding of compressed air at the nozzle

Thin-flow process



- Dry mix (oven dry to earth moist)
- **Ready to use shotcrete or gunite**
- Conveyed in air stream (flying)
- Water / accelerator added at the nozzle

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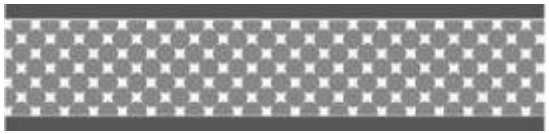
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166

DENSE-FLOW VERSUS THIN-FLOW

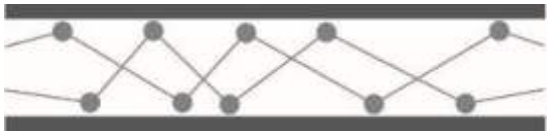
Pump:

- High spray output
- Low air demand
- Low dust generation
- Constant input product quality
- Best system for fibers
- Lower wear compared to dry process
- More complicated start & stop
- Less flexibility
- Higher requirements on mix design



Rotor:

- High flexibility (mix design, stop & go, space,…)
- Simple start and stop operations
- Lower requirements for mix design
- Less investment
- Higher early strength compared to wet
- Repair or dewatering work
- Lower out-put (productivity)
- Higher rebond compared to wet
- Inconsistent quality (nozzelman depency)

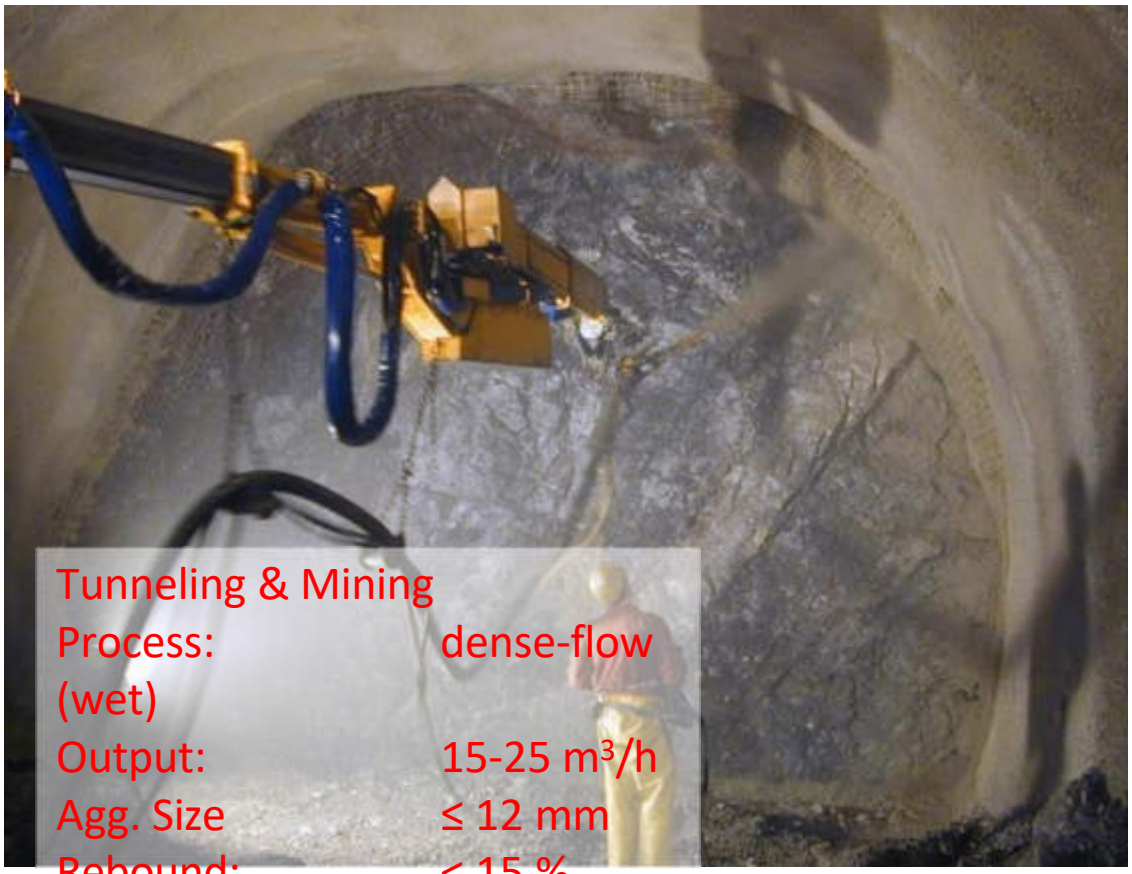


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167

SHOTCRETE / SPRAYED CONCRETE AS UNDERGROUND SUPPORT

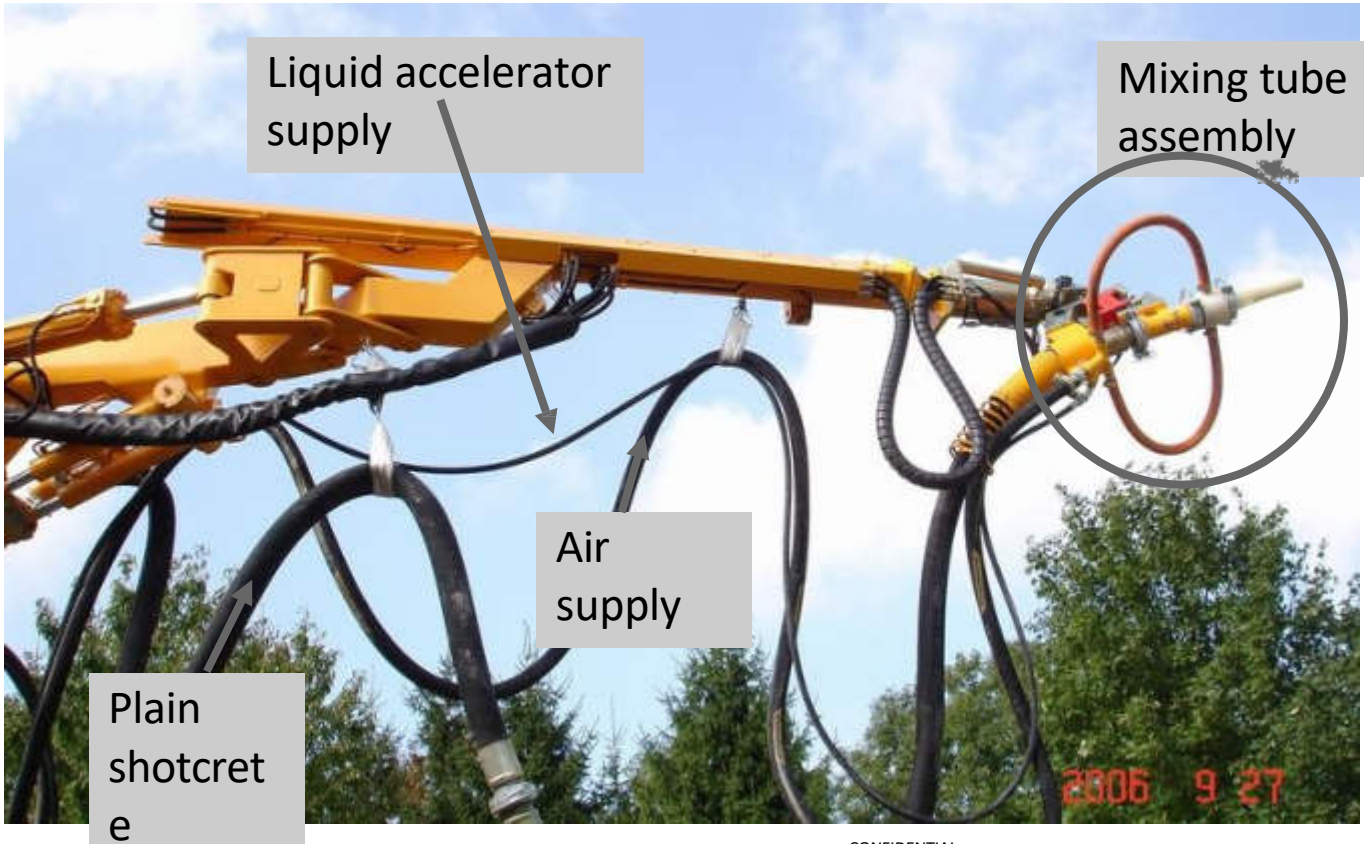


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168

DENSE-FLOW METHOD

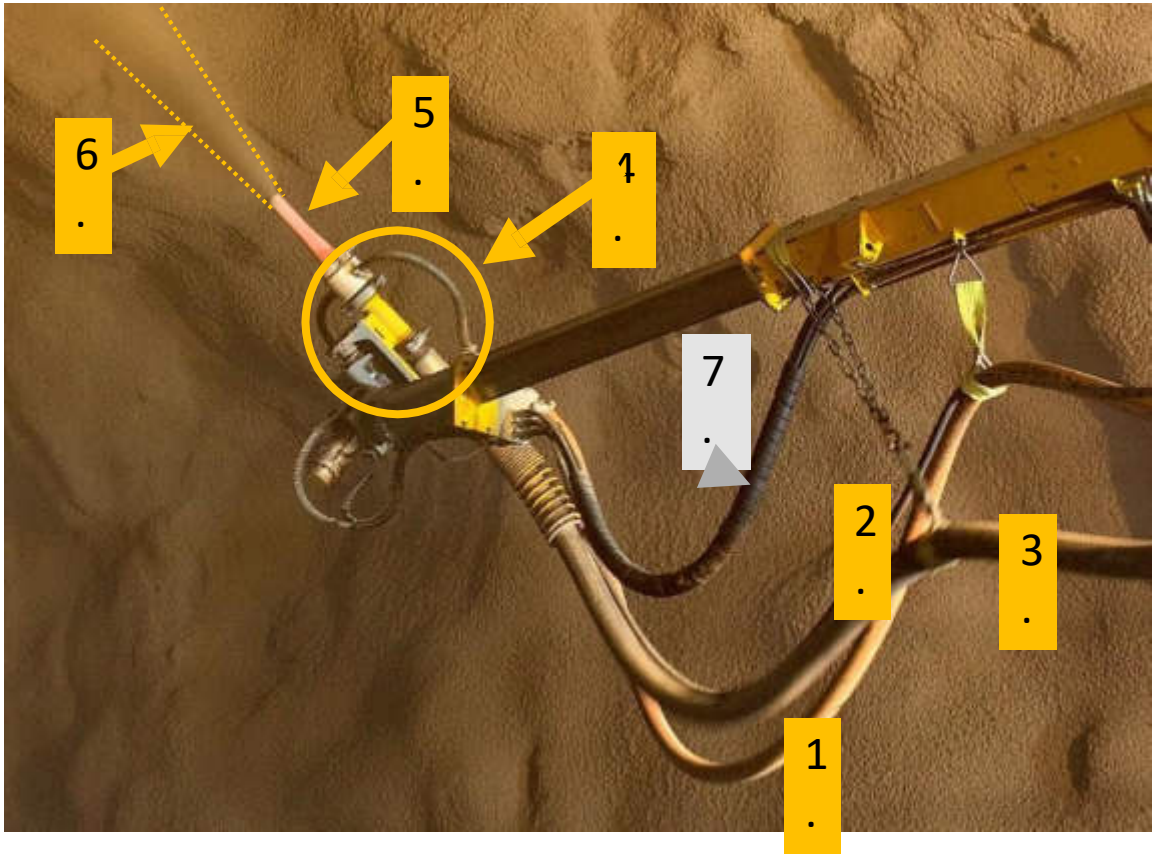


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DENSE-FLOW METHOD



1. Air under pressure supply
2. Liquid accelerator supply
3. Plain shotcrete supply
4. Mixing tube assembly
5. Nozzle
6. Accelerated sprayed concrete

7. Hydraulic pipes

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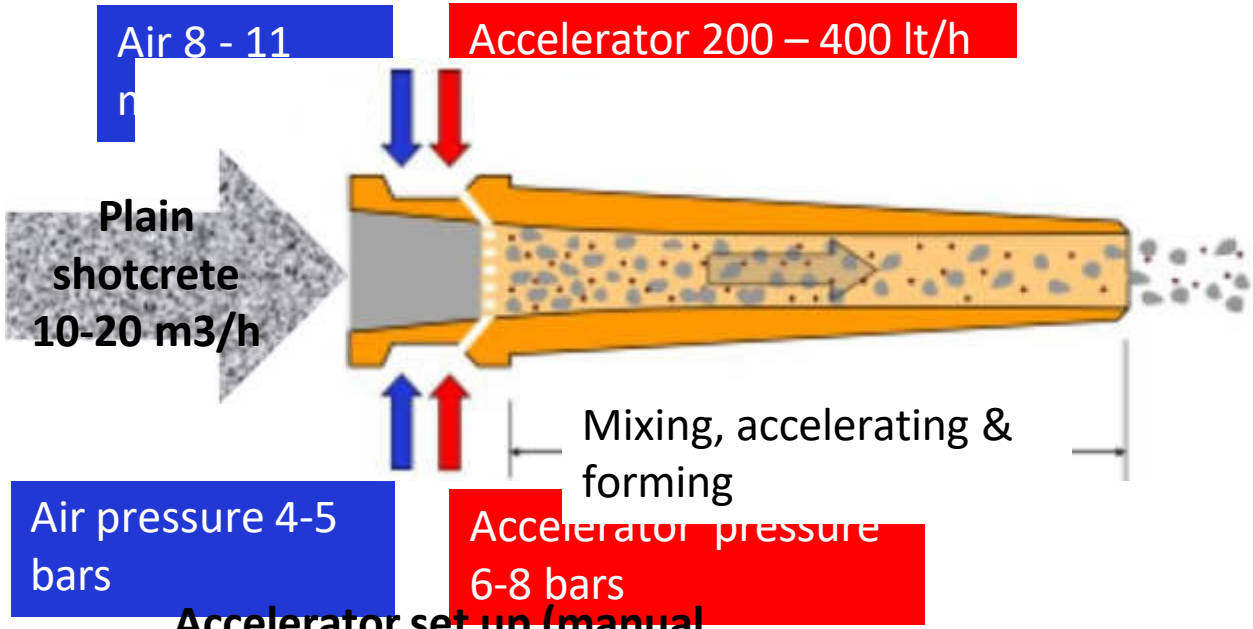
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170

DENSE-FLOW METHOD

The nozzle design is important:

- To feed and properly mix the air with the accelerator
- To achieve an optimal spray jet, e.g. to achieve optimum compaction
- For good handling for nozzleman



Example:

Output: 15 m3/h

Cement content: 450 kg/m3

Accelerator set up (manual mode)

15 m3/h x 450 kg/m3 x 5% = ~

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GLOBAL SHOTCRETE TREND

Ordinary Portland cement	→ Blended cement	CO_2 and cost reduction
500 kg/m³ binder	→ 400 kg/m³ binder	CO_2 and cost reduction
Alkaline Accelerator	→Alkali free Accelerator	EH&S and durability
Solutions	→Suspensions	Cost efficient
Steel mesh	→Fibre reinforced	Cost efficient
Steel fibres	→Macro synthetic fibres	Cost efficient / durability
General Solution	→Customized Solution	Sharper optimization
Penetrometer / Hilti	→Ultra sound	Reliability

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172

CHALLENGES IN THE FUTURE

New Mix Designs → Reduced clinker content

- Sustainability (CO_2 reduction)
- Tunnel maintenance (sintering)
- Cost reduction (clinker, high costs)

→ High activities in development of shotcrete admixtures and additives

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173

SHOTCRETE PRODUCTION & APPLICATION

MATERIALS FOR SPRAYED CONCRETE **USED DURING PRODUCTION OF WET MIX BY RMC PLANT**

Cement | Sand / Aggregate | Water | Admixtures | Additives



MATERIALS FOR SPRAYED CONCRETE **USED DURING APPLICATION**





Accelerator

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174

SHOTCRETE PRODUCTION & APPLICATION

SIKA PRODUCTS

SIKA PRODUCTS FOR SPRAYED CONCRETE **USED DURING PRODUCTION OF WET MIX BY RMC PLANT**

Cement

Sand / Aggregate

Water

Admixtures

Additives

Sika® ViscoCrete®
Products used to reduce the water demand of the concrete, control the workability and increase the durability of sprayed concrete. Strength development is positively influenced by the dual actions of the superplasticizer and the acceleration effect.

SikaTard®
Admixtures developed to regulate the hydration of sprayed concrete which enable an extended workability time, so that continuous spraying with fresh mixes can continue without difficulty for defined periods of time as required.

SikaTell®
Rebound reduction and improvement of the cohesion of the shotcrete can be achieved by the use of SikaTell®.

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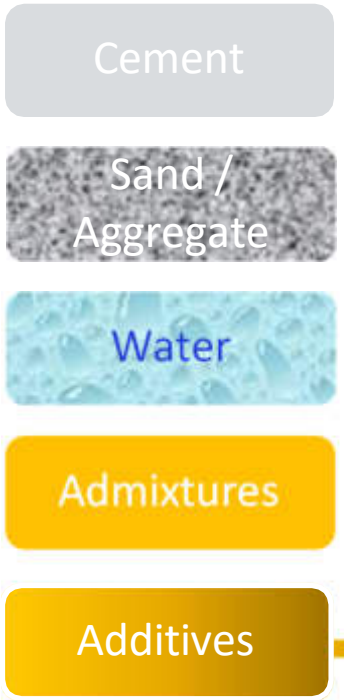
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175

SHOTCRETE PRODUCTION & APPLICATION

SIKA PRODUCTS

SIKA PRODUCTS FOR SPRAYED CONCRETE **USED DURING PRODUCTION OF WET MIX BY RMC PLANT**



SikaFiber®
All types of fibers, including macro synthetic, polypropylene fibers, micro-PP fibers, fiber-mesh and steel fibers.

SikaFume®
SikaFume® strongly enhances the durability of the concrete by improving all around mechanical strength, reduce permeability. Reduces rebound formation during spraying process.

SikaPump®
Special pumping aids to improve pumpability under difficult circumstances and/or long transport distances**.**

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176

SHOTCRETE PRODUCTION & APPLICATION

SIKA PRODUCTS

SIKA PRODUCTS FOR SPRAYED CONCRETE **USED DURING APPLICATION OF WET MIX AT SITE**

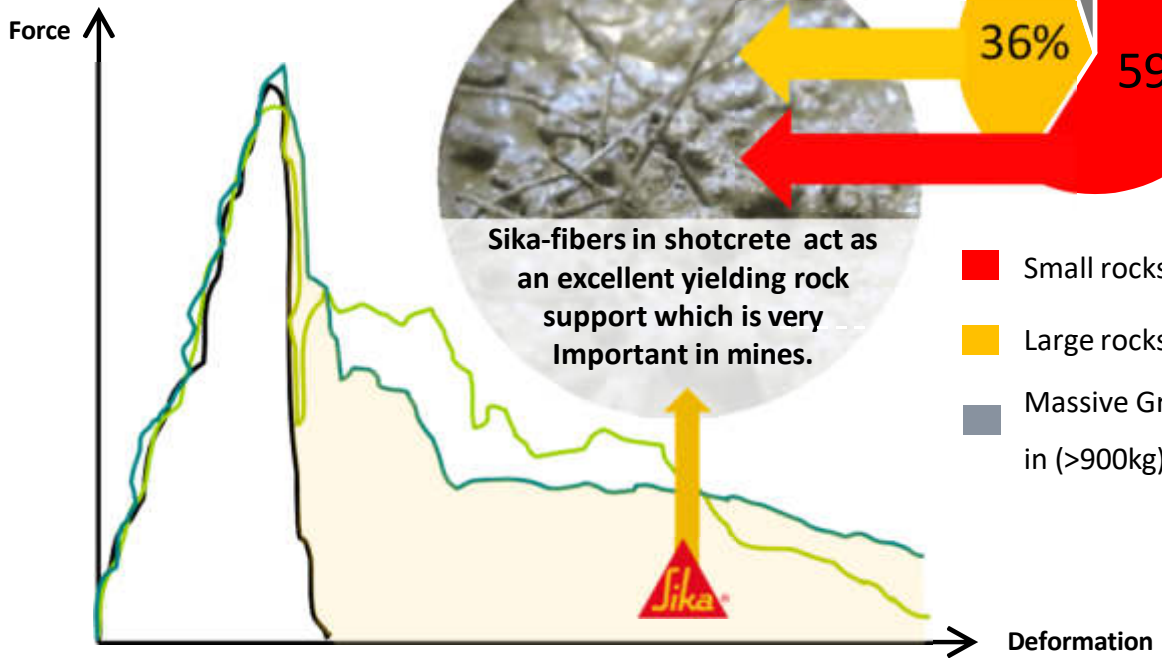
Accelerator

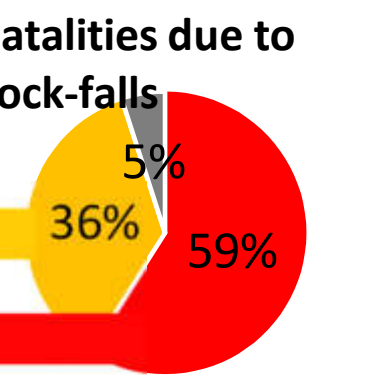
Sigunit®

Shotcrete accelerators for high early strength requirements. Sigunit® is available as a liquid solution or suspension or as a powder to meet the challenging logistics for mining operations.

SHOTCRETE PRODUCTION & APPLICATION

SIKA PRODUCTS - FIBERS



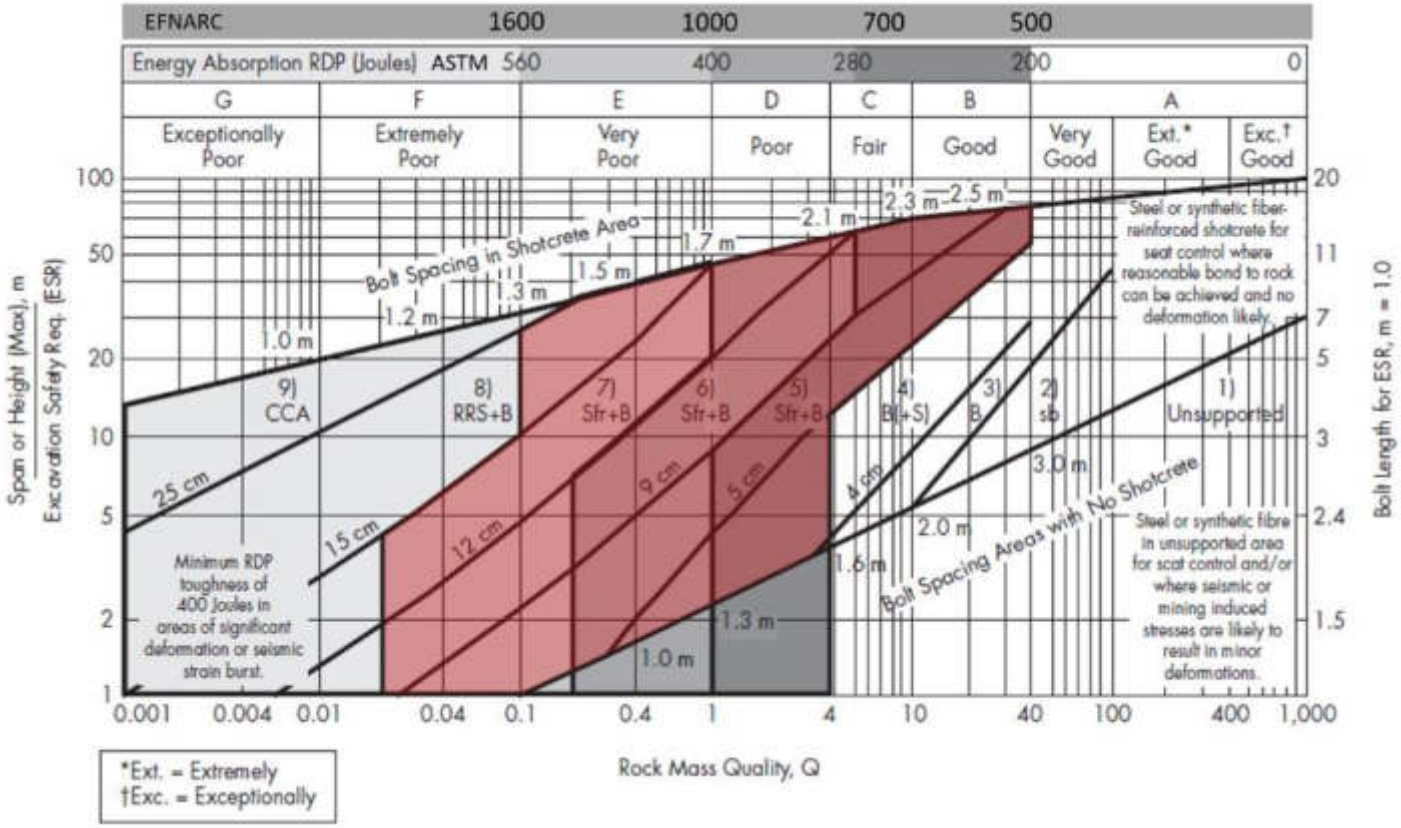




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SHOTCRETE PRODUCTION & APPLICATION

SIKA PRODUCTS - FIBERS





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179

SHOTCRETE PRODUCTION & APPLICATION

SIKA PRODUCTS - FIBERS





Old underground support

Modern, fiber-reinforced shotc support



180

SUBSTRATE PREPARATION AND SHOTCRETE APPLICATION



Cleaning the substrate

- Cleaning and removing loose material, dust, fines from the substrate thanks to water under pressure
- Cleaning direction from top (crown) to bottom (invert)



Filling the overbreak

- The Nozzelman should consider after having observed the area to be sprayed, to initially spray the localize spot of which it is obvious that the final thickness layer would be more than the design spray thickness

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181

SUBSTRATE PREPARATION AND SHOTCRETE APPLICATION



Initial spraying

- In case of layer thicker than 20 cm final, 1st shotcrete layer should be sprayed (min 5 cm)
- Purpose is first excavation stabilization and adhesive bridge for the 2nd shotcrete layer



Preparation for additional thickness layer

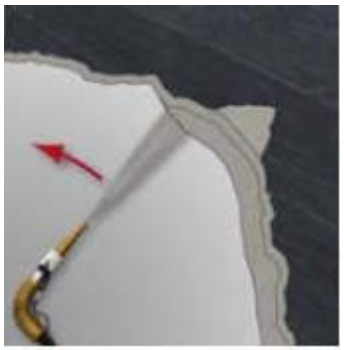
- Cleaning and prewetting the substrate in case of extended time between initial spraying and secondary spraying is required in order to avoid poor bonding of the second layer and cold joint formation

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182

SUBSTRATE PREPARATION AND SHOTCRETE APPLICATION



Secondary additional spraying

- Installation of wire mesh and 2nd shotcrete layer – excavation stabilization.
- Or 2nd shotcrete layer containing SikaFiber® - Macro PP or Steel Fibers



Spraying distance and spraying angle

- The angle of the sprayed shotcrete (cone) shall be as close as 90O to the substrate.
- The spraying distance may vary from 1 to 2 meters.
- Factors influencing the spraying distance:
  - Spraying location (overhead, benches)
  - Spraying out put
  - Spraying pressure

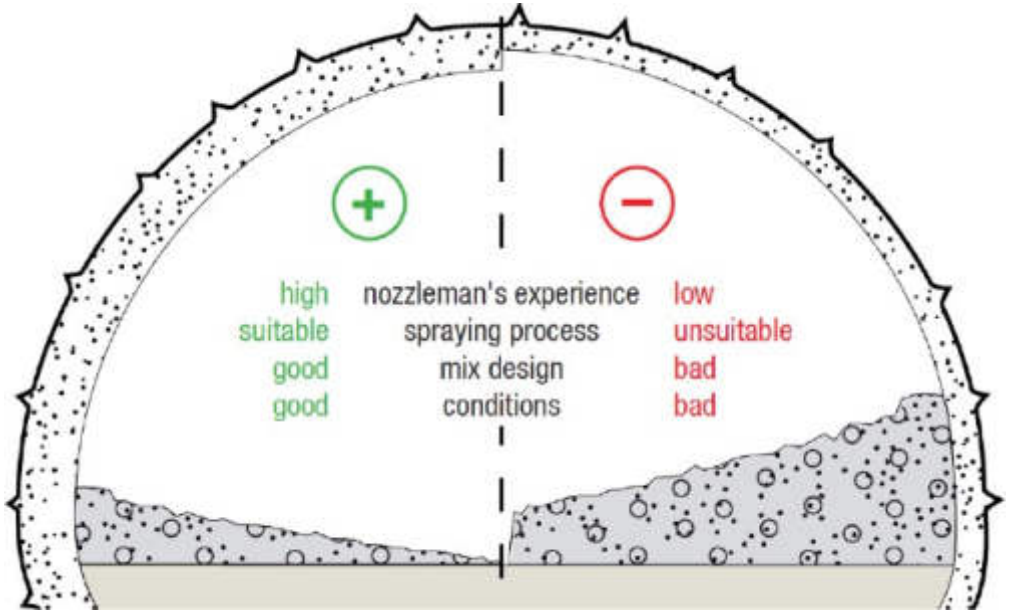
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183

SHOTCRETE APPLICATION

Parameters influencing rebound formation



pumped through the spraying and that ricochets off the receiving surface.
Loose material from the shotcreting process which has not adhered to the excavated surface of shotcrete being formed.

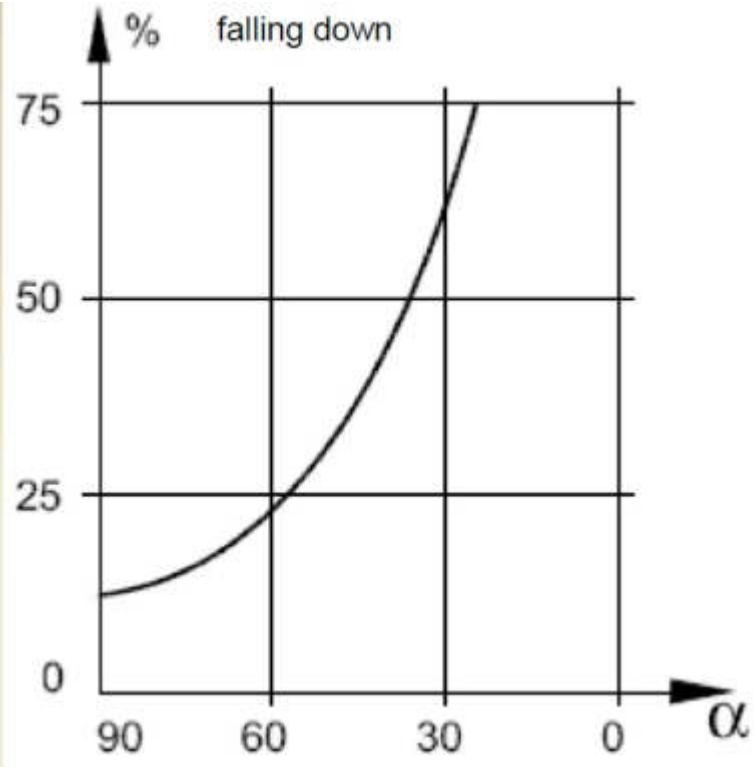
Shotcrete rebound occurs when the **stream of sprayed concrete hits a solid wall and partially bounces off** [1]. The rebound phenomenon not only affects the spraying process and wastes concrete but also affects the quality of the project and presents potential safety hazards.

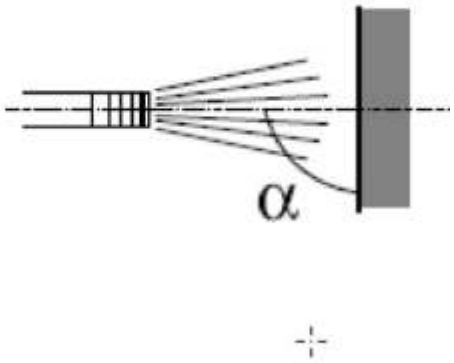
Nevertheless, if it can be reduced

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SHOTCRETE APPLICATION

Relationship between sprayed angle to the substrate and rebound formation



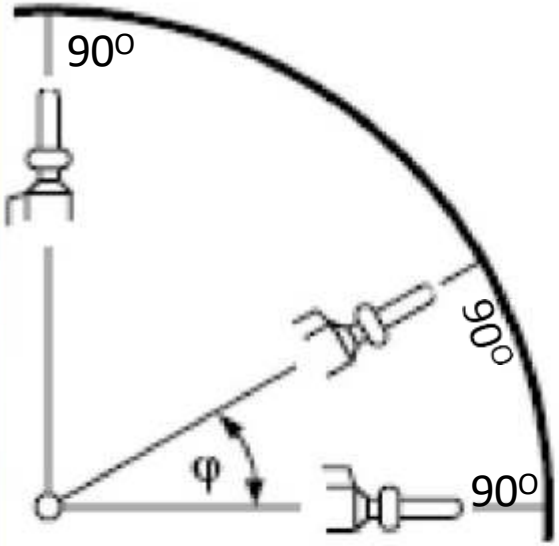


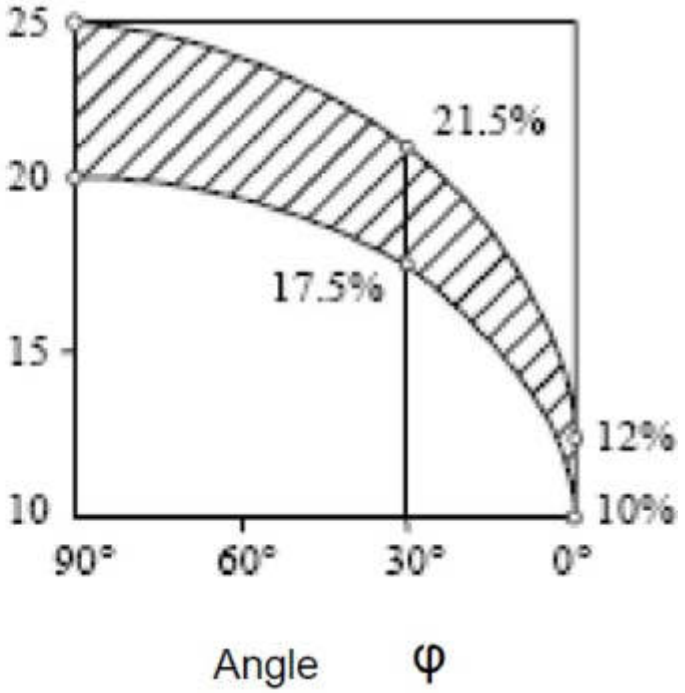
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SHOTCRETE APPLICATION

Relationship between spraying direction and rebound formation



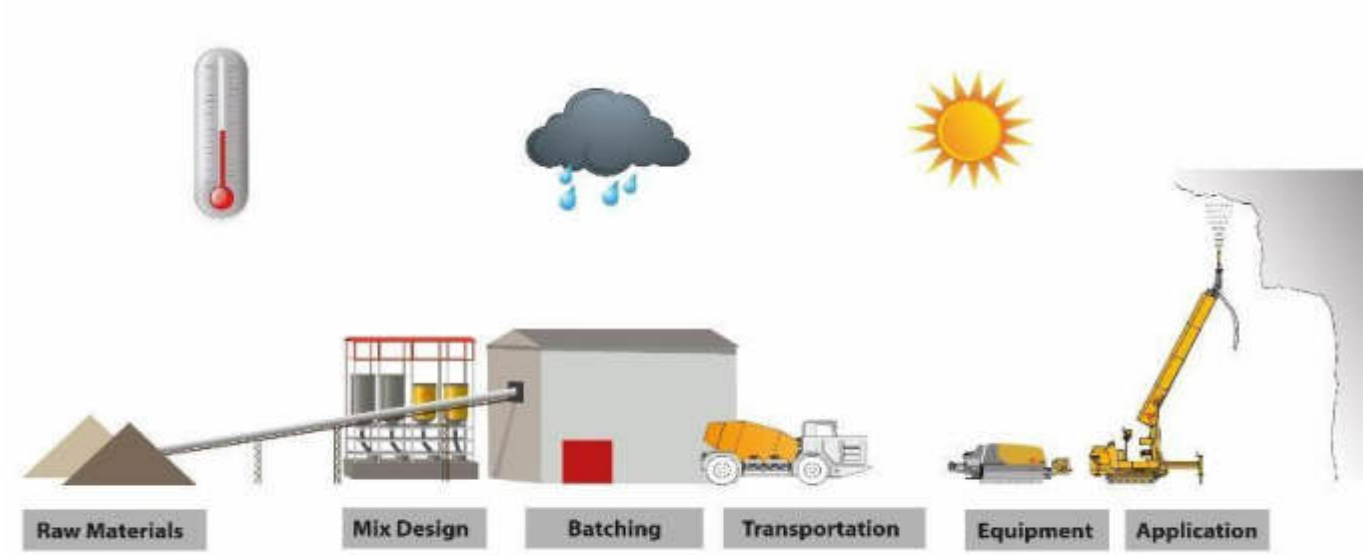


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TROUBLESHOOTING OF HIGHLY OPTIMIZED SYSTEMS



CONCLUSIONS

Shotcrete is a technology in change regarding mix design and quality measurement.

Clients demand for more than only products. They want solutions and support.

As more optimized the technology as more sensitive against fluctuating parameters

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SIKA CHEMICAL BOLTING PRODUCTS

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CURRENT BOLTING SYSTEMS ON THE MARKET

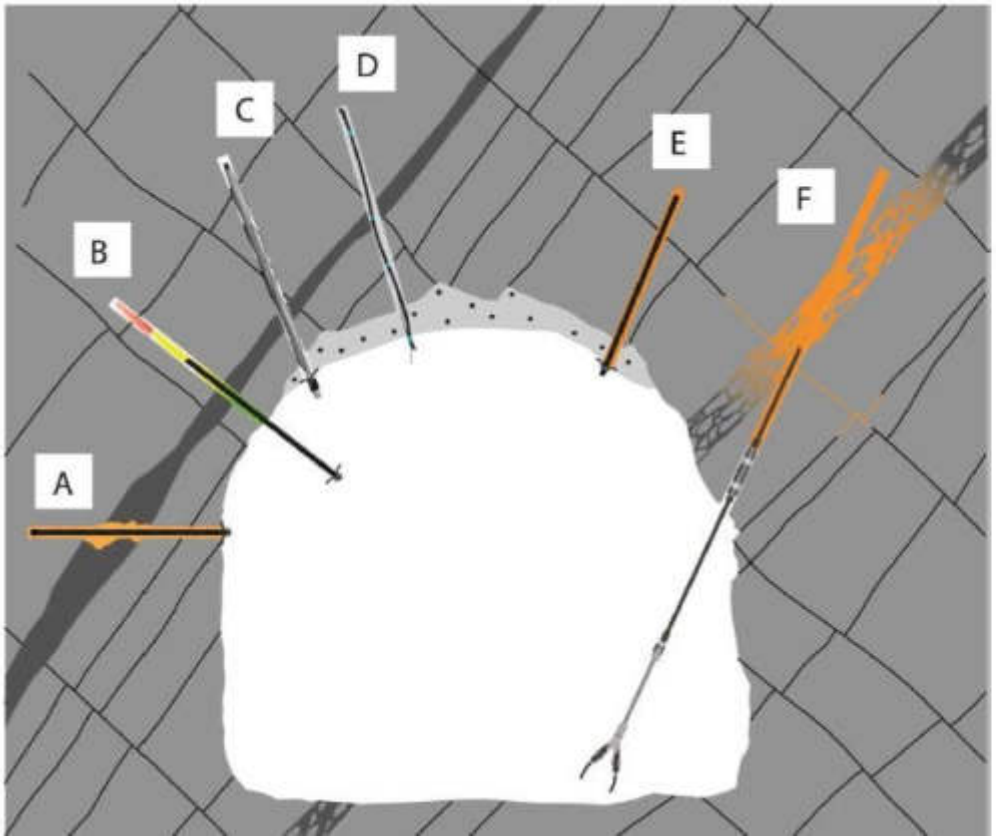


Fig.1

Current bolting systems used in mines:

A: Self drilling bolts, fully resin grouted or cementitious grouted

B: Polyester cartridges

C: Swellex bolts

D: Fully grouted cable-bolts using a cementitious grout

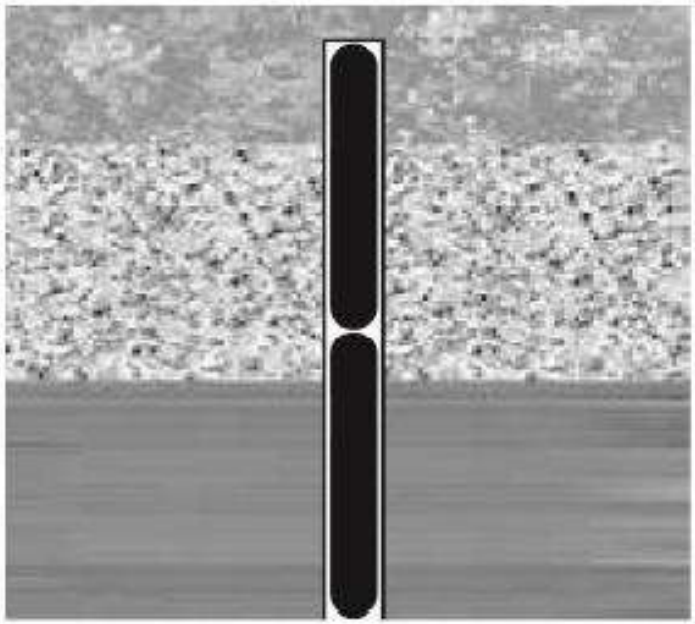
E: Chemical resin grouted bolt (silicate, PU etc.) using similar technologies as for chemical rock injections **(F)**

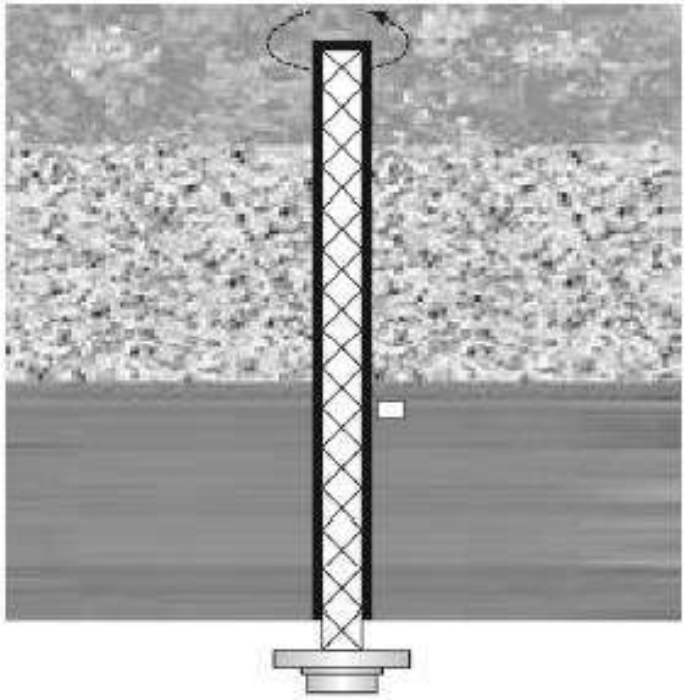
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BOLTING CARTRIDGES FOR MINING AND TUNNELING





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HOW TO TEST PERFORMANCE

First drill the whole with a Drill rig (Drill Jumbo, Boomer)



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HOW TO TEST PERFORMANCE

Place the cartridges in the drilled hole (minimum depth of 600mm)





HOW TO TEST PERFORMANCE

Place the cartridges in the drilled hole (minimum depth of 600mm), mechanical placing:



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HOW TO TEST PERFORMANCE

Place the anchor (standard Re-bar anchor of 25mm diameter, spinning time of at around 30-45 seconds at 150 rpm). Manual placing:





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HOW TO TEST PERFORMANCE

Testing the bolt with a hydraulic cylinder

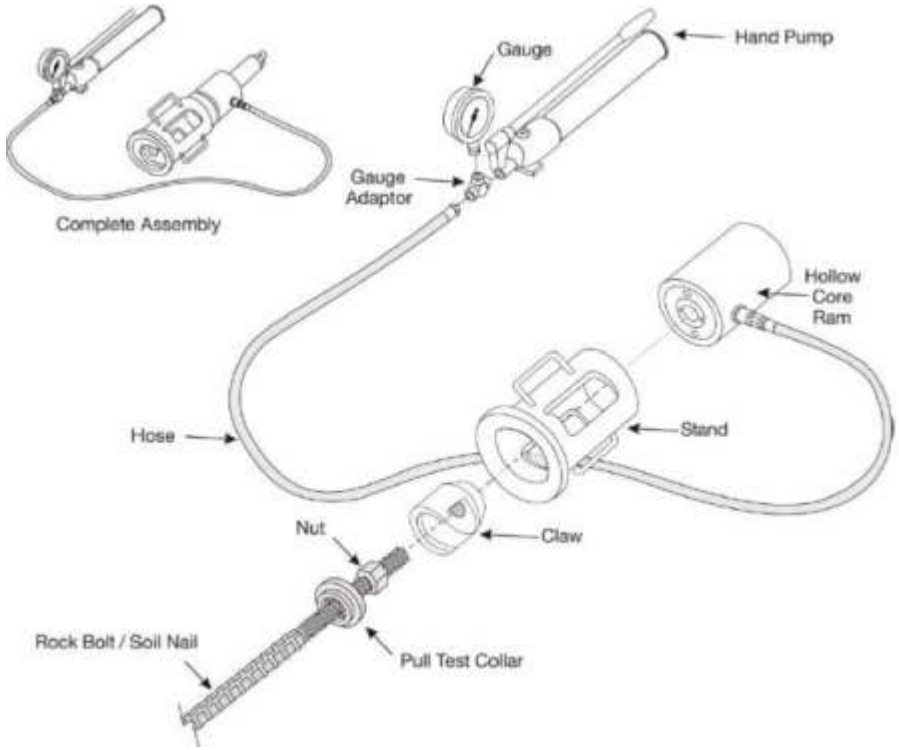




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HOW TO TEST PERFORMCANCE

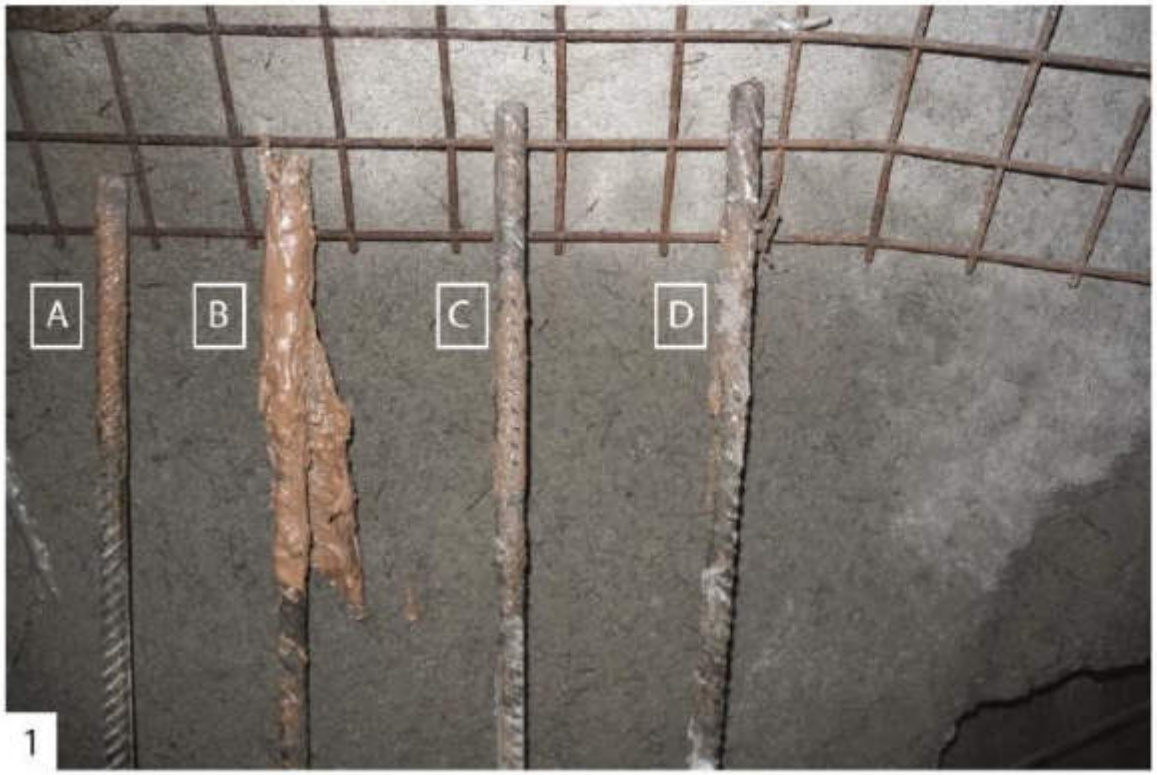


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WHAT CAN GO WRONG





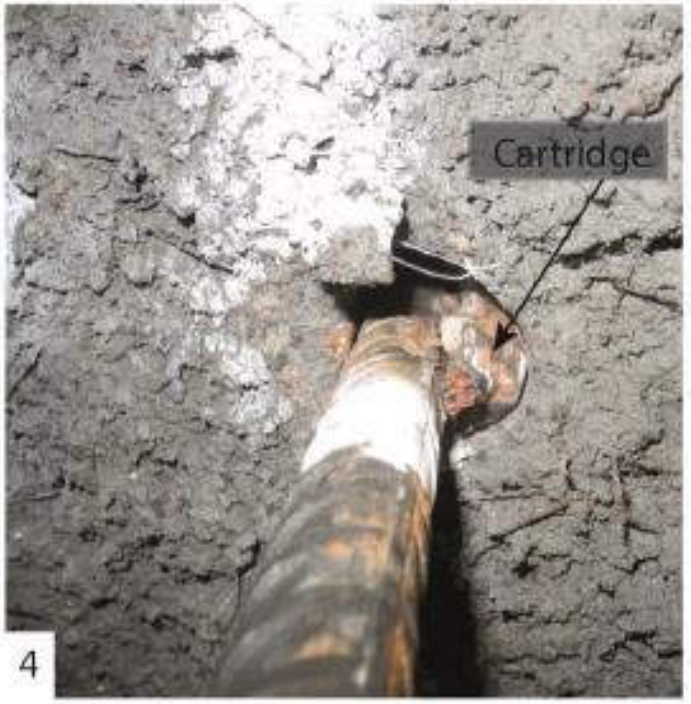
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WHAT CAN GO WRONG





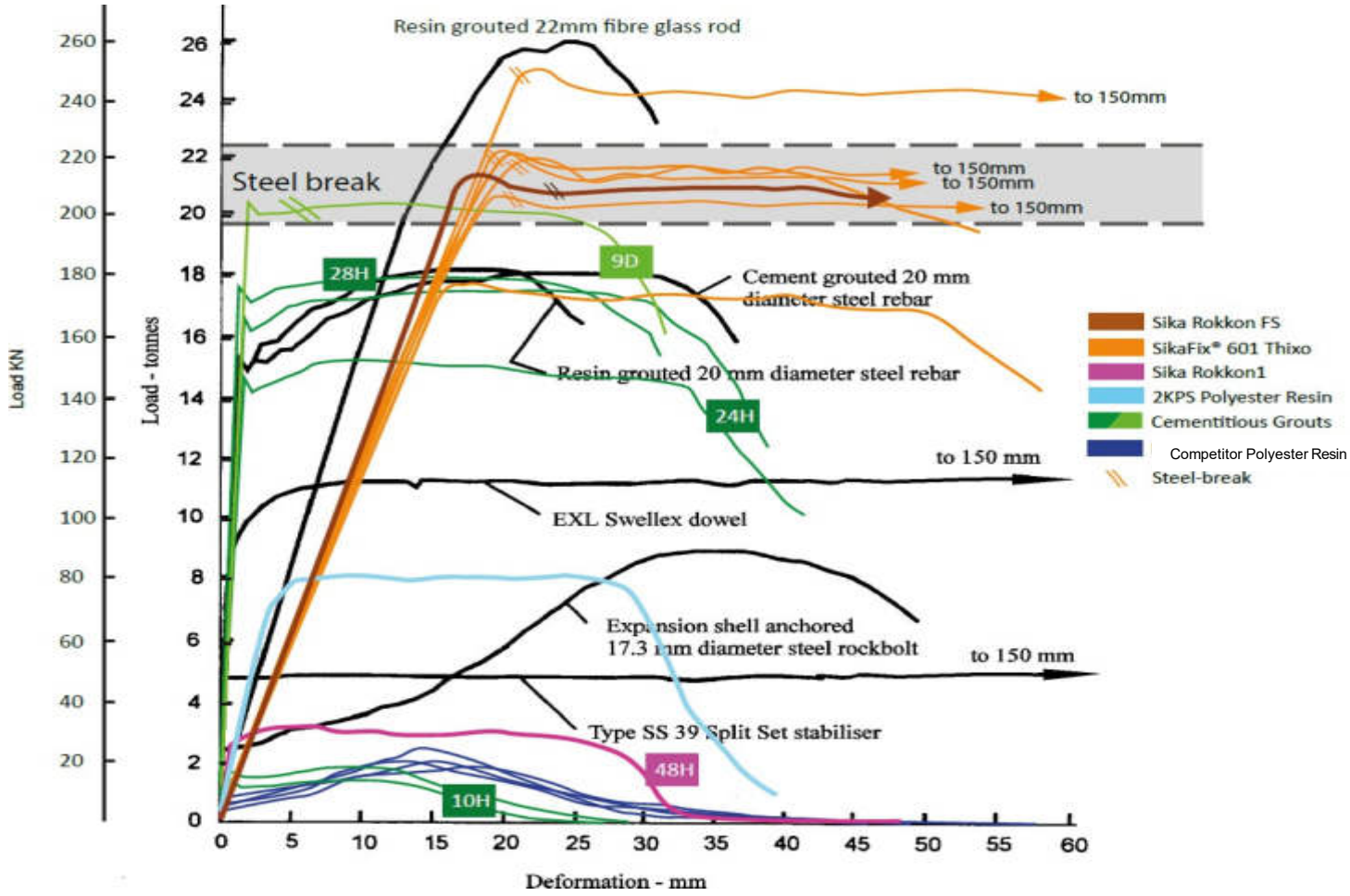
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SIKA ROKKON FS





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SIKA PRODUCT FOR CHEMICAL ANCHOR

CEMENTITIOUS BASE

SikaInject® MG 10 NA (MasterRoc® MG 10 NA)
Thixotropic, high yield, and shrinkage compensated grout used for placement of cable and rock bolts in tunnels and underground mines.
Pre-grouted application

SikaInject® MG 57 (MasterRoc® MG 57)
Thixotropic, high yield, shrinkage compensated grout that exhibits rapid strength gain designed for rock reinforcement for placement of cable and rock bolts in civil tunnels and underground mines.
Pre-grouted application

Sika® Rokkon- C / Rokkon®-1
Cementitious fast setting, shrinkage compensating mortar supply in sausage for the grouting of rock bolts, soils nails.
Pre-grouted application

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SIKA PRODUCT FOR CHEMICAL ANCHOR

RESIN BASE

ROKKON® R (FS)
Polyester very fast setting resin, supply in sausage for the grouting of rock bolts, soils nails.
Pre-grouted application

SikaInject®-601 THIXO 2 (SikaFix®-601 THIXO 2)
Two-component, CFC-free, halogen-free and solvent-free silicate resin with a fast reaction time for solidification and stabilization of soils.

SikaInject®-680 (formerly MasterRoc® RBA 380)
Fast reacting, two-component, instantly-thickening, solvent free polyurea silicate anchoring resin specifically designed for rock bolting. Its thixotropic feature facilitates overhead applications without additional sealing.

SITE MIXES

SikaRapid®-700 L
Liquid hardening accelerator for grout, mortar and concrete. Suitable for Cable and bolt grout
Post-grouted application

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BACK TO PRESENTATION

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SIKAFIBER® FOR SHOTCRETE

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WHY RATING THE ROCK QUALITY IN MINING AND TUNNELING?

- Geology is the foremost controlling factor for the drillting and excavation process in mining and tunneling
- In mining, the rating of the surrounding rock-mass is particularly important due to the absence of a massive inner lining of the excavated infrastructure and the reliance on a stable primary support
- Many metrics rely on the dimensioning of the primary support, including the time and costs of the drifting process, safety conditions, maintenance efforts (including hydrological situation of the UG infrastructure)
- In mines, time and production losses are the largest operating cost factor and the main reason for heavy write downs.

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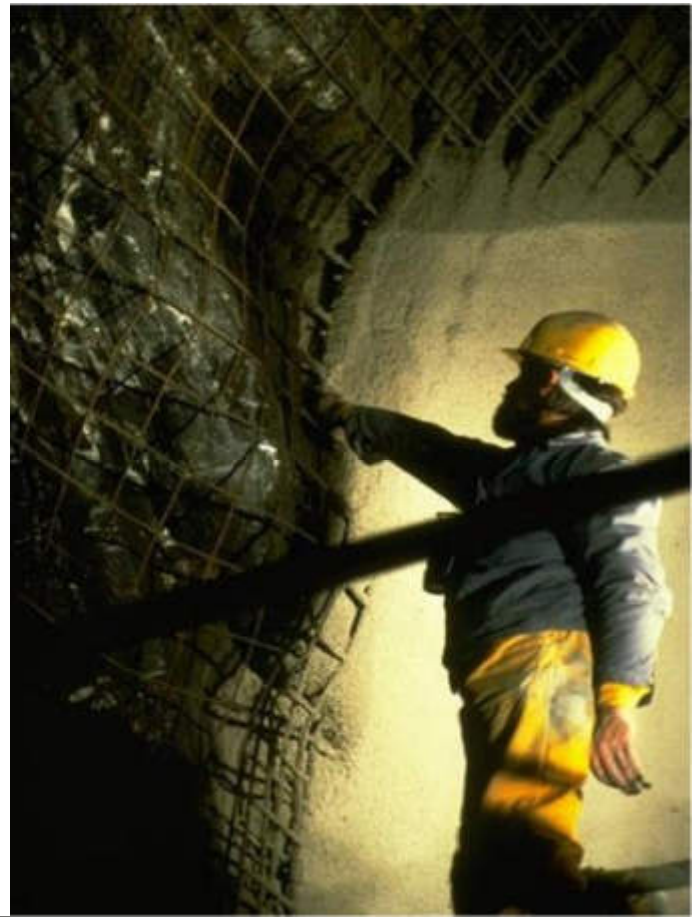
ADVANTAGES OF FIBERS VERSUS STEELMESH

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PROBLEMS WHEN SHOTCRETING WIRE-MESH





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207

PROBLEMS WHEN SHOTCRETING WIRE-MESH

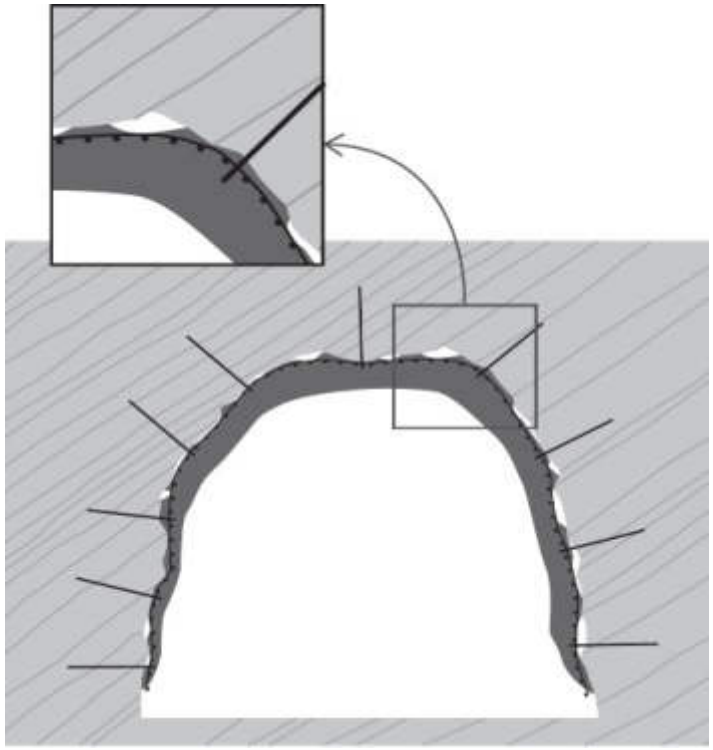
- Steel ribs or wire mesh are difficult to place and often require increased volumes of shotcrete to provide minimum cover over the steel
- Installation of mesh is time consuming and work is conducted under unsupported roof.
- Automated installation of mesh requires special equipment
- Placing of mesh is unregular and in different positions within ths shotcrete liner
- Shotcreting over mesh results in higher rebound (up to 20% more) compared to the sprayed fibercrete due to caused vibration of the mesh while spraying
- Pressure shadow and the creationof voids behind the mesh

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PROBLEMS WHEN SHOTCRETING WIRE-MESH



Application of steel mesh

Steel bar or wire mesh are difficult to place and often require increased volumes of shotcrete to provide minimum cover over the steel

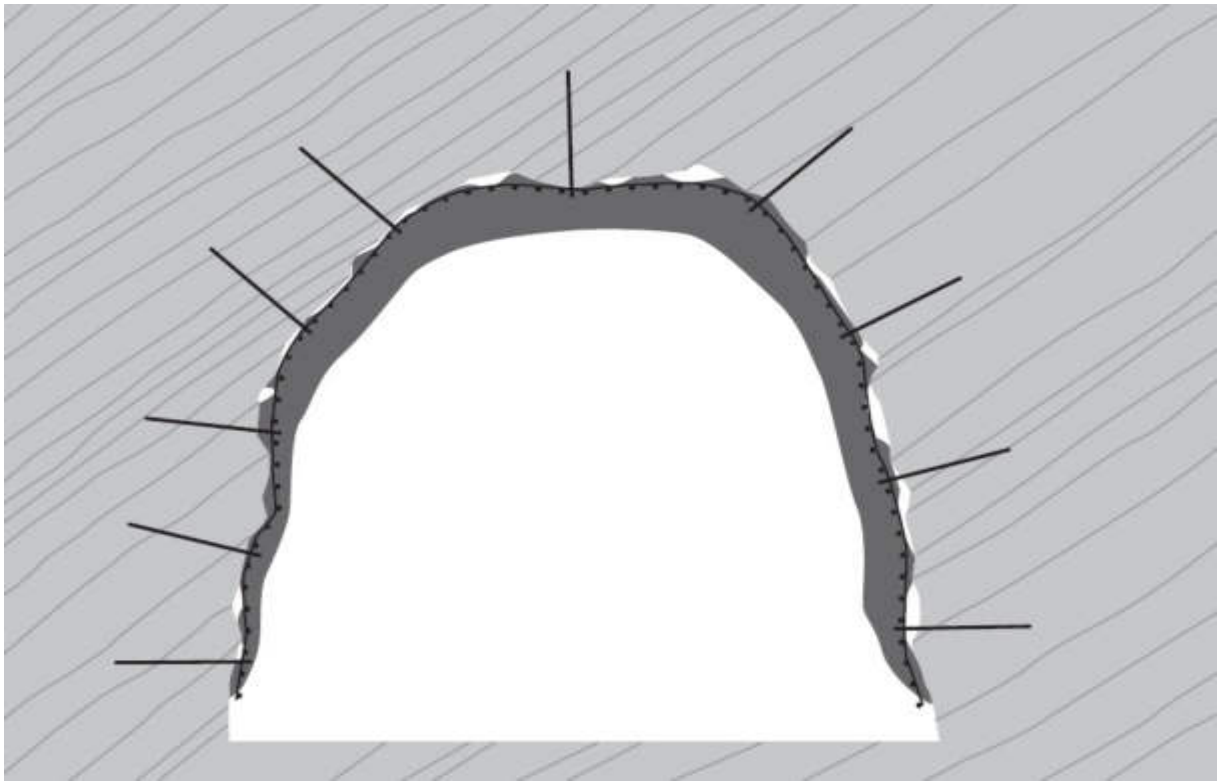
High risk of voids behind the mesh

Can cause excessive vibration while spraying increasing the rebound of shotcrete and consequently higher costs

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ADVANTAGES OF FIBERS VERSUS STEELMESH

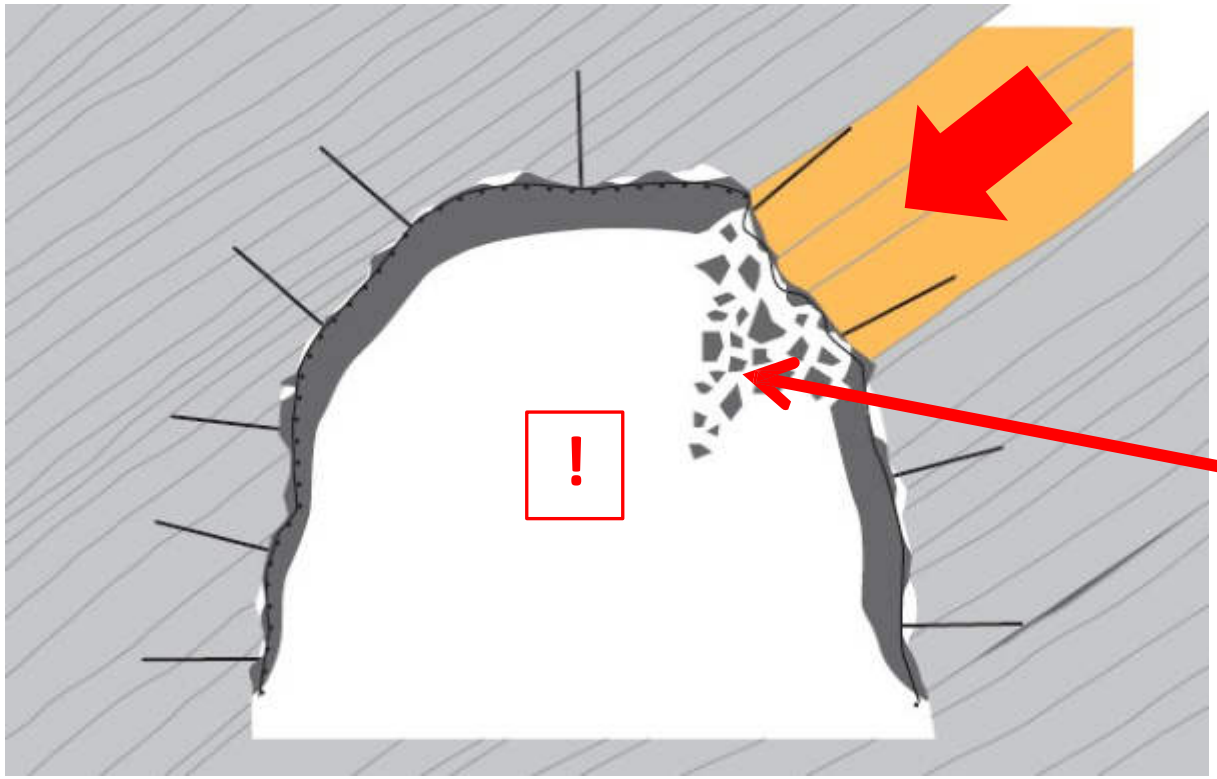


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210

PROBLEMS WHEN SHOTCRETING WIRE-MESH



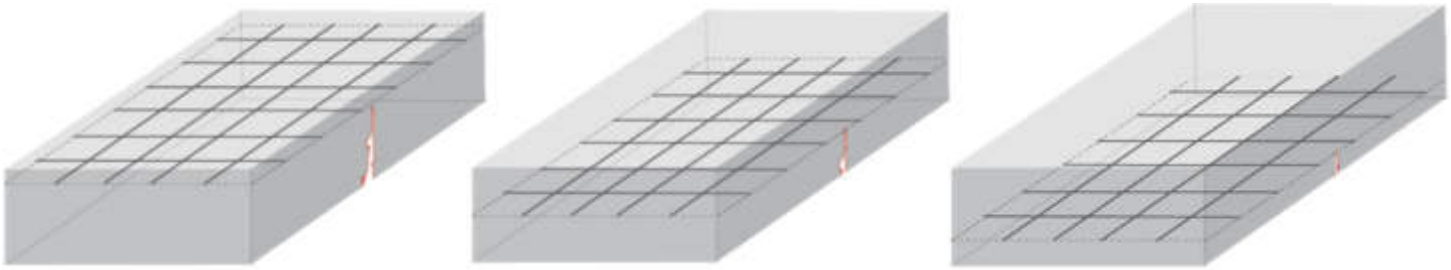
Shotcrete liner above Mesh will fail and fall down when deformation occurs

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ADVANTAGES OF FIBERS VERSUS STEELMESH IN SHOTCRETE?

KEY ADVANTAGES – APPLICATION



Application of steel mesh

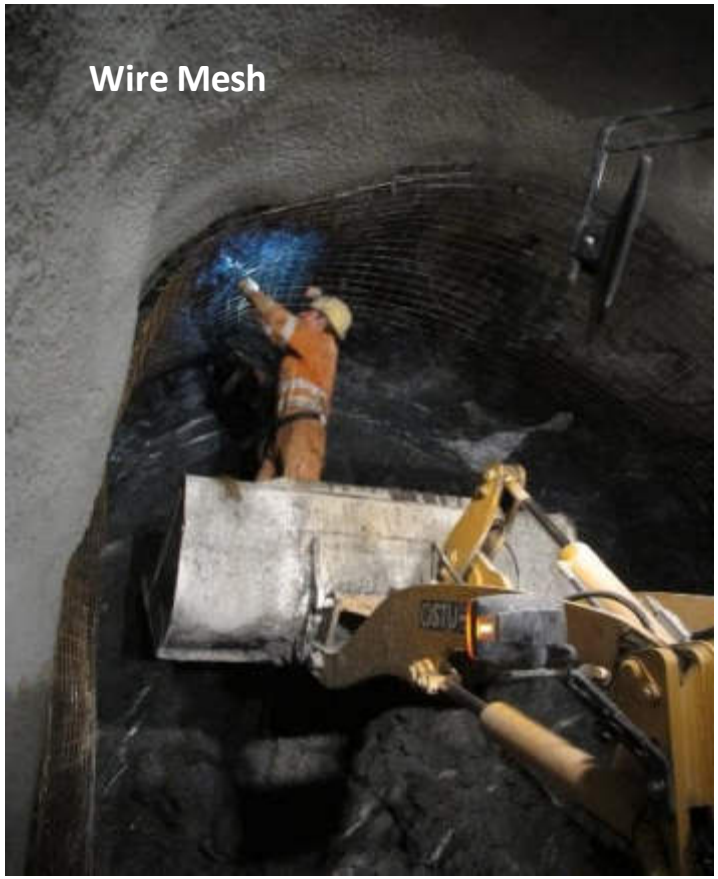
Depending on the final position of the steel mesh inside the reinforced shotcrete arch, **different evolutions of the cracking control** process could occur.
Therefore, the shape of the excavated rock area as substrate of the support arch and the shotcreting itself generate an **heterogeneous system** with different responses to the loads and deflections.
From the design point of view and in order to avoid risks, this deviation usually drives to an extra safety layer of shotcrete and **higher costs**.

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212

CHANGING PRIMARY SUPPORT METHODS IN MINES





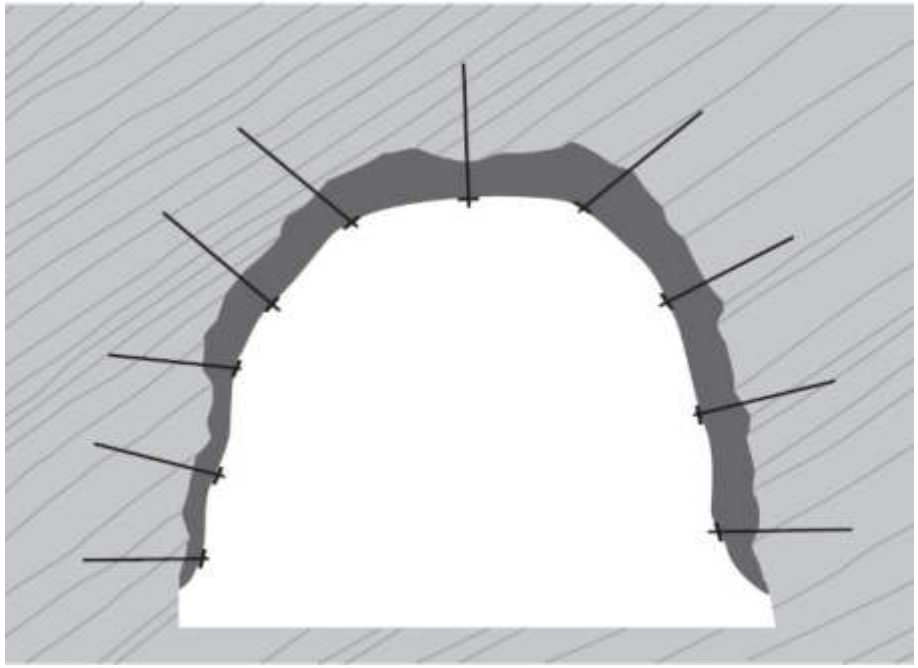
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ADVANTAGES OF FIBERS VERSUS STEELMESH IN SHOTCRETE?

KEY ADVANTAGES – APPLICATION



Application of fiber reinforced shotcrete

Fibers will reduce fall-out and rebound, safer to use - less risk

Less shotcrete used /less cost

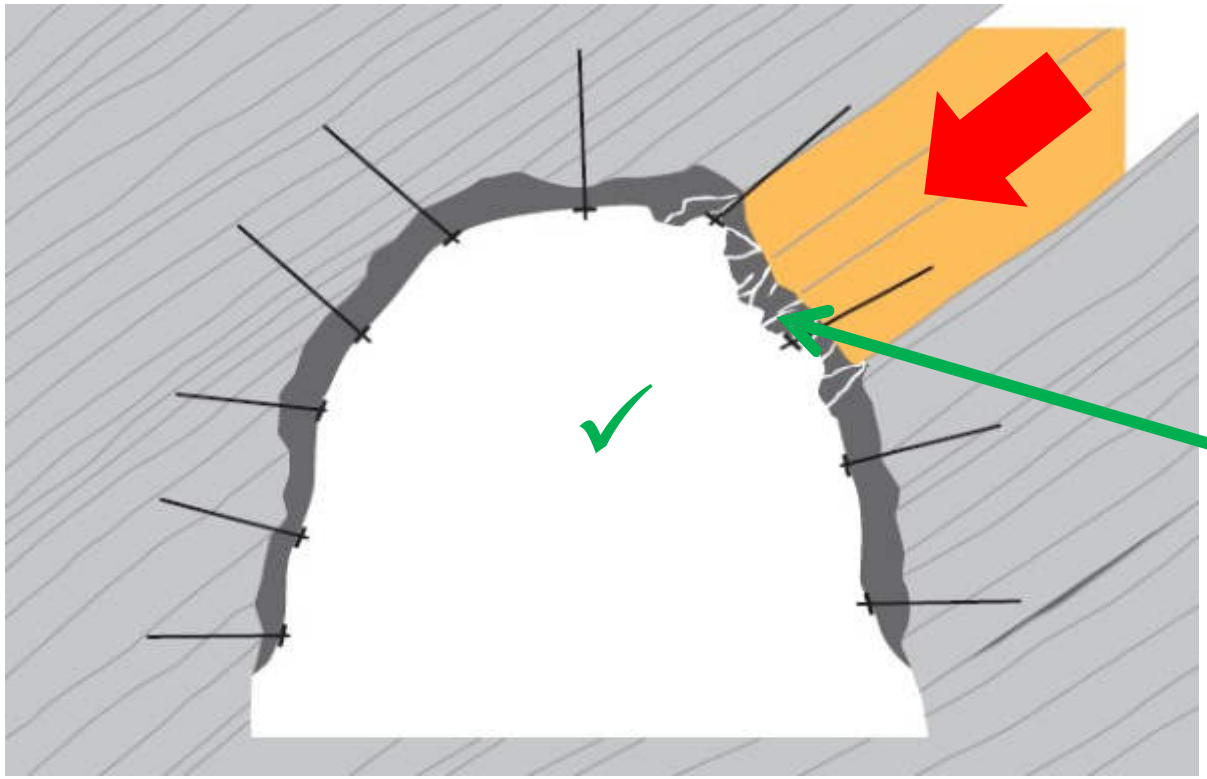
Less time to place/concreting

Fibers can be added at the concrete batch plant

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FIBERS AN SHOTCRETE - AN EXCELLENT YIELDING SUPPORT



Shotcrete lining steel or Macro PP fibers will stay in place an take up dynamic load.

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215

TEST PROCEDURE

For the beam tests fibers (macro, length: 60mm) were used as well as a standard mesh with 100mm spacing.

- 3 plain batches were filled into 3 formworks in total in 9 beams:
- **3xmesh up**, **3xmesh middle**, **3xmesh down**

- 3 Fiber batches were filled into 3 formworks in totoal in 9 beams:
- Fiber dosage **3x6kg/m3**, **3x8kg/m3, 3x10kg/m3**





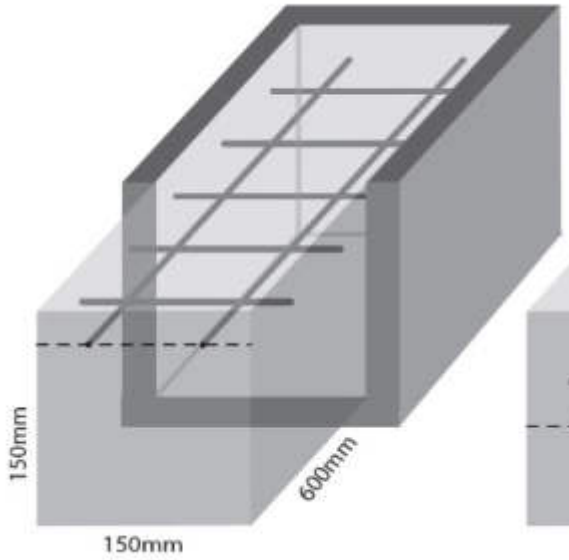
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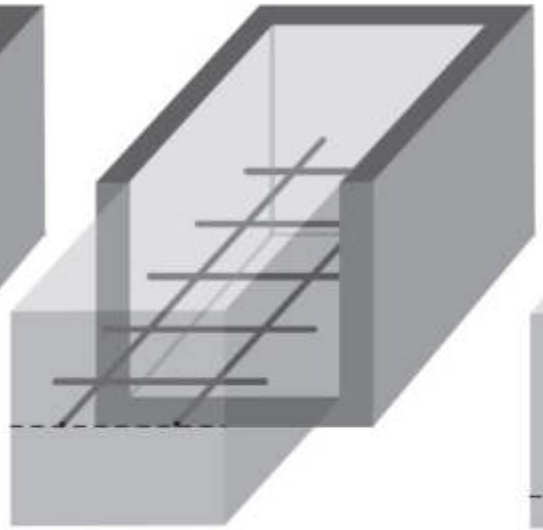
216

TEST PROCEDURE

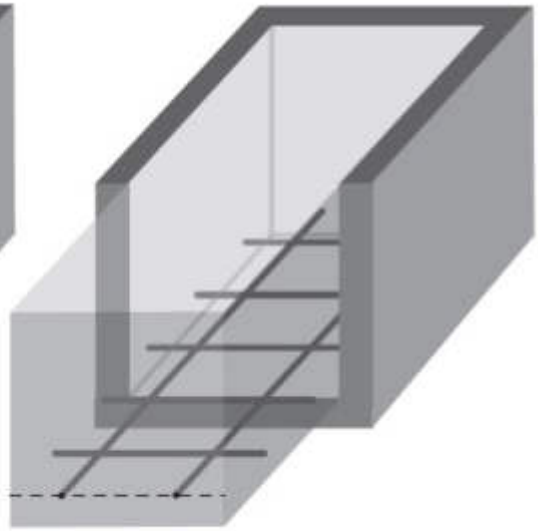
«Mesh-Up»



«Mesh-Middle»



«Mesh-Down»



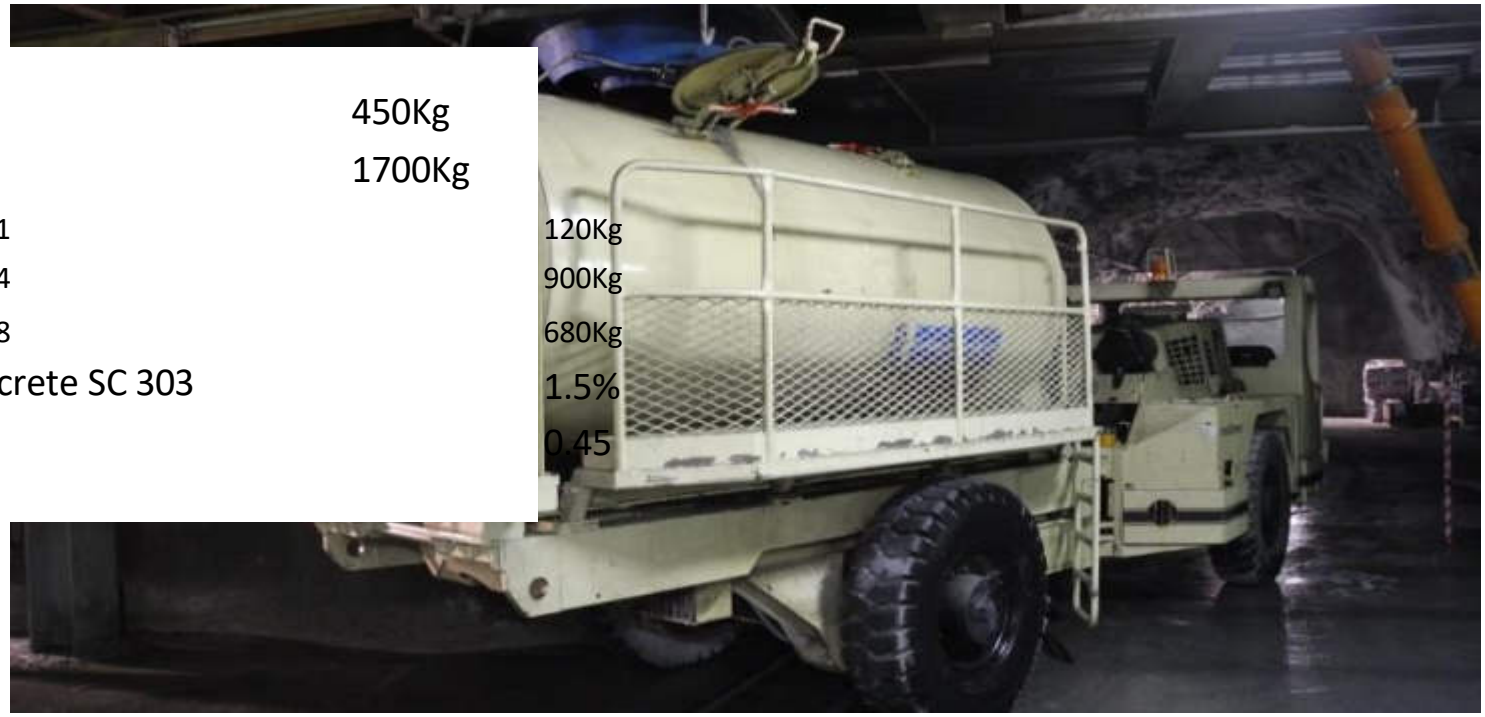
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217

TEST PROCEDURE

- Mix design: Shotcrete mix design



CEM-1		450Kg
0-8mm		1700Kg
	0-1	120Kg
	1-4	900Kg
	4-8	680Kg
Sika Viscocrete SC 303		1.5%
W/C		0.45

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ADVANTAGES OF FIBERS VERSUS STEELMESH

SUMMARY

- Fibers are distrubuted homogeneously in the shotcrete mix and after spraying the fiber content in the shotcrete layer is constant
- The steelmesh is hard to mount on the rough tunnel surface. This results in different positions of the mesh in the shotcrete layer with different effects on the structural performance of the shotcrete layer
- Spraying on the steelmesh causes higher rebound as the mesh vibrates
- Behind the steel rebars spraying shadows can occur which may lead to bad effects on the durability of the structure
- Improved efficiency on site due to time saving for installing of steelmesh, cost saving in less material and less rebound
- Highly improved safety on site as there is no need to work in unsupported area for mounting steelmesh

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ADVANTAGES OF FIBERS VERSUS STEEL MESH IN SHOTCRETE?

KEY ADVANTAGES – HEALTH & SAFETY / DURABILITY

Health & Safety

The aim is to avoid the risky labors during placing the steel mesh in the unsupported area with a high danger of rock wedge falling.

Durability

Synthetic fiber were already proven as the best option when highly corrosive environments are expected while steel reinforcement suffers oxidation where water appears.

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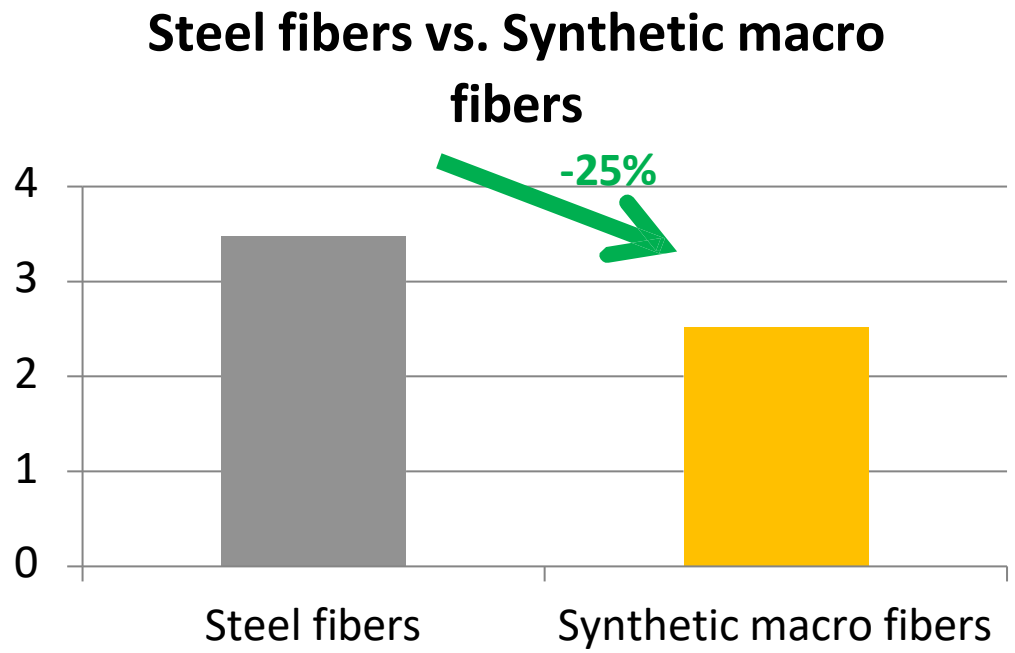
SYNTHETIC FIBERS VERSUS STEEL FIBERS

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HOW TO REPLACE STEEL FIBERS BY SYNTHETIC FIBERS?

SOME COMMENTS & REMARKS

Cost Analysis Example:



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ADVANTAGES OF SYNTHETIC FIBERS VERSUS STEEL FIBERS

SUMMARY

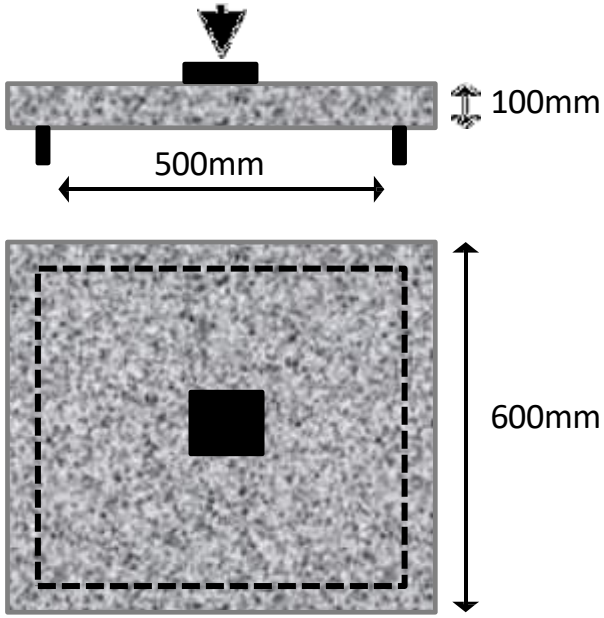
- Steel fibers in the shotcrete mix cause higher abrasion on mixing, transportation, pumping and spraying equipment, which is usually 20% higher than without fibers. This increases the wear costs of the whole machinery
- Synthetic fibers show the same adrasion as normal shotcrete without fibers, therefore the wear costs with synthetic fibers do not increase.
- As the density of synthetic fibers is much lower than the one of steel fibers the equivalent dosage is usually 5 times lower with synthetic fibers
- This causes less weight which needs to be transported to the batching plant
- For tunnel projects where a waterproofing membrane is installed steel fiber reinforced concrete can cause damages on the membranes and a protection layer without fibers needs to be sprayed on top
- Synthetic fiber reinforced shotcrete does not harm the waterproofing membranes and there is no need for an additional spraying layer

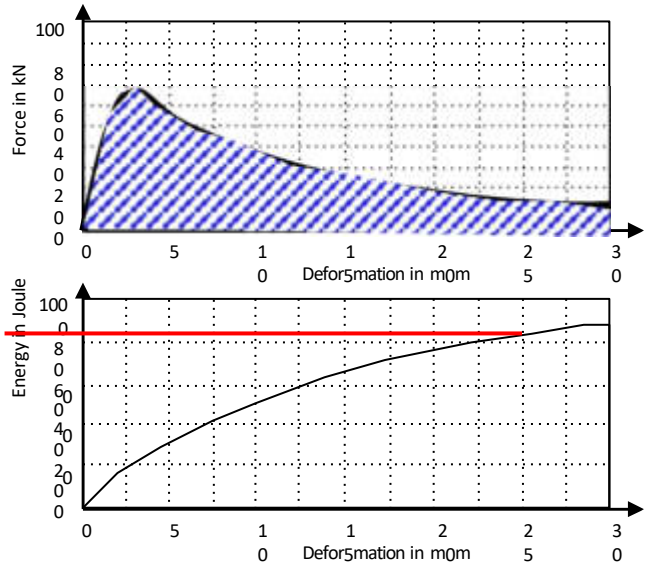
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TESTING SHOTCRETE PANELS

PANEL TESTS EN 14488-5 / ENERGY ABS. CLASSES EN 14487-1





Energy absorption class	Joule until w=25mm	Synthetic Fiber	Steel Fiber
E 500	500	4-6	20-25
E 700	700	5-8	25-35
E 1000	1000	>8	>35

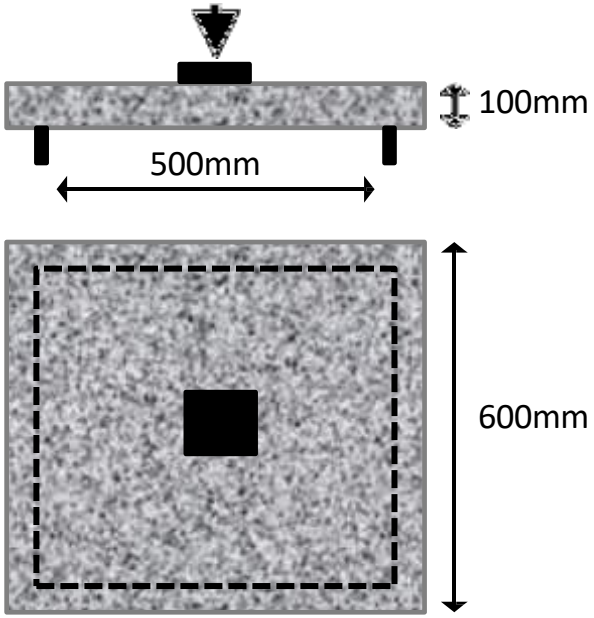
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TESTING SHOTCRETE PANELS

PANEL TESTS EN 14488-5 / ENERGY ABS. CLASSES EN 14487-1





Energy absorption class	Joule until w=25mm	Synthetic Fiber	Steel Fiber
E 500	500	4-6	20-25
E 700	700	5-8	25-35
E 1000	1000	>8	>35

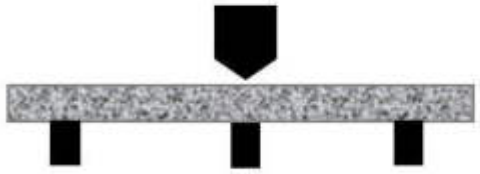
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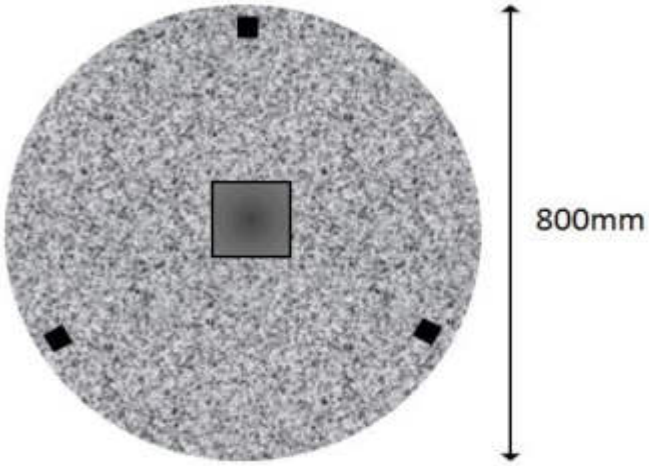
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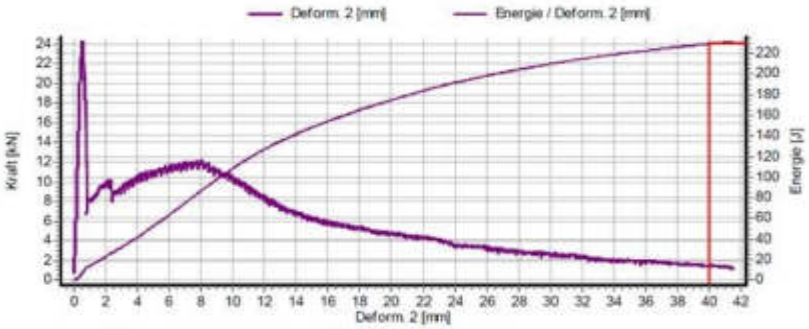
225

TESTING SHOTCRETE PANELS

PANEL TESTS ASTM C-1550









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TESTING SHOTCRETE PANELS

COMPARISON EFNARC VS. ASTM

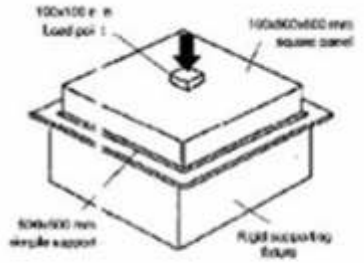


Figure 8. EFNARC panel test [7].

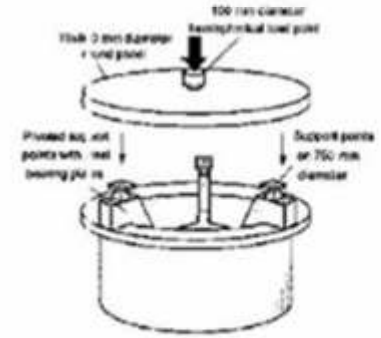
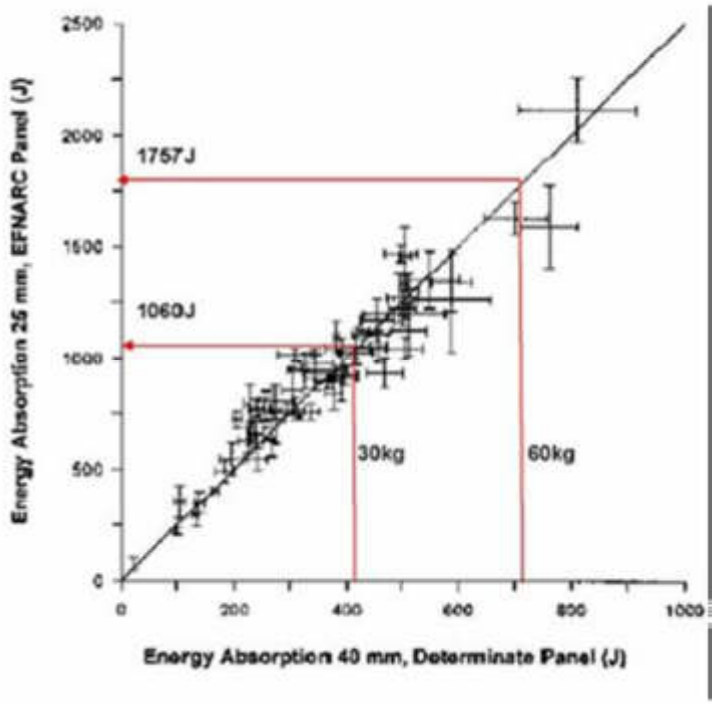


Figure 10 Round determinate panel test.



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PRACTICAL ISSUES

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FIBER REINFORCED SHOTCRETE

PRACTICAL ISSUES PUMPING

General

Blockages often result of:

- High pumping pressure → segregation
- Reduction pipes → 1 inch reduction = up to 2 x pumping pressure
- Short fiber → starts to rotate and blocks
- Rubber hoses → high friction between side wall and concrete matrix
  - Rubber hoses vs Steel pipes = up to 3 x friction value
- Viscosity of concrete more important compared to slump or slump flow

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FIBER REINFORCED SHOTCRETE

PRACTICAL ISSUES PUMPING



What is important on site?

- Mix design
- Workability
  - If possible check influence of fibers on the workability
  - Fiber concrete has to be pumped and has to pass the mesh on the hopper!

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FIBER REINFORCED SHOTCRETE

PRACTICAL ISSUES PUMPING



What is important on site?

- Right machine setup
  - Spraying volume (not too large)
  - Accelerator dosage for spaying testing specimens (low dosage for plate tests ~3%)
  - Check pumping pressure
  - Check air pressure
  - Check accelerator pressure, dosing unit

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FIBER REINFORCED SHOTCRETE

PRACTICAL ISSUES PUMPING



What is important for spraying boxes or plates for testing?

- Right positions
- Fix position of plates (fix angle , fix distance to nozzle)
- Spray 90° angle to plate
- Note positions for further tests in future

FIBER REINFORCED SHOTCRETE

PRACTICAL ISSUES PUMPING





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SIKAFIBER® FOR SHOTCRETE APPLICATION

- Length: 48 to 65 mm
- Equivalent diameter: ~0.8mm
- Stiff fiber
- Embossing optimized for shotcrete
- All fibers are available in water soluble pucks for automatic dosing systems and better distribution in the concrete mixer
- Fiber pucks are available in watersoluble bags with 5kg each or in bigbags



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BACK TO PRESENTATION

BACK TO PRESENTATION “UNDERGROUND SUPPORT”

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ULTRA HIGH STRENGTH CONCRETE
FOR SLAB BY MEAN OF HEAVY PRECAST ELEMENTS

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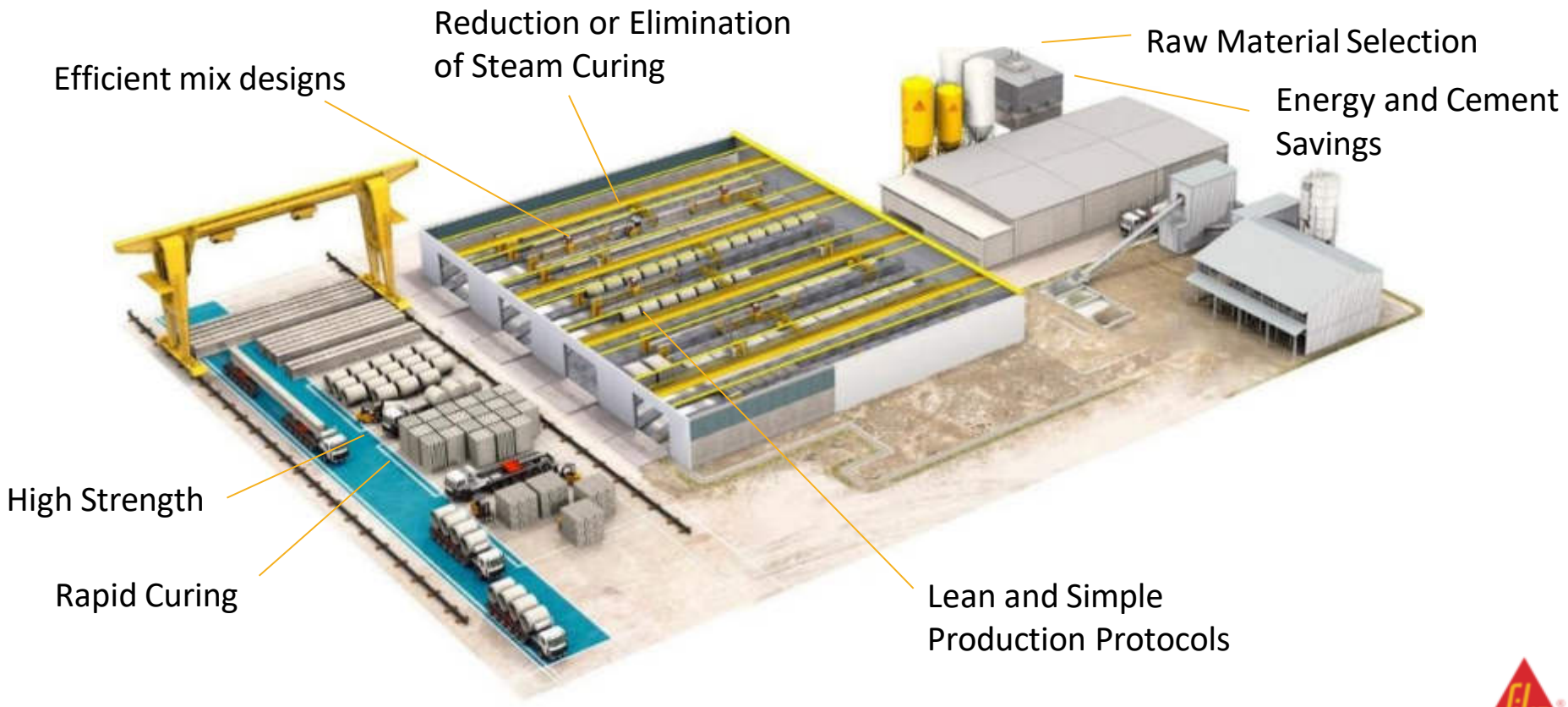
SIKA - THE GLOBAL LEADER IN WET PRECAST PRODUCTION



- Sika has decades of heavy precast experience from sites and pre-cast plants around the globe
- Very experienced with challenging raw materials and difficult site conditions
- Leading edge for efficient materials supply
- Clever, all-inclusive, pre-bagged solutions for Ultra High Strenght (UHS) precast applications

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WITH SIKA FROM MOLD TO MINE – FAST AND RELIABLE



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CRITICAL ASPECTS IN HIGH STRENGTH, HEAVY PRECAST PRODUCTION

- High early strength for rapid demolding
- Fulfil final strength requirements UCS* of 140-160 MPa
- Good workability as per mix concept
- Reduce steel reinforcement by fibers
- Elimination of heat- or steam curing
- High final quality and maximum durability
- Simplified production procedure
- As lean logistics as possible
- Sustainable and environmental friendly
- production at maximum CO_2 saving



* Uniaxial Compressive Strength

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PRECAST OPTION FOR UHPC * PRODUCTION



* Ultra High Performance Concrete

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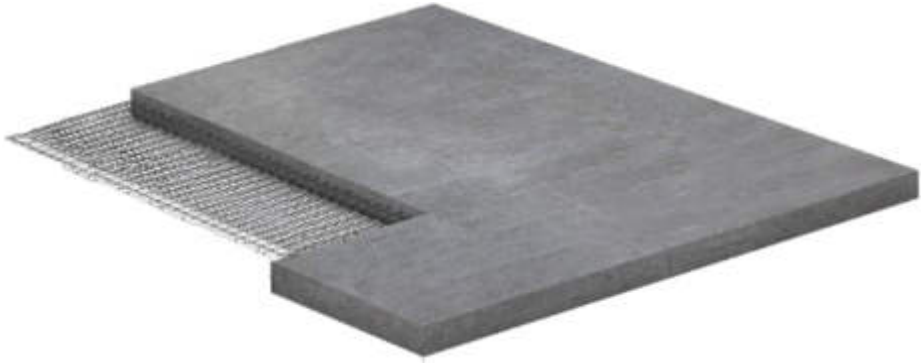
240

PRECAST OPTIONS FOR UHPC* PRODUCTION

Requirement:

Pre-cast elements of 4 x 5 x 0.4m of UCS** between 140-160 MPa

Use: Ground slabs for haulage levels, extraction drives, draw points



* Ultra High Performance Concrete ** Uniaxial Compressive Strength

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OPTION 1	OPTION 2	OPTION 3
Pre-cast plant using on site produced concrete	**Pre-packaged, Ultra-High Performance Concrete topping with conventional grade concrete base**	**Pre-packaged, Ultra-High Performance Concrete Mix**
	  Sika/King UP-F3-Poly	 Sika/King UP-F3-Poly
+ Sika® Viscocrete® + Sika® Rapid® + SikaFiber® + SikaCrete® L Slurry (liquified silica fume, made locally) + Imported Cement and good quality aggregates/Sand	+ Sika® Viscocrete® + Sika® Rapid® + SikaFiber® + SikaCrete® L Slurry (liquified silica fume) + Sika/King UP-F3 Poly pre-packaged Ultra-high Performance Concrete topping	+ Sika/King UP-F3 Poly pre-packaged Ultra-high Performance Concrete
		
-Most sensitive option with regards to quality issues and variations in used raw materials -Batching and production procedure is critical -Good know-how in pre-cast needed -Excellent admixtures needed	-Use of local raw materials possible for base concrete structure -Pre-bagged topping would provide quality assurance and ease of application -Two stage manufacturing process of slabs	-Easiest and fastest way to produce the slabs but likely also the most expensive -Quality assured at all times -Doable with limited pre-cast experience -Excellent product mix needed

Application characteristic

© Sika Services

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OPTION 1

Full pre-casting on site with local or imported raw materials

Pros/cons

- Quality of raw material limited
- Reaching designed strength difficult
- Maintain quality throughout produced volume will be difficult
- Less expensive option

OPTION 2

Full pre-casting on site using local raw materials and pre-packaged ultra-high performance concrete topping with lower grade base concrete

Pros/cons

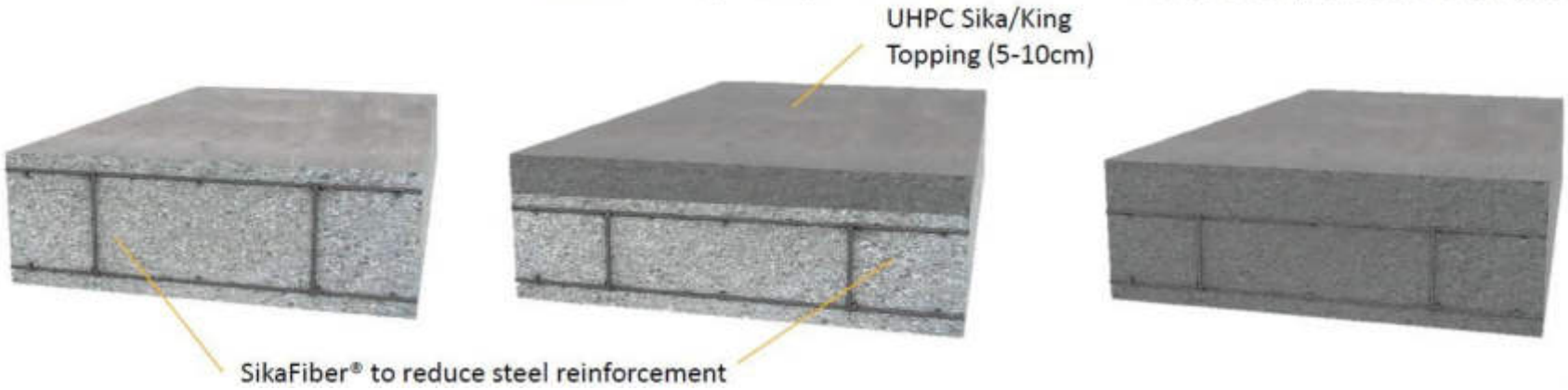
- Main slab could be of conventional grade concrete strength (C50 concrete ?)
- Topping (10cm) would be Sika/King UHS concrete made of a pre-bagged imported mix
- Second less expensive option

OPTION 3

Full pre-casting on site using pre-bagged Sika/King UP F3 Poly UHPC mix

Pros/cons

- Full casting of the slab with Sika/King UP F3 Poly UHPC mix
- Potential to reduce steel reinforcement, thickness and weight of the slabs using this UHPC mix
- Most durable, leanest and fastest option



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243

OPTION 1 - POTENTIAL MIX DESIGN FOR ON SITE PRODUCTION

- CEM1 42.5N or higher 700-800 Kg/m³
- SikaCrete® L Slurry* 350 Kg/m³
- Sikament® 1200N 2-4% by weight of cement

- Sand (0-4mm) 650-700Kg/m³
- Gravel (4-8mm) 700-800Kg/m³

- SikaFibre® PP 10mm 2 Kg/m³
- SikaFibre® Steel 40 Kg/m³

- Water/Binder Ratio 0.2

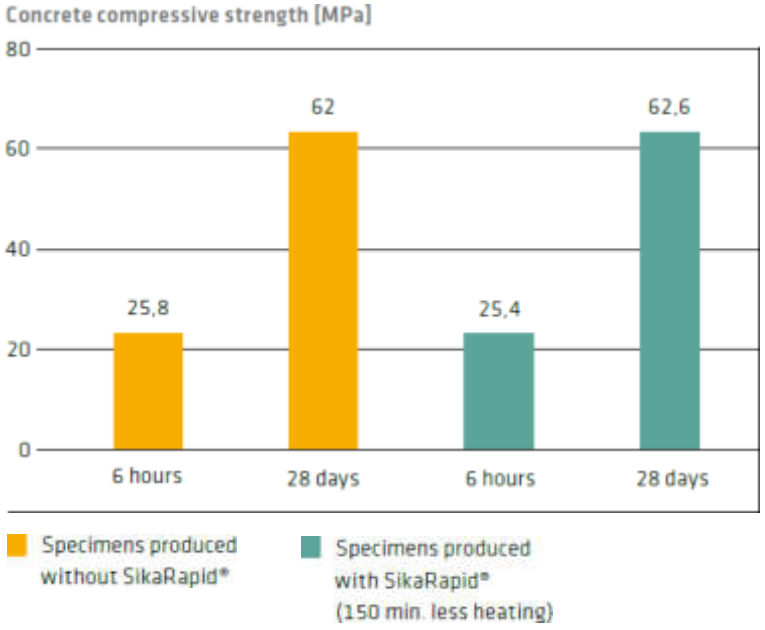
* Containing 143Kg SilicaFume powder and 200Kg Water (50% solids)





OPTION 1 -SIKA RAPID FOR FAST DEMOLDING

Elimination of Steam Curing

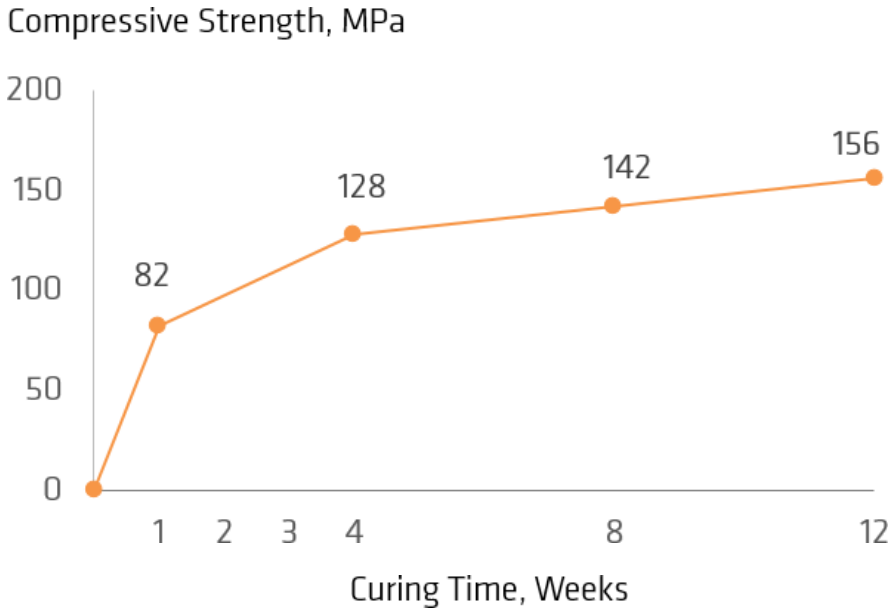




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OPTION 1 - POTENTIAL MIX DESIGN FOR ON SITE PRODUCTION

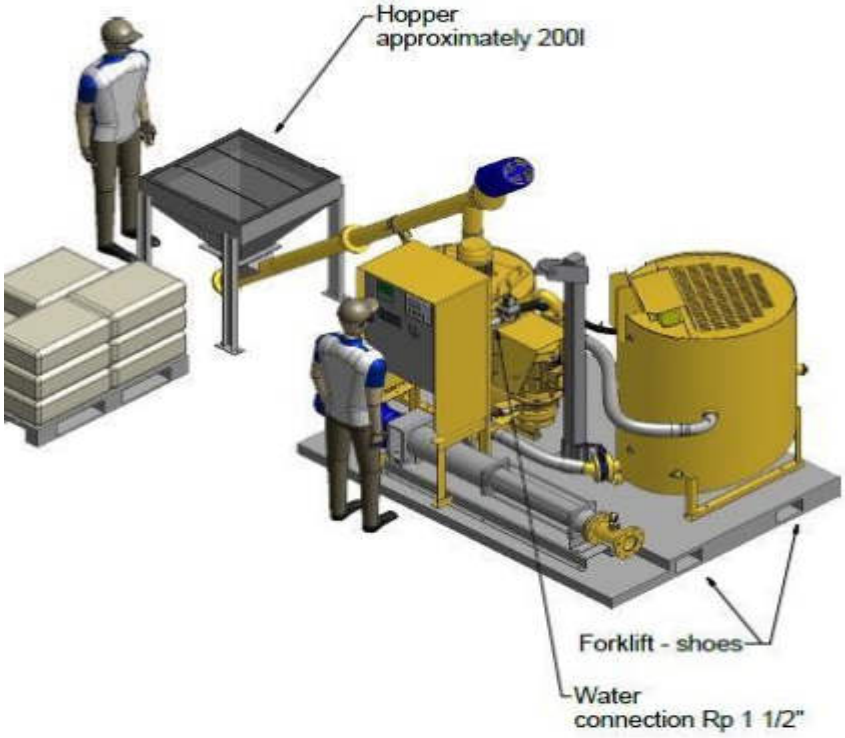




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OPTION 1 – PROVEN SIKACRETE® L SLURRY PRODUCTION AT SITE

- Sika has established equipment at mining site for the production of a liquified Silica Fume Slurry, namely SikaCrete® L Slurry



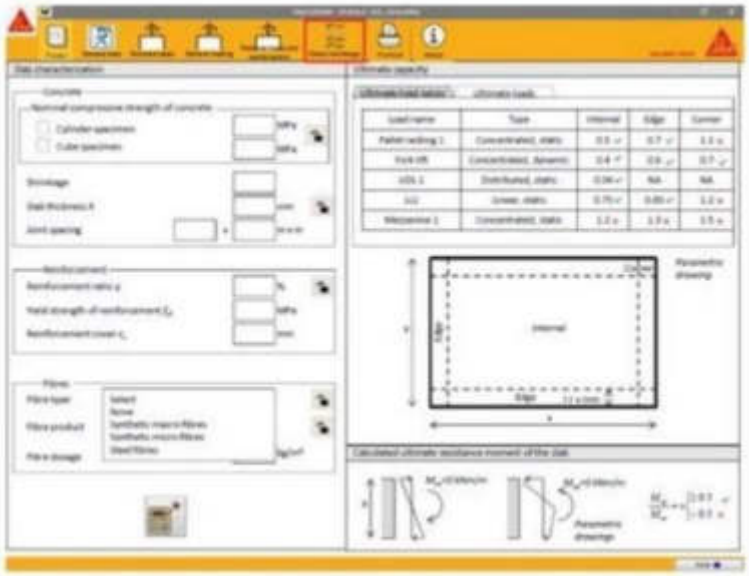
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OPTION 1 – DESIGN CALCULATION AND MIX OPTIMIZATION

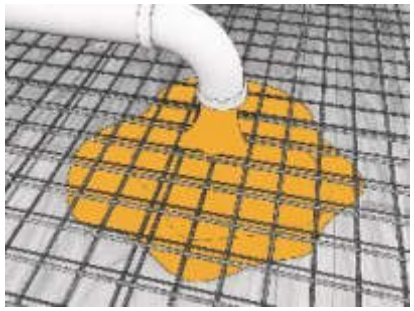


Sika Design Software to minimize steel reinforcement in «Slab on Ground» elements by the use of Sika Polypropylene Macrofibres (SikaFiber®)



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OPTION 1 – COVERING ALL DETAILS



Workability Improvement

- High range water reduction combined with required workability time

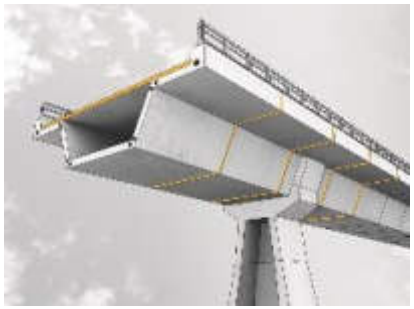
- Sikament® 1200N
- SikaCrete® L Slurry (liquified silica fume produced on site)



High Early Strength

High range water reduction is is the best pre-condition for fast early strength and high final strength. Early strength can be boosted by the use of accelerators

- Sikament® 1200N
- Sika Rapid®-1



High Durability and Strength

- High final strength of concrete for maximum durability, impact- and abrasion resistance and low permeability
- Shrinkage & crack reduction
- Freeze-thaw resistance

- Sikament® 1200N
- SikaCrete® L Slurry
- SikaFiber®
- SikaControl®



Damage Repair

- Repair of broken edges or smooth out irregular surfaces and fix cracks

- Sika® MonoTop® repair mortar
- Sikadur® crack repair resin

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OPTION 2/3 – UHPC READY BAGGED MIX SIKA/KING UP-F3 POLY

- Density: 2400Kg/m3
- UCS: >140 MPa after 28 days

- Excellent freeze-thaw resistance
- Designed to rapidly cure at 21 degrees WITHOUT steam curing
- Very high impact and abrasion resistance
- Excellent bond to parent concrete
- Self compacting



Excellent workability, fast strength gain, no steam curing needed

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A FEW WORDS ON UHPC (ULTRA HIGH PERFORMANCE CONCRETE)

THAT IS 226 MPA !

900 kg/m3	SPC (SikacreteNT-OPC)*
225 kg/m3	Flyash
135 kg/m3	Water (w/b = 0,12)
714 kg/m3	Sand (Ferro nickelslag)
549 kg/m3	Gravel
39 kg/m3	Steel Fibers
19,7 kg/m3	Sikament® 1200N (1,75%)





*Mixed cement with liquid nano-silica

UHPC - DEFINITION

High Performance Concrete is a concrete type with a high resistance against physical and/or chemical loads

	W/B* ratio	UCS**
▪ Ordinary Concrete	> 0,40	20 – 50 MPa
▪ High Performance	0,40 – 0,30	50 – 80 MPa
▪ Very High Performance	0,30 – 0,20	80 – 150 MPa
▪ Ultra High Performance	< 0,20	> 150 MPa

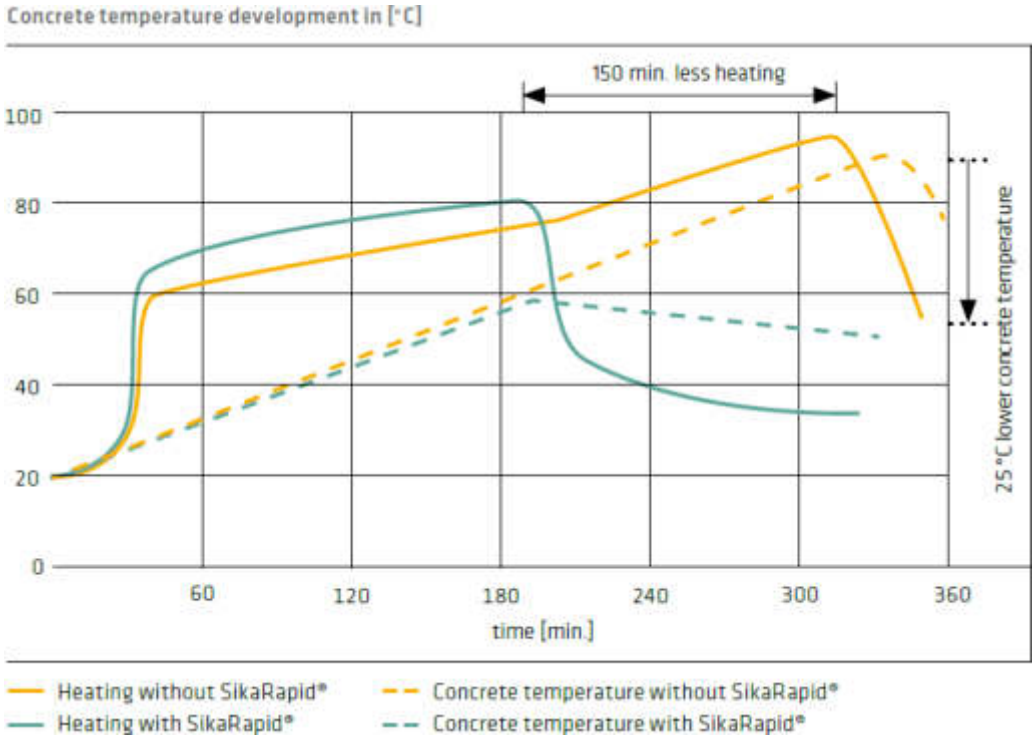
*Water to Binder Ratio ** Uniaxial Compressive Strength

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252

SIKA RAPID –THE GO TO SOLUTION FOR RAPID PRECASTING



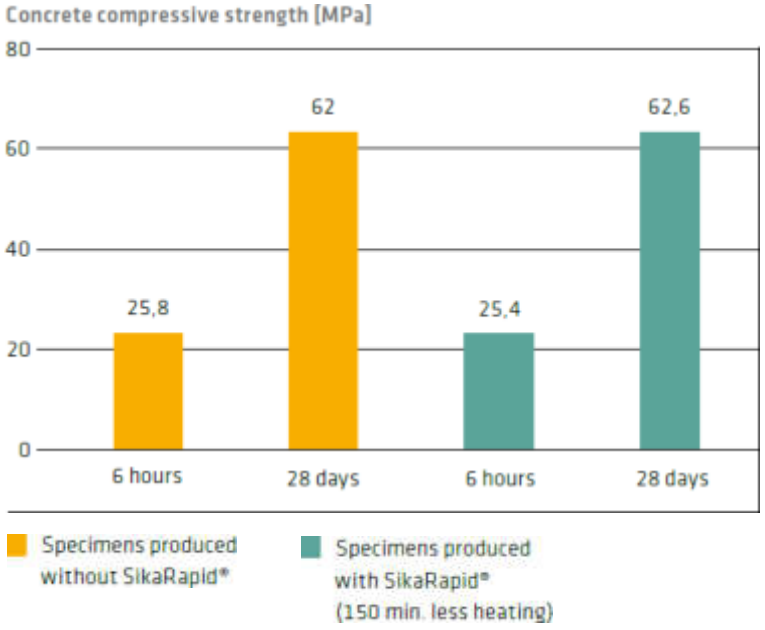


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253

SIKA RAPID –THE GO TO SOLUTION FOR RAPID PRECASTING

Elimination of Steam Curing



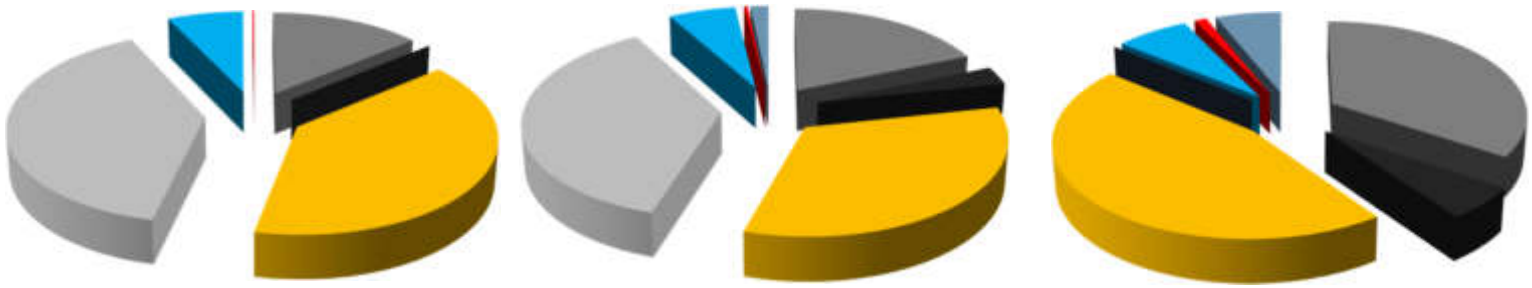


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ULTRA HIGH PERFORMANCE CONCRETE – MIX DESIGN

Concrete **HPC*** **UHPC****



Cement
SCM
Water
Sand
Gravel
Admixtures
Fibers

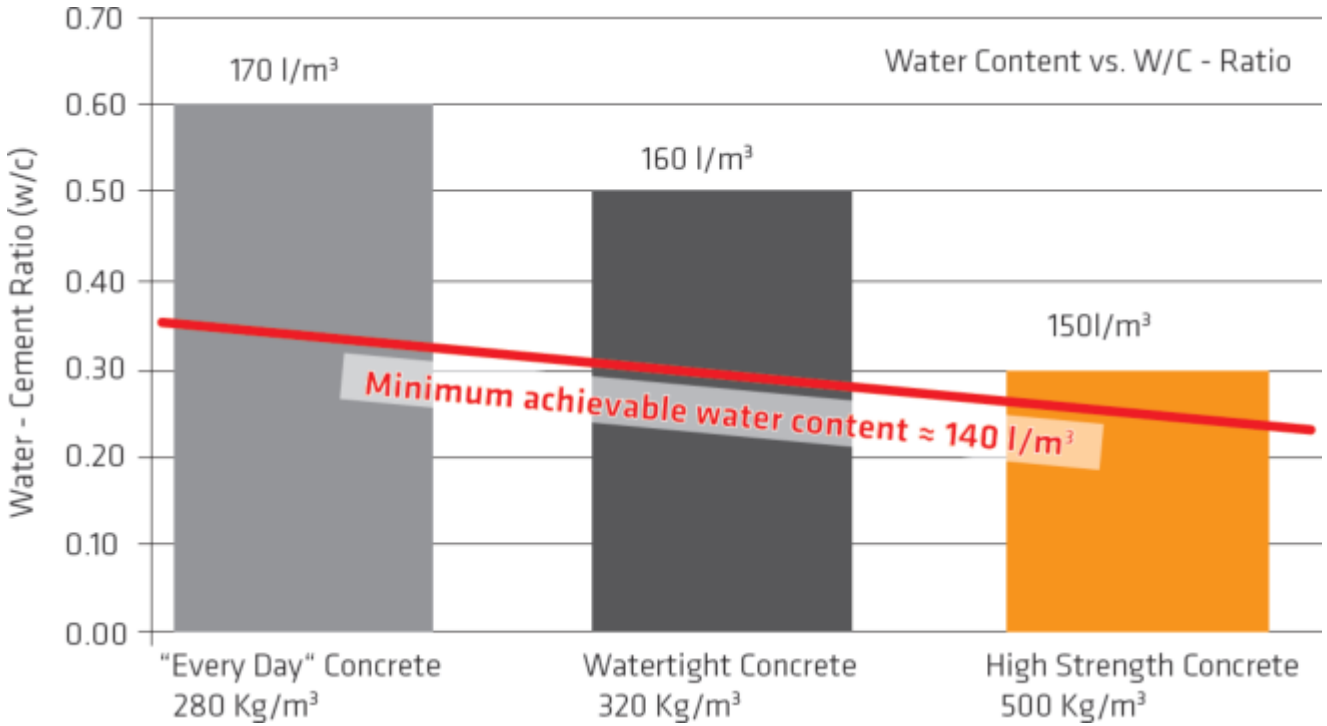
*High Performance Concrete **Ultra High Performance Concrete

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ULTRA HIGH PERFORMANCE CONCRETE – MIX DESIGN



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256

ULTRA HIGH PERFORMANCE CONCRETE – BINDER CONCEPT

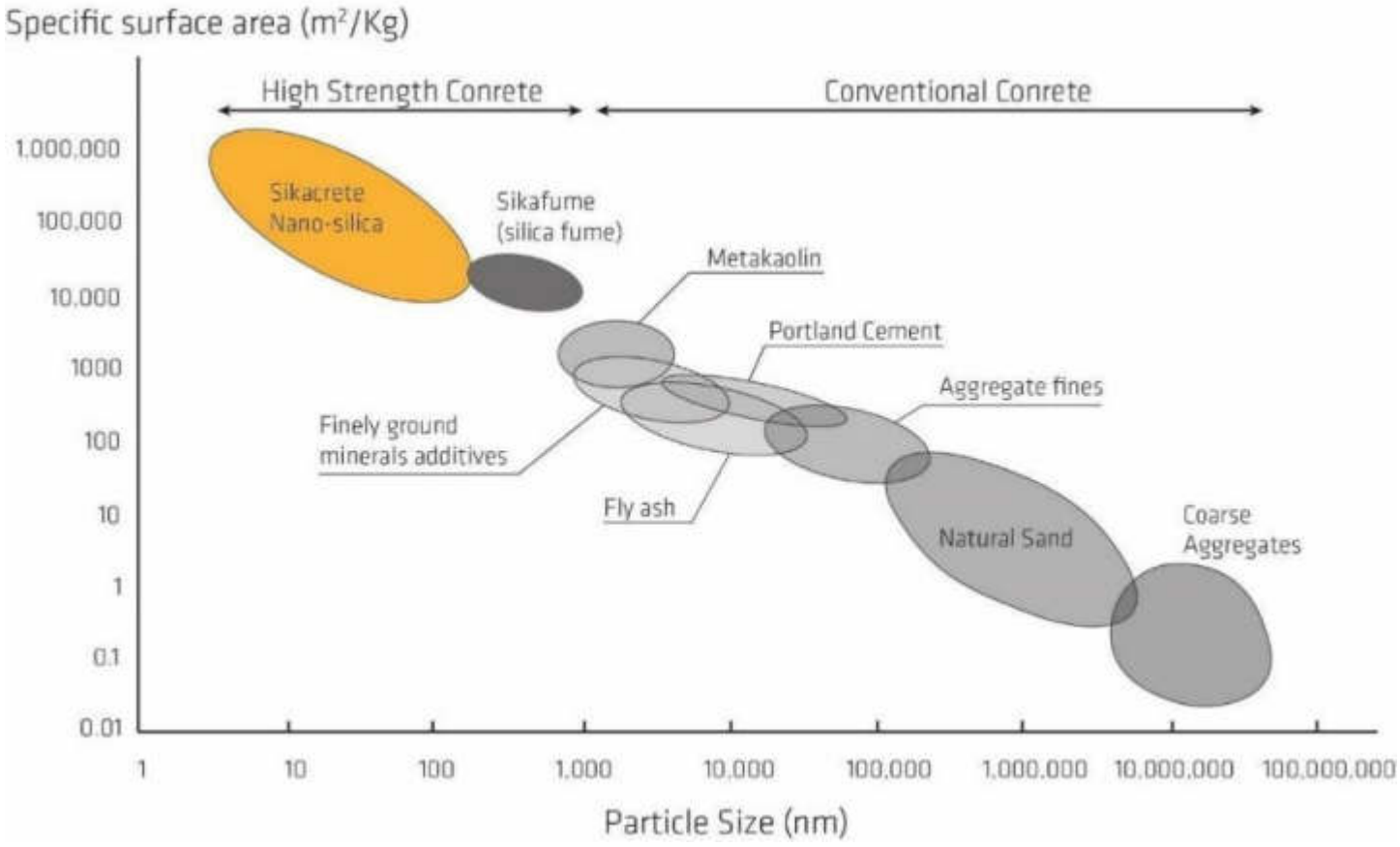
- **Type** — Portland Cements or SF, *GGBFS, *FA
- **Content** — Not strength class is key criteria, workability requirement define minimum content of cement and influences the "best combination" of Binders
- **Powder Additives**

Type	Content	Influence strength (by replacement)
MS Microsilica	**3% to 8%**	**20% to 50%**
FA Flyash	**10% to 33%**	**5% to 20%**
GGBFS Blast Furnace Slag	**10% to 50%**	**10% to 30%**

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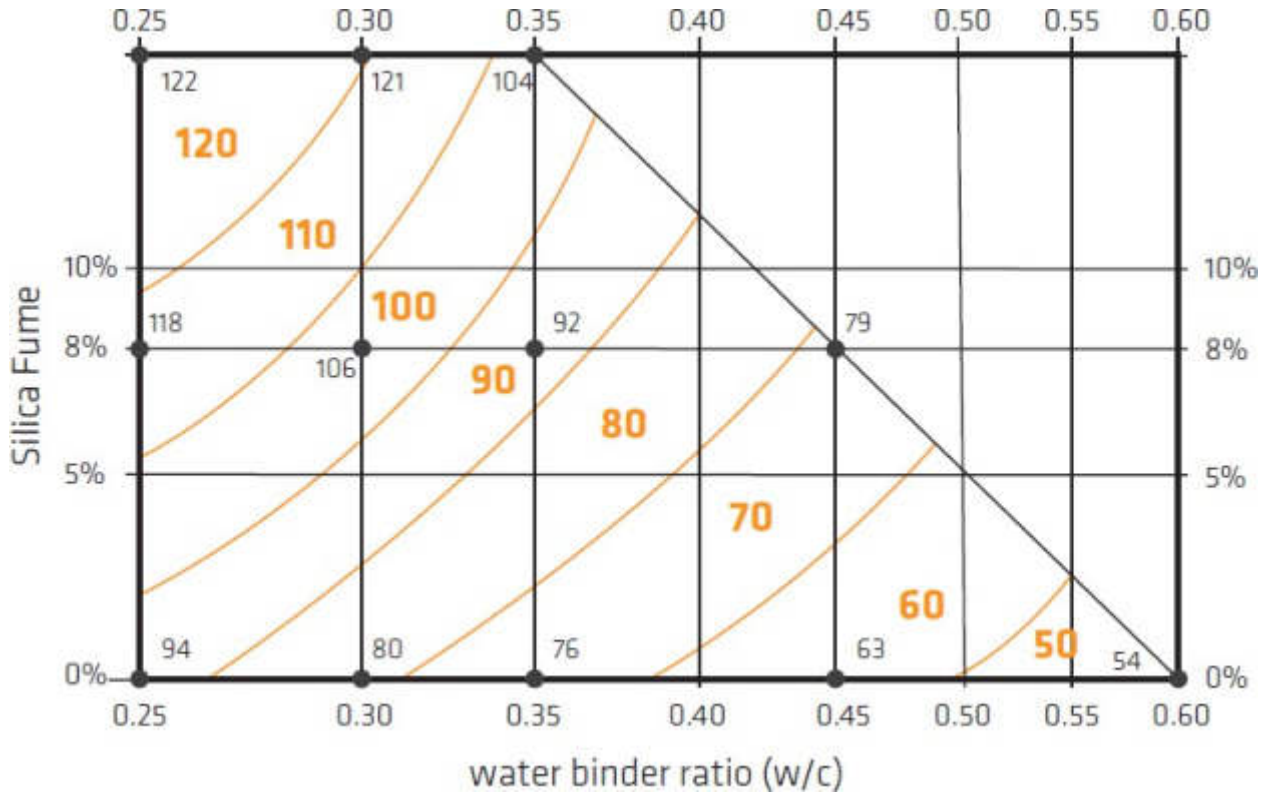
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WHY SILICA FUME IS IMPORTANT FOR UHPC DESIGNS



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WHY SILICAFUME IS IMPORTANT FOR UHPC DESIGNS



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ULTRA HIGH PERFORMANCE CONCRETE – AGGREGATE CONCEPT

▪ **Quality**	Limestone	80 to 180 MPa
	Gneiss	160 to 280 MPa
	Crushed Basalt	UHPC > 200 MPa
	Crystalline Quartz	UHPC > 200 MPa
▪ **Characteristics**	Crushed, at least coarse aggregates	
	Absolute clean (no silt on the surface)	
	No structural cracks (inner)	
▪ **Sieve curve**	Depend on the application	most < 16mm
	Reduced content of Sand	total amount
	Reduced amount of fraction	2mm to 4mm
	Increased amount of fraction	0,25mm to 1mm

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AGGREGATES ARE A KEY COMPONENT!

H_2O 140 l/m^3 ≈ 0.33 w/b



Concrete matrix failed

H_2O 120 l/m^3 ≈ 0.28 w/b



Aggregates failed

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261

INDEPENDENT TESTING (CANADIAN MILITARY @ -70)

Additional information on the impact resistance of UHPFRC since it is related to the purpose of the application. The Royal Military College of Canada carried out a detailed impact resistance test on a 38mm section of UHPFRC. Full report is available on request.

The Sika/King UHPFRC was deemed practically indestructible (up to 2000 Joules without failure).

Some of the tested slabs were frozen to -70 *C before the impact load (to simulate extrema harsh climates).



International Journal of Impact Engineering

Cold temperature effects on the impact resistance of thin, lightweight UHPFRC panels

Matthew Beirnes*, Marc-André Dagenais, Gordon Wight

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BACK TO PRESENTATION

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ROCK INJECTION

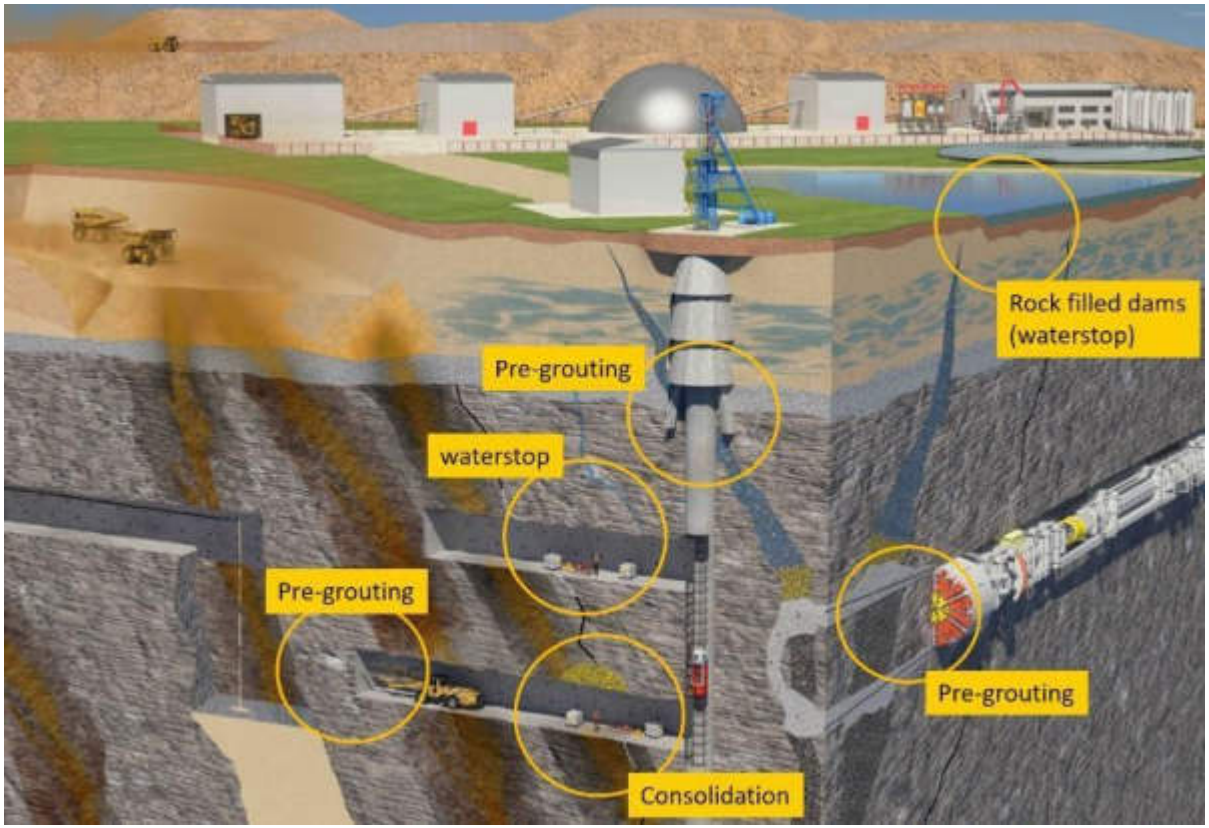
INJECTIONS FOR TUNNELING AND MINING

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UNDERGROUND CONDITIONS

GROUND IMPROVEMENT & CONSOLIDATION, DEWATERING



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UNDERGROUND CONDITIONS
SAFETY, ENVIRONMENT AND COSTS

Safety

Biggest problem in tunnels is uncontrolled water ingress

- Immediate flooding
- Ground collapse with water ingress
- Reduce risk of accidents / Improve working safety

Environment

- Contaminated water has to be managed accordingly
- Lowering of groundwater has to be prevented in most areas (Avoid environmental misshape)
- Limited pumping capacities

Costs

- Pumping costs
- Reduced development speed due to water
- Complete halt of operation due to water

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UNDERGROUND CONDITIONS
CHALLENGES

Problems with geology:

a) Unstable rocks
- Rock burst / failure (= collapse)
- Convergence (= deformation at the section)
- Surface subsidence (= deformation at surface)

b) Cavities, overbreaks

c) Water ingress

Appropriate action:

a) Stabilization by
- Rock bolting
- Shotcrete
- Injection

b) Backfilling, cavity filling

c) Waterproofing by
- Injections
- Shotcrete in combination with a drainage system

Injection works mostly done due to SAFETY, ENVIRONMENT & COST (reduction of downtime)

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UNDERGROUND CONDITIONS

TYPICAL CHALLENGE IN TUNNELING: WATER INGRESS



Case study: tunnel access road for a HPP project:

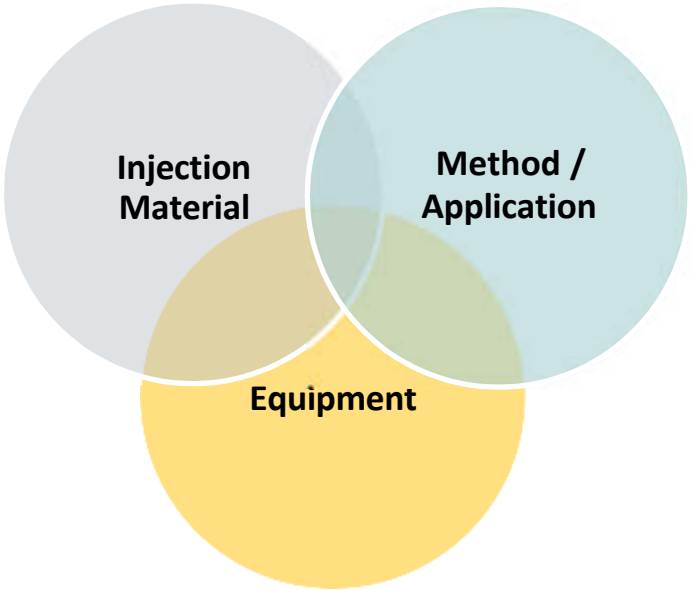
- The injection process has been implemented onto both faces of this geological accident.
- The length of this geological accident to be "fixed" was about 120 m.
- It took about 6 weeks to solve the issue.
- 250 tons on injection resins were injected (several supplier)

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KEY SUCCESS FACTORS
FOR INJECTION IN TUNNELING & MINING



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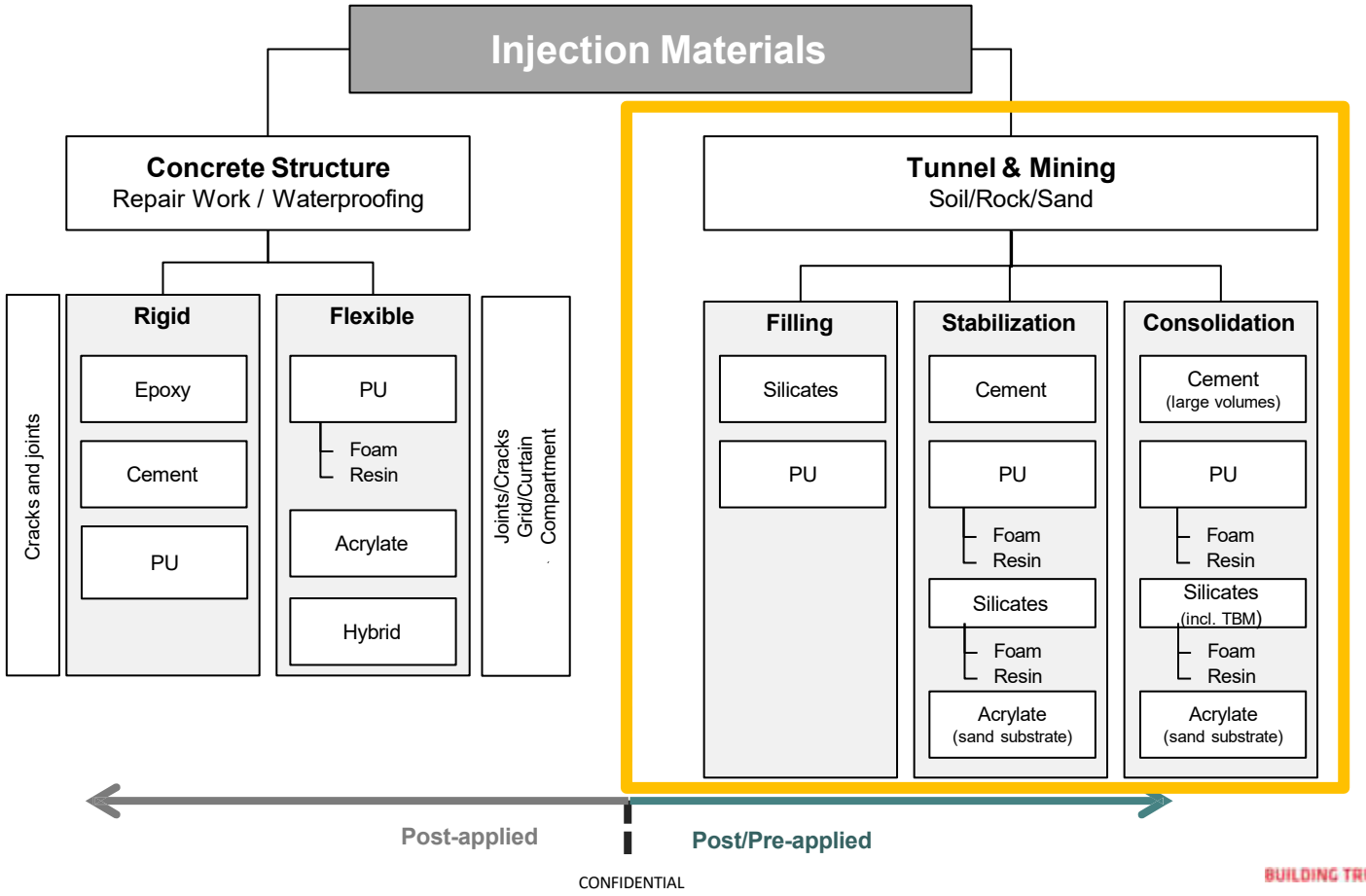
INJECTION MATERIAL

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INJECTION MATERIAL

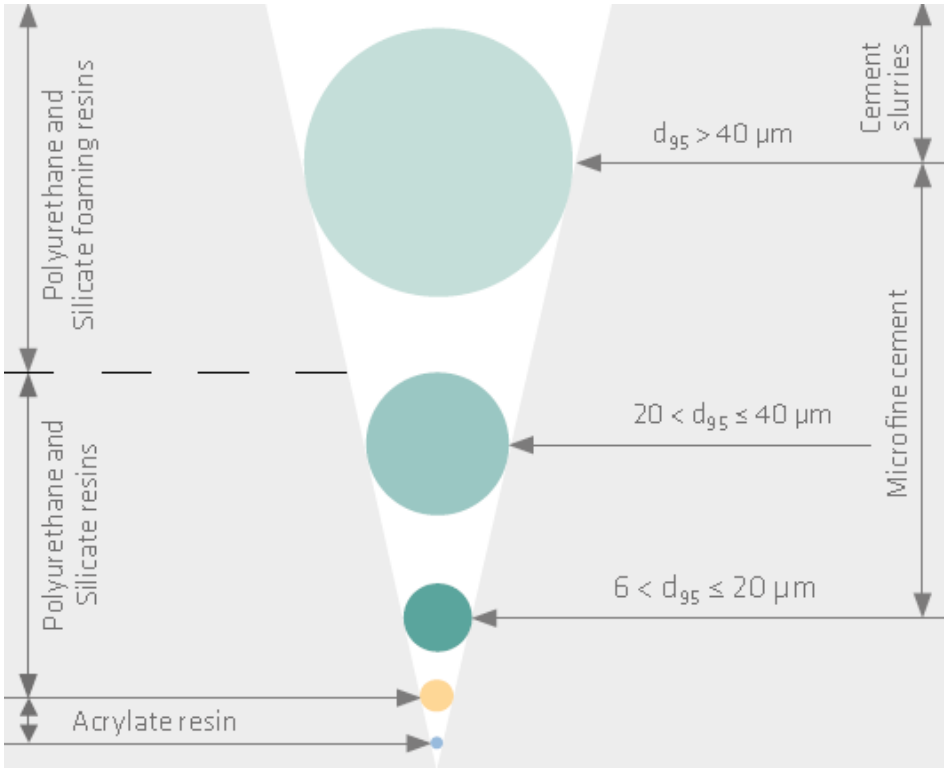
TECHNOLOGY AND USE



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INJECTION MATERIAL
WHY VARIOUS TECHNOLOGIES ARE NEEDED



- Cements are the most economical material but can not be used for all applications
- The smaller the particles, the better the penetration into difficult grounds such as sandy substrates, fault zones
- If fast reaction required (e.g water pressure or to reduce downtimes), polymer based injection resins are used

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INJECTION MATERIAL
WHY VARIOUS TECHNOLOGIES ARE NEEDED

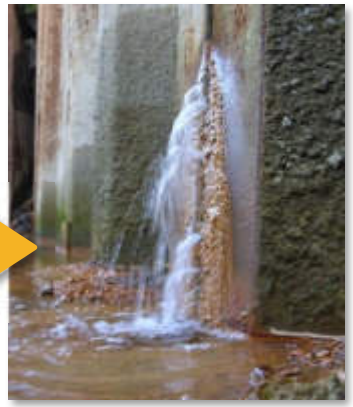
PU FOAM

- Sika® Injection-101 RC → **SikaInject®-102**
- MasterRoc® MP 355 1K DW → **SikaInject®-155 DW**
- Sika® Injection-107 → **SikaInject®-107 DE**

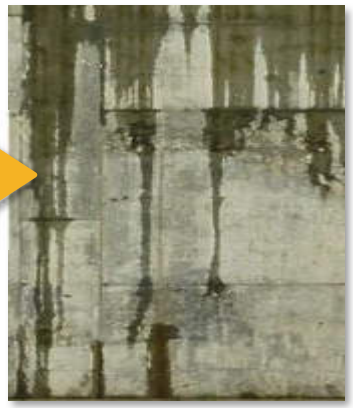
PU RESIN

- Sika® Injection-201 CE → **SikaInject®-201 DE**
- SikaFix®-210 → **SikaInject®-210 DE**
- MasterRoc® MP 355 → **SikaInject®-255**
- Sika® Injection-216 → **SikaInject®-216 DE**

For PRESSURIZED WATER



For WET CRACKS



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INJECTION MATERIAL
MICROFINE CEMENTS

MICRO CEMENT

- Sika® Microcement xxx → **SikaInject®-8xx**

- Various Blaine surface avalaible
- > 6000 cm2/gr - > 16,000 cm2/gr



INJECTION MATERIAL
POLYURETHANE RESINS

SikaInject®-210 DE (fomerly SikaFix®-210)

- Fast reacting (40 sec) resin with *high final strength* (80 N/mm2)
- Foams in contact with water up to 3 times.
- Accelerator **SikaInject® AC-21** (<10 sec)
- In combination with SikaInject® AC-21, SikaInject®-210 DE is reacting as a foaming waterstop system.

Use:

- to stop high water ingress underground



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INJECTION MATERIAL
WHY VARIOUS TECHNOLOGIES

ACRYLATE RESIN (gels)

- Sika® Injection-304 → **SikaInject®-304 DE**
- Sika® Injection-304 PS → **SikaInject®-315 PS**
- Sika® Injection-307 → **SikaInject®-307**
- SikaFix®-301 → **SikaInject®-301 DE**
- Masterseal® 901 → **SikaInject®-901**

SILICATE FOAM

- SikaFix®-501 → **SikaInject®-501 DE**
- MasterRoc® MP 367 Foam → **SikaInject®-567**

SILICATE RESIN

- SikaFix®-601 → **SikaInject®-668**
- SikaFix®-601 THIXO 2 → **SikaInject®-601 TX**
- MasterRoc® RBA 380 → **SikaInject®-680**



SikaInject® -315 PS

SikaInject® -301 DE



For CONSOLIDATION OF SOIL WITH LOW PERMEABILITY



For FILLING BIG CAVITIES, STABILIZING LOOSE ROCKS



UMBRELLAS, STABILIZING, UNDER WATER APPLICATIONS

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276

INJECTION MATERIAL
SILICATE FOAMS

SikaInject®-501 DE (fomerly SikaFix®-501)

- Fast foaming (30-fold) resin with short reaction times (15 sec.).
- Reacting with itself, in case of water no change of technical properties
- *Low reaction temperature (important for mining ~100°C)*
- *Cured foam is cuttable*
- *Cheaper but more mixing energy required compared to PU*

Use

- to stabilize disintegrated rocks with high cavity content
- filling of big cavities
- TBM Heading consolidation



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277

INJECTION MATERIAL
SILICATE RESINS

SikaInject®-668 (fomerly SikaFix®-601)

- Non-foaming, non-flammable resin with short reaction times (40 sec.)
- *Very good adhesion even on damp surfaces*
- Reacting with itself, in case of water no change of technical properties
- *Low reaction temperature (important for mining ~100°C)*

Use

- Ground consolidation
- Bolting



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278

INJECTION MATERIAL

ACRYLATE

SikaInject®-301 DE (fomerly SikaFix®-301)

- a quick-reacting resin with very low viscosity (5 Mpa*s) and short, adjustable reaction times (2-15 min).
- Slight swelling factor up to 10%
- Cleaning with water



Use

- to seal layers with low infiltration rates
- for consolidation soil with low permeability
- to grout cracks, fissures and voids of well below 0,2 mm

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279

INJECTION MATERIAL

ACRYLATE

SikaInject®-306 DE (fomerly Sika® Injection-306)



- **Elastic polyacrylic injection resin used for permanent watertight sealing**
- Very low viscosity, elastic polyacrylic injection resin with a versatile and adjustable reaction time.
- Suitable for use in hot and tropical climatic conditions.
- Permanently elastic, can absorb limited movements
- Capable of reversibly absorbing (swelling) and releasing (shrinking) moisture
- Cured Sika® Injection-306 is insoluble in water and hydrocarbons and resistant to acids and alkalis

Use

- Sika® Injection-306 can be used to seal water-bearing cracks and voids.
- For consolidation soil with low permeability.
- To grout cracks, fissures and voids of well below 0,2 mm

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INJECTION MATERIAL

ACRYLATE

SikaInject®-310 US (fomerly Sika® Injection-310 US)

- Polyacrylic, 1-component, powder based injection resin which is formulated to contain all the active parts in one powder.
- The all in one, ready to use powder, only requires mixing with water.
- After the addition of water the chemical reaction is activated producing a very low viscosity resin which cures to form a tough-elastic gel.



Use

- to seal layers with low infiltration rates
- for consolidation soil with low permeability
- to grout cracks, fissures and voids of well below 0,2 mm

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INJECTION MATERIAL

SELECTION OF INJECTION MATERIAL

Injection Material	Clay	Silt	Sand: fine	Sand: medium	Sand: coarse	Gravel	Pre-injections	Water stopping	Ground consolidation
Cement					days		X		X
Micro-cement					hours		X		X
SikaInject®-210 DE				10 seconds				X	
SikaInject®-501 DE				15 seconds			X	X	X
SikaInject®-688				40 seconds			X		X
SikaInject®-3xx				2 – 15 minutes			X	X	X
Grain size (in mm)	0.001	0.01	0.1	1	1.5	10			
Permeability K (m/s)		10^{-7}	10^{-6}	10^{-5}	10^{-4}	10^{-3}			

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EQUIPMENT

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283

PUMPS FOR PU/SILICATE
2-COMPONENT

- This type is an air-driven, 2-comp. injection pump with a fixed mixing ratio (1:1 parts by volume).



Physical properties	
Injection pressure	up to 240 bar
Pump capacity	approx. 13 l/min
Air requirement	6 m^3/min
Weight	approx. 118 kg
Suitable for	PU / SIL

Complete w. special mixing head, hoses. etc

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EQUIPMENT AND ACCESSORIES

PUMPS FOR PU/SILICATE 2-COMPONENT



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285

EQUIPMENT AND ACCESSORIES
INFLATABLE PACKERS

Inflating Packers	
Material	Plastic core , alternative steel core with rubber tube
Diameter	ø 40 mm
Maximum expansion	ø 65 mm
Length	approx. 325 mm
Max. flow rate	19 l/min
Opening pressure	approx. 30-40 bar





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EQUIPMENT AND ACCESSORIES

PACKER AND COUPLER FOR RESIN MATERIAL



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EQUIPMENT AND ACCESSORIES
WATER STOPPING USING PU RESIN



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PUMPS FOR PU/SILICATE 2-COMPONENT



- Do not change technologies before **PERFECTLY** CLEANING the pump !





APPLICATION METHODS

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290

APPLICATION METHODS

PRE- VS. POST-INJECTIONS

Pre-Injection

- Mostly cementitious injection

Benefits of pre-injection

- Reduce risk of uncontrolled water inrushes
- Improved stability through poor ground «Dry» working conditions in the tunnel



Post-Injection

- Dominated by the chemical injection
- Almost all tunnels require some form of post-injection works either with micro-cements, PU's or acrylate for leak sealing, repair or ground stabilization



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APPLICATION METHOD: PRE-INJECTION
DRILLING PATTERN



Typical lengths of drilling in hard rock:

- Exploratory holes:
  25-30 m
- Cement-grouting holes in Rock: **18-24 m**
- Chemical Injection against hydrostatic water pressure**: 2-9 m**

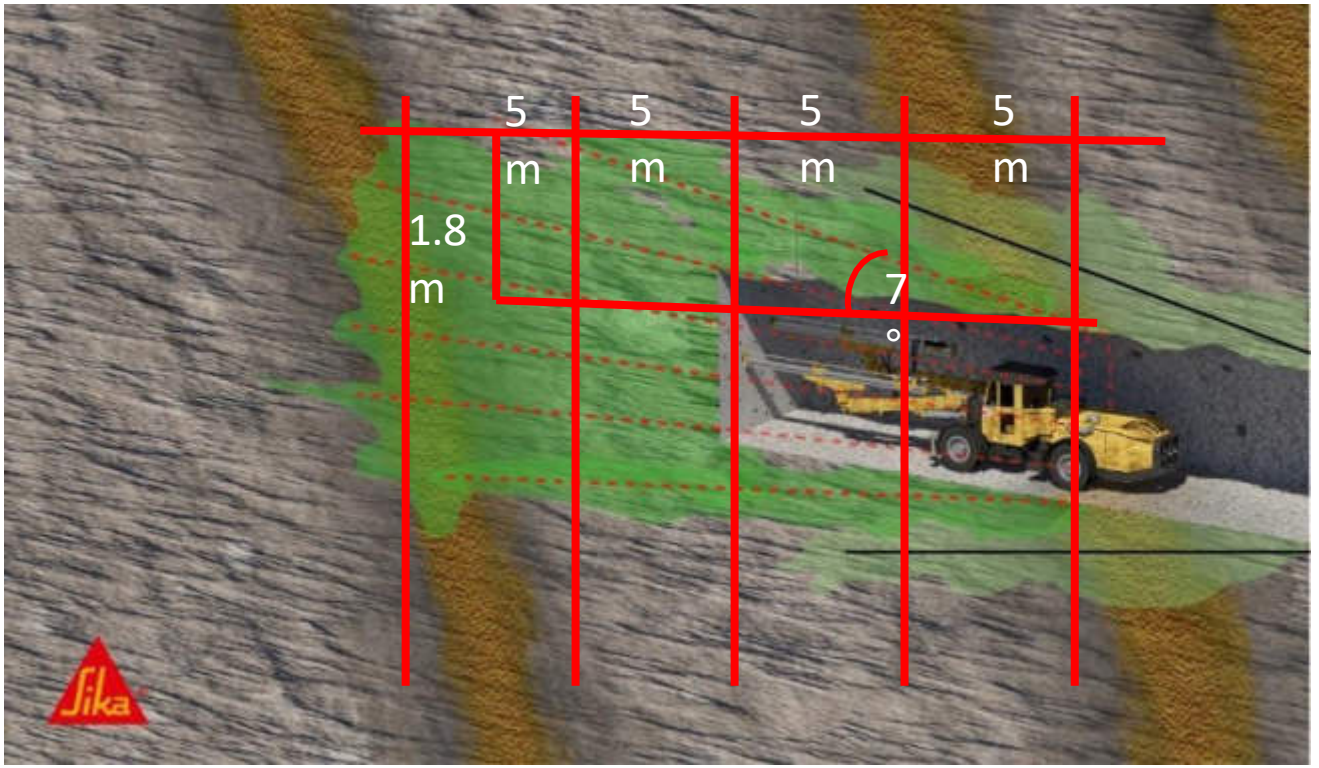
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APPLICATION METHOD: PRE-INJECTION

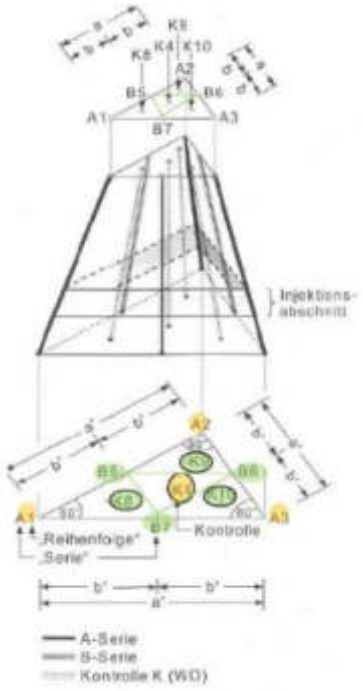
STEP 3



APPLICATION METHOD: PRE-INJECTION

FACE STABILIZATION

Borhole distance and check (basic concept)



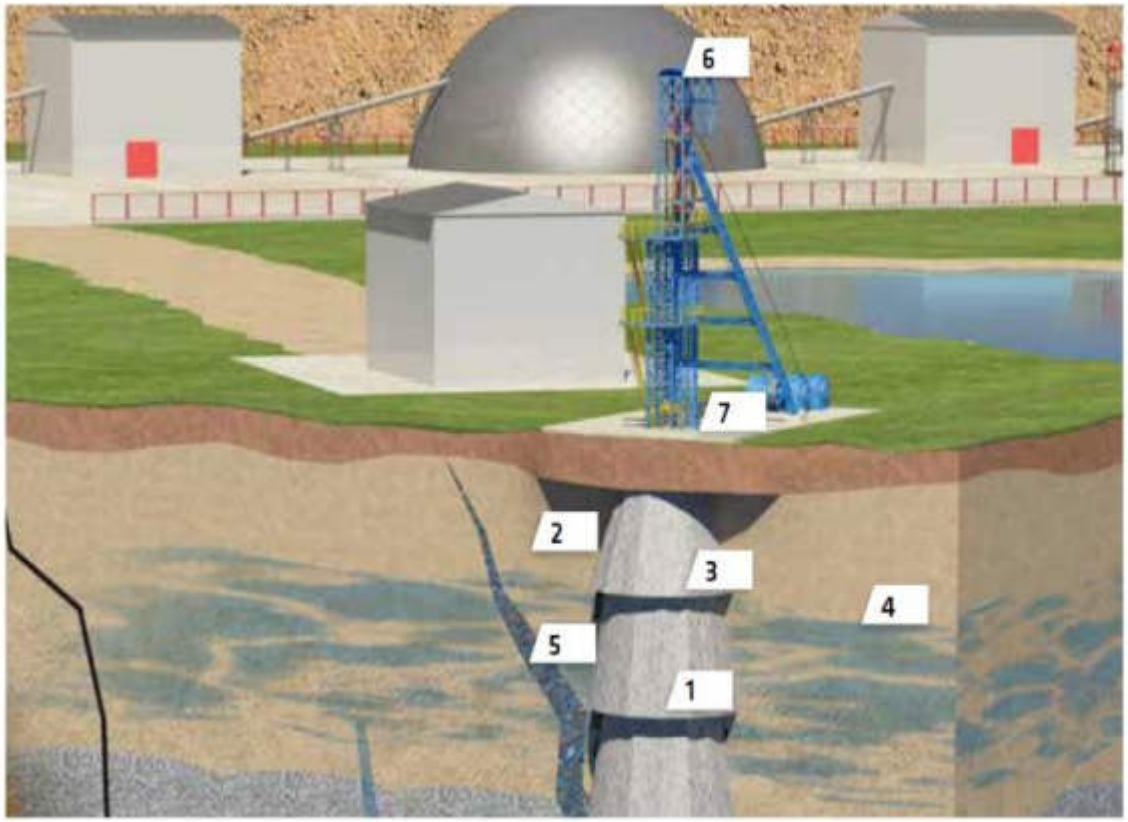
- Testing injection area: 2-3 m of a tunnel
- Injectionborehole spacing:
  - a’ = 1-3 m
- A1 – A2 – A3 primery injection boreholes
- K4 Controll borehole after injection

- If Controll shows not good results of injection material penetration start with secondary injection
- B5-B6-B7 – secondary injection boreholes
- K8-K9-K10 Conroll boreholes

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EXAMPLE PRE-INJECTION
SHAFTS



1
Umbrella like injection prior to shaft sinking

2
Quaternary cover above hard rock

3
Pre-injected with SikaFix®-800 series

4
Ground water horizon

5
Fault zone

6
Head frame

7
Shaft collar

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APPLICATION METHOD: POST-INJECTION

SIKAFIX® INJECTION RESINS

High-performance, 2 comp. polyurethane, silicate and 3 comp. acrylate resins for tunneling and mining

- Control water ingress underground
- Control unconsolidated geology
- Dealing with high water pressures
- Control overblasting
- Waterproofing rock-filled dams
- Waterproofing structures with low permeability
- Fill cavities



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APPLICATION METHOD: POST-INJECTION
EXAMPLE SIKAFIX®-210 WITH SIKAFIX® AC-21

- To counter water ingress under high water pressure, SikaFix®-210 in combination with SikaFix® AC-21 accelerator has been used
- Reaction time below 10 sec. when accelerated (for heavy water ingress)
- Foaming only triggered in presence of water



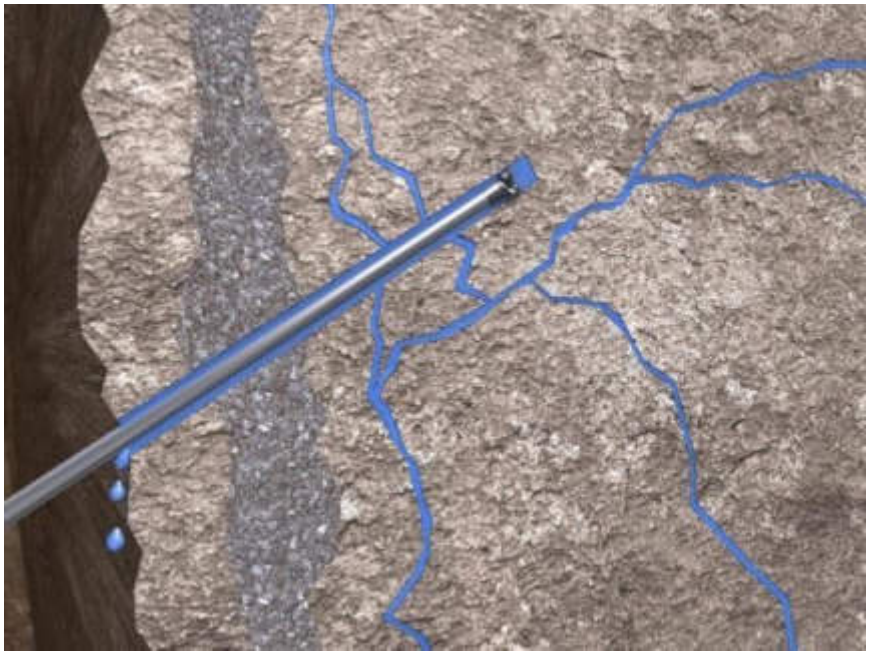
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APPLICATION METHOD: POST-INJECTION
DO'S AND DON'TS

- Drill the hole



- Place the injection lance incl. packer



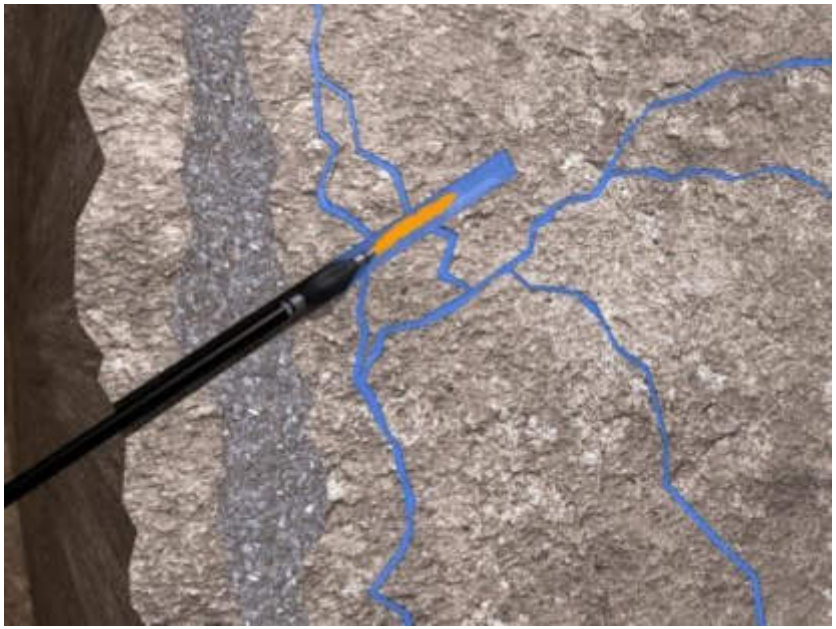
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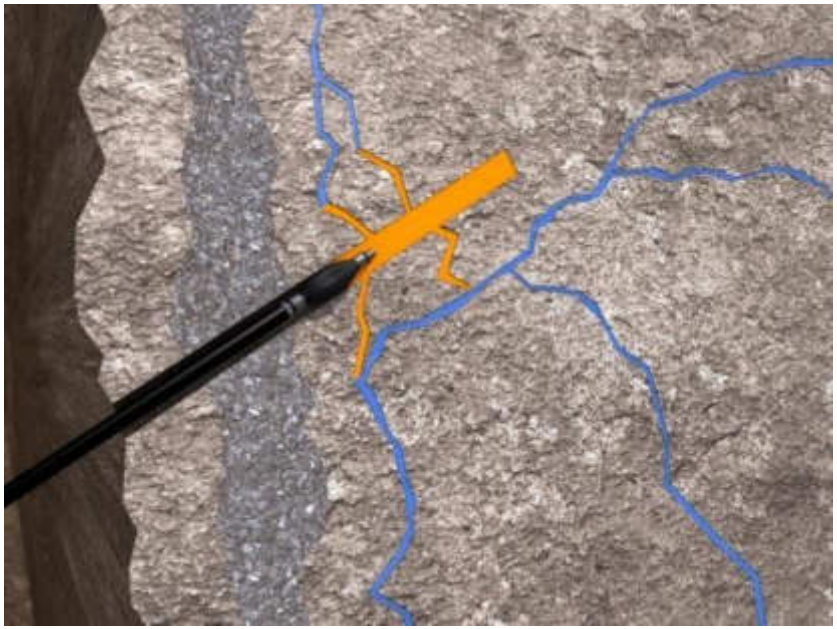
298

APPLICATION METHOD: POST-INJECTION
DOS AND DON'TS

- Resin blows up the packer and enters the fault zone



- Resin starts to foam and blocks water



APPLICATION METHOD: POST-INJECTION
DOS AND DON'TS

- Wrong: Packer placed in the faulty zone. Packer not thight and no pressure can be build-up



- Wrong: packer placed at the end of the borehole.

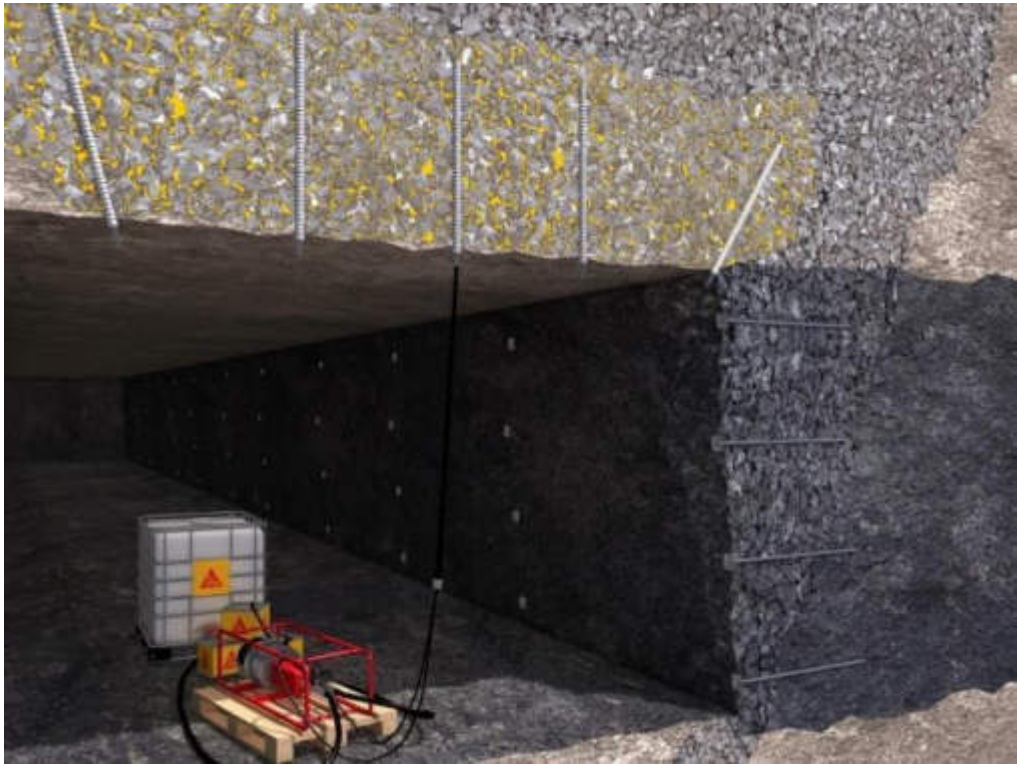


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APPLICATION METHOD: POST-INJECTION
CONSOLIDATION USING SIKAFIX®-501/601



- In a coal mine, only silicate products shall be used due to lower exothermic reaction

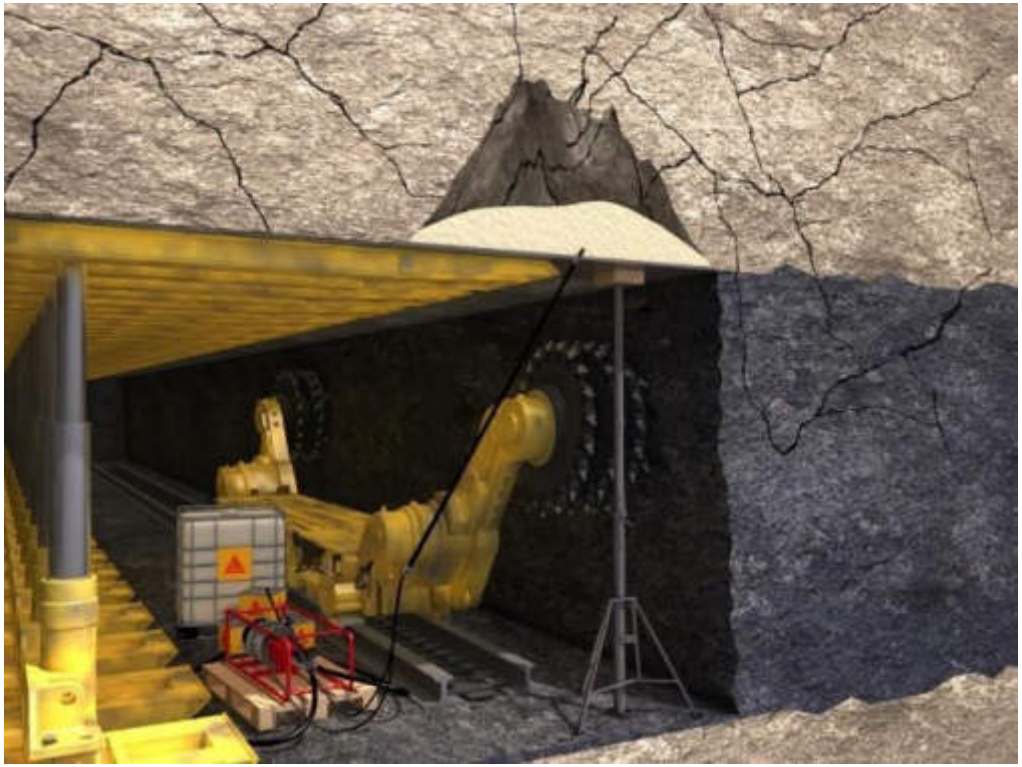
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APPLICATION METHOD: POST-INJECTION

CAVITY FILLING USING SIKAFIX®-501 / COAL MINE



- Step 1: filling of cavities with silicate foam above coal vein
- Step 2: drilling through injected area and stabilize ground with silicate resin

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APPLICATION METHOD: POST-INJECTION

CAVITY FILLING USING SIKAFIX®-501



- Example of cavity filling with silicate foam SikaFix®-501

APPLICATION METHOD: POST-INJECTION

CONSOLIDATING ROCK-FILLED DAMS USING SIKAFIX®-301

- SikaFix®-301 has adjustable reaction times of 2-15 min.
- Water-like rheology and excellent penetration capabilities in soils and sands with low permeability.
- Right choice to waterproof rockfilled dams such as tailings dams and decant ponds.
- Slight swelling factor of 10%.



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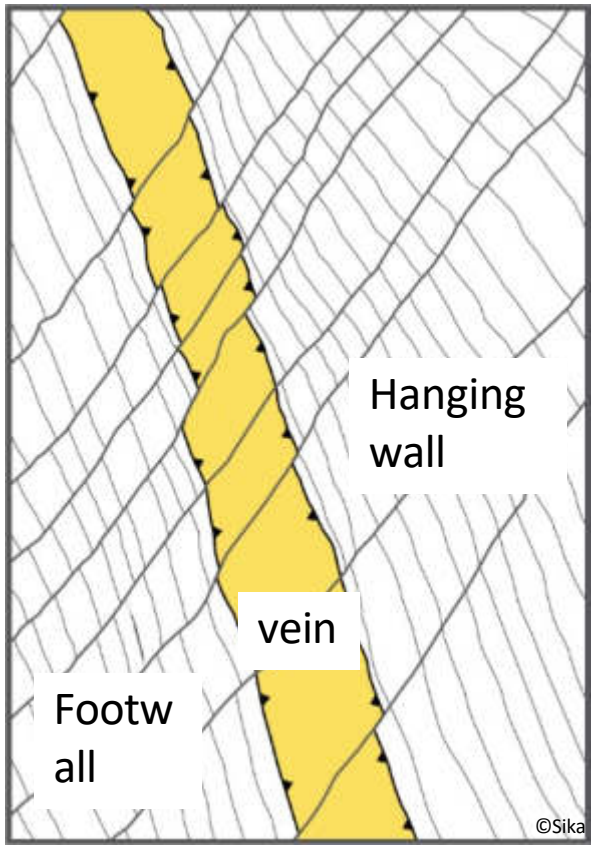
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APPLICATION METHOD: POST (PRE) -INJECTION

CONTROLLING OVERBREAK WITH SIKAFIX®-601

- Controlling external dilution in narrow, high grade vein deposits is among the largest driver for the profitability of the operation
- Overblasting caused an increase of ore dilution and a decrease of the head grade that goes to the mill



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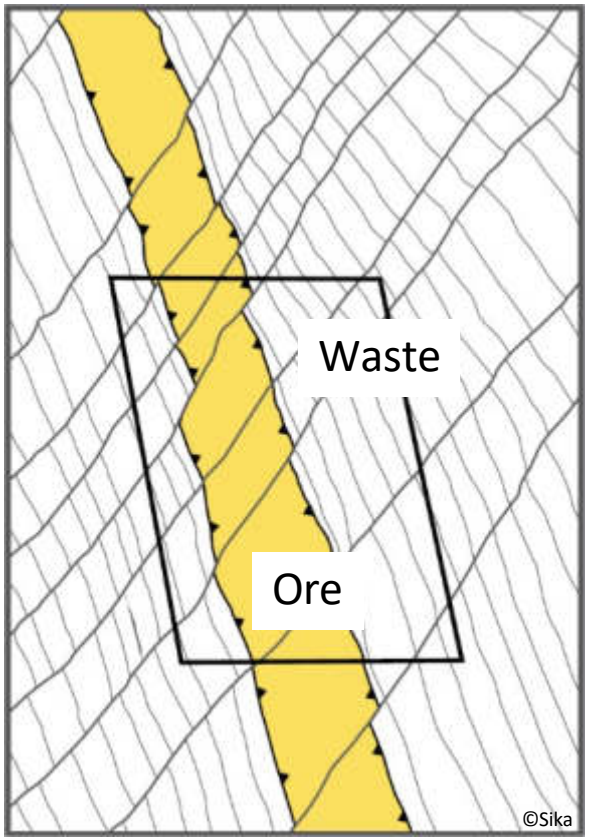
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APPLICATION METHOD: POST (PRE) -INJECTION

CONTROLLING OVERBREAK WITH SIKAFIX®-601

- Ore to waste ratio is key for maintaining planned mill-feed and to stay within the mine-plan
- Size and shape of the stope is strongly effecting ore to waste ratio



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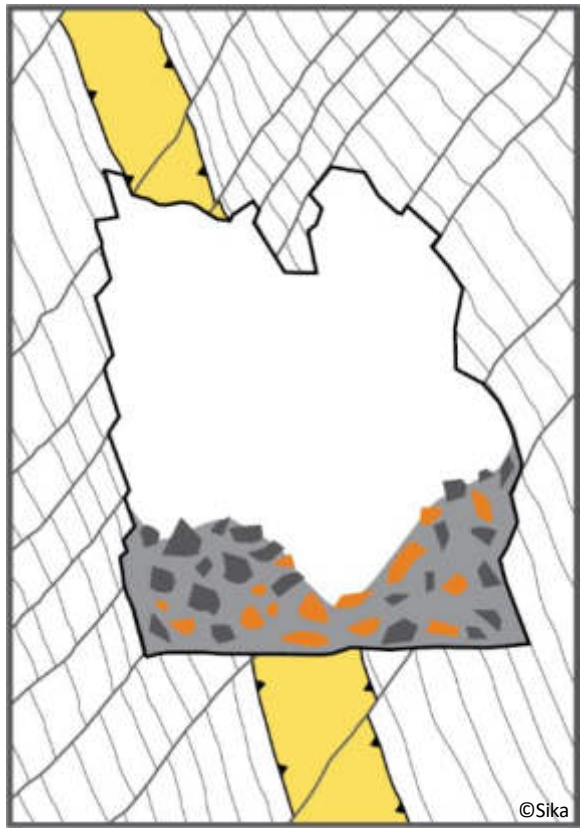
APPLICATION METHOD: POST (PRE) -INJECTION

CONTROLLING OVERBREAK WITH SIKAFIX®-601

Without stabilization:

Heavy overbreak causes a range of issues including

- increased ore dilution
- reduced stability of the excavation
- excessive need for fortification
- impact on adjacent stopes
- Increased cycle times



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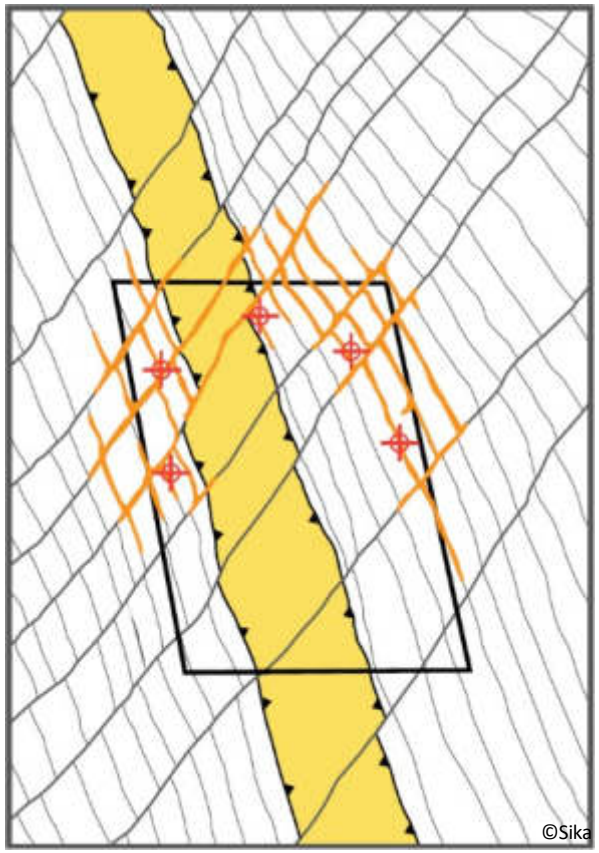
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APPLICATION METHOD: POST (PRE) -INJECTION

CONTROLLING OVERBREAK WITH SIKAFIX®-601

The injected resin SikaFix®-601 will migrate through cracks, faults and non-consolidated ground and stabilise the rock-mass efficiently

As curing of the resin is fast, the time used for the injection process is limited.



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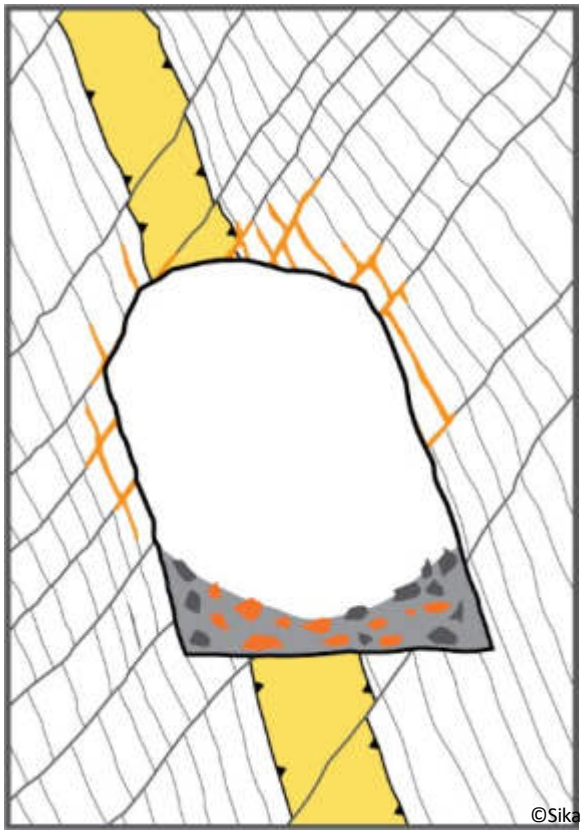
APPLICATION METHOD: POST (PRE) -INJECTION

CONTROLLING OVERBREAK WITH SIKAFIX®-601

When injection is done properly, overbreak will be limited to the injected perimeter after blasting the tunnel face.

Leaving behind a stable stope outline

Potential to reduce the support measures such as bolts and mesh.



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APPLICATION METHOD: POST-INJECTION

CONTROLLING OVERBREAK WITH SIKAFIX®-601



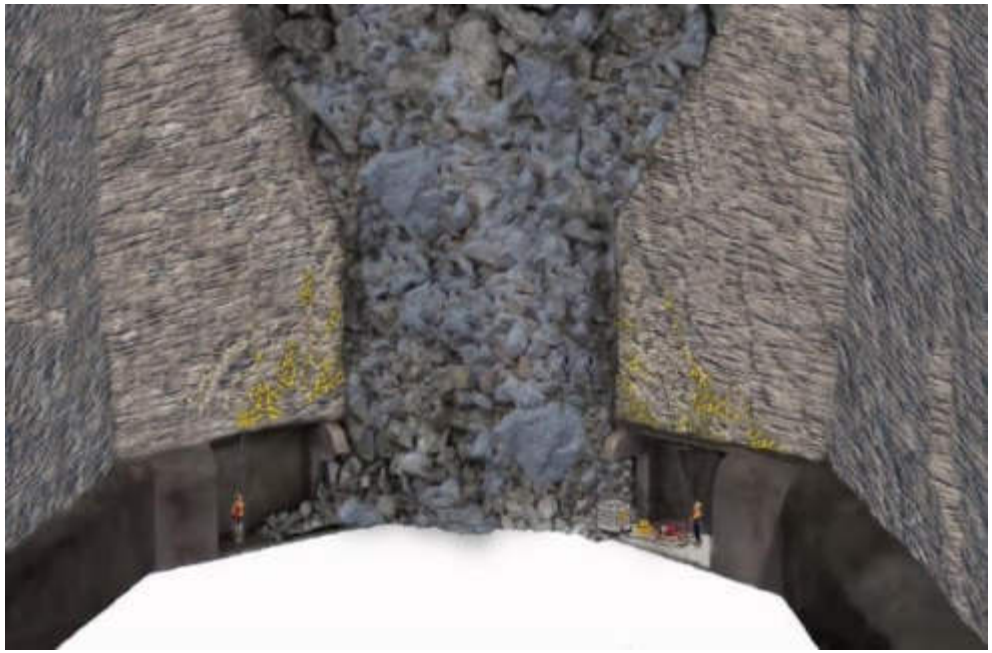
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GROUND CONSOLIDAITON

DRAW PANEL STABILIZATION





ADVANTAGES OF USING SIKA 2K RESINS

- Easy to apply
- Highly flexible, small equipment
- Air-driven, no electricity needed
- Powerfull immediate effect
- No expensive pre-investments
- No expensive, easy to source hardware (packers, lances, Y-pieces etc)
- Easy one to one chemical components
- Can significantly reduce expensive re-habilitation works
- Well-trained Sika teams that can be utilised for application trainings

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ROCK CONSOLIDATION, STABILIZATION & WATERSTOPPING
SUMMARY

Key success factors are:

- selection of appropriate injection material depending on ground conditions
- Related application procedure
- Right equipment and accessories
- The injection Team!!

Resins and Foams are used in presence of water

A specialist (geologist, geotechnical engineer) shall be consulted to define water fractures and indicate drilling pattern

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BACK TO PRESENTATION

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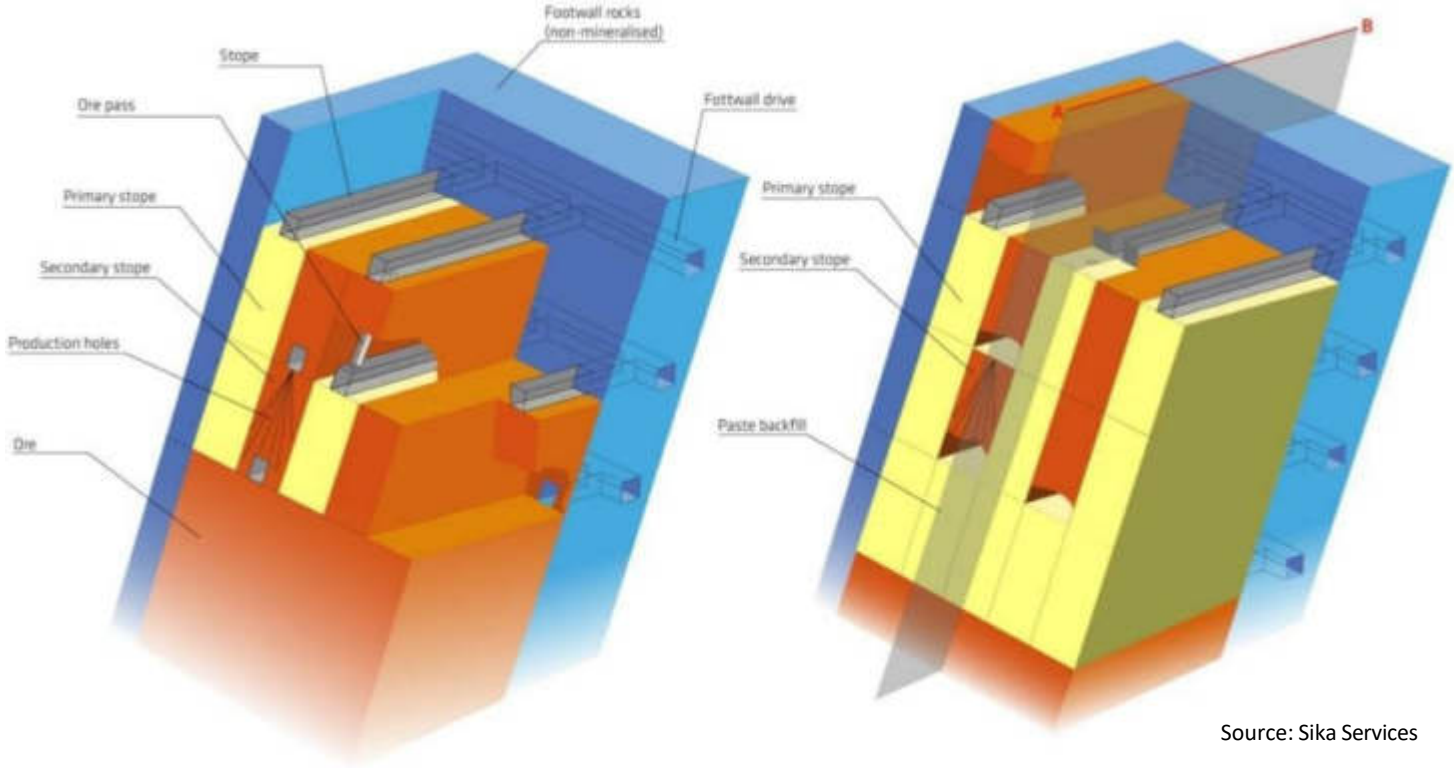


SIKA MINE PASTE FILL SOLUTIONS

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MINE PASTE BACKFILLING

EXAMPLE: LONG-HOLE MINING AND FILL

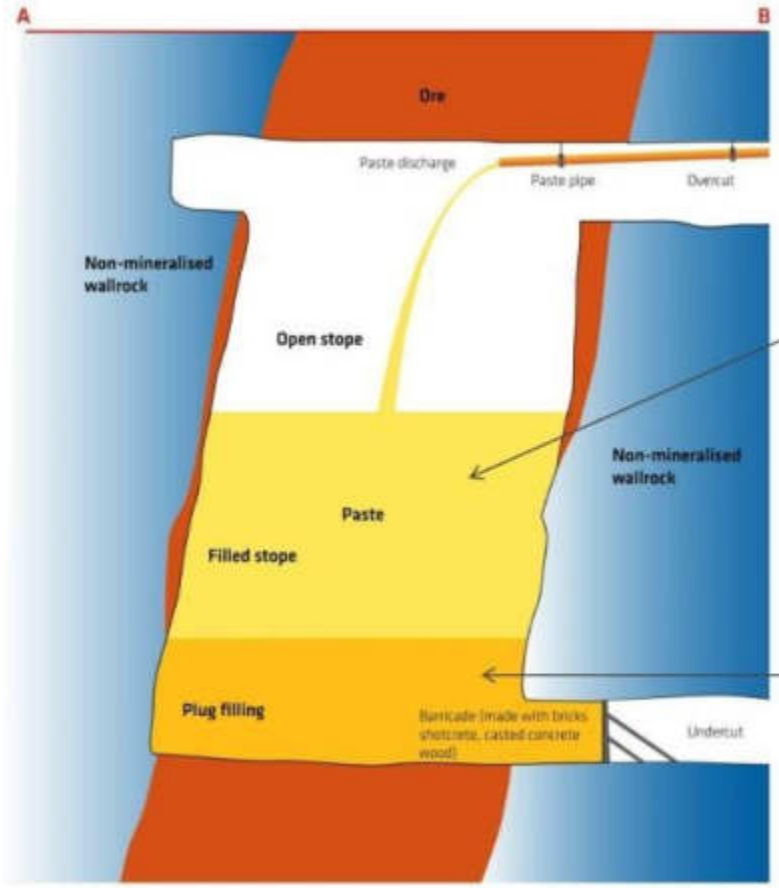


Source: Sika Services

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HOW DOES IT WORK



Lower cement content (2-5%) for secondary pours

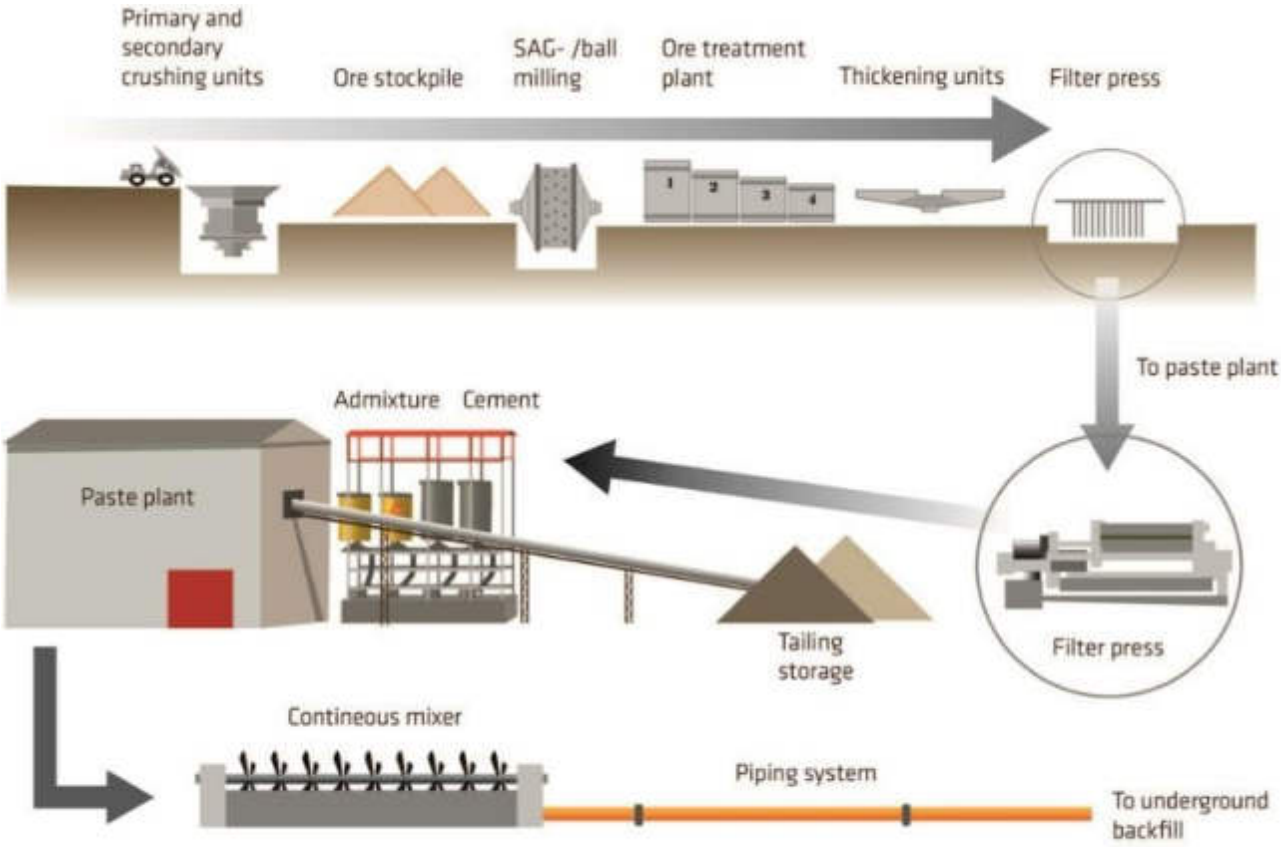
High cement content (up to 10%) for primary plug filling

Source: Sika Services

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BACKFILLING - FLOWSHEET



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RULES OF THUMB FOR MINE PASTE BACKFILL

- Mining Back fill is used, if a) ore body is rich in minerals or b) regulations require it.
- 10-20% of OPEX of a cut and fill operation are related to the backfilling infrastructure and maintenance
- Cement responsible for 75% of the total costs in cemented paste backfill operations.
- Typical binder content ranges between 3-8% by weight of m3 paste material.
- Typical material costs of backfill in mines range between US$ 2-US$ 20 per cubic meter
- For active ground support, the backfill needs at least 5 MPa compressive strength
- Free standing fill applications require usually less than 1 Mpa of compressive strength
- Typical W/C-ratio ranges between 3 - 8
- Slump usually between 10-25 cm
- Solid content ranges in the order of 70-85%
- Material density of paste in the order of 2600 kg/m3
- Specific density of paste around 1900-2100 kg/m3
- Fines content in pastes should be at least 15% by weight (<45 microns in size)
- Pipe diameter for backfilling is usually 100mm and 200mm

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WHY USING BACKFILLING ADMIXTURES?

- Improve cost performance of the paste fill mix by reducing the binder content (cement, slag, fly-ash)
- To improve the rheological properties of the cemented paste fill during the contineous mixing, pumping and paste fill process
- Enhance workability of the paste
- Reducing pumping pressure for the paste transport
- Reduce wear and tear of machinery, equipment and paste lines
- To reduce segregation and bleeding of the placed paste fill
- Reduce risk of paste line blockages
- To improve in-situ strengths at both, early and final stage
- Improve curing time
- Increase solid content of the paste mix
- To change cement types (to cheaper versions)

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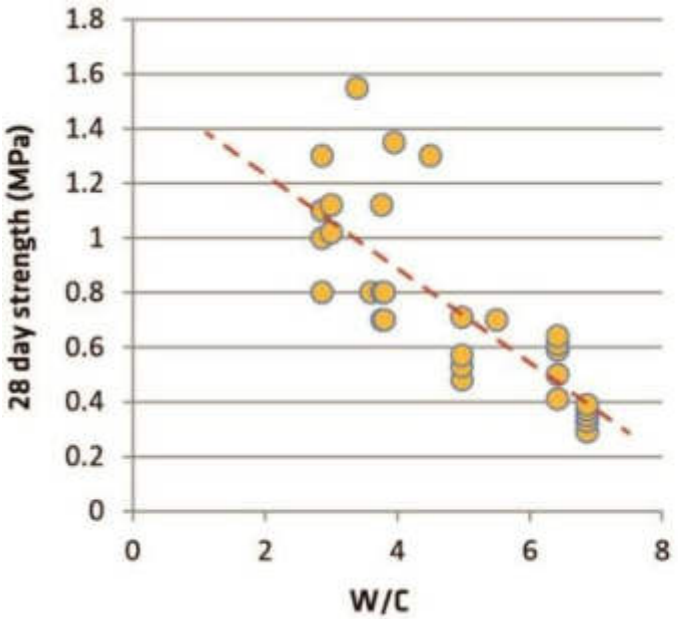
USING SIKA CONCRETE TECHNOLOGY FOR MINE PASTES

Per m3	Concrete (CVC)	Paste Fill
Cement	400 kg	80-200 kg
Water	200 kg	100-400 kg
Water type	Mainly fresh	Mainly process water
W/B ration	0.4-0.5	1-5
Aggregates	Sand/Gravel Ø ≤ 38 mm 1,750 kg	Tailing: Ø ≤ 50-300 µm 1,900 kg
Density of harden mix	2,200-2,400 kg/m3	2,500-2,700 kg/m3
Strength	20 to 60 MPa	0.3-5 MPa
Additive in use	Yes (fly ash, Slag)	Yes (fly ash, Slag)
Admixture	HRWR (type G)	HRWR (type G)
Dosage	0.5-2.0 %	0.5-2.0 %
Testing	Slump, Cubes	Slump (small cone), Cylinder, Bars

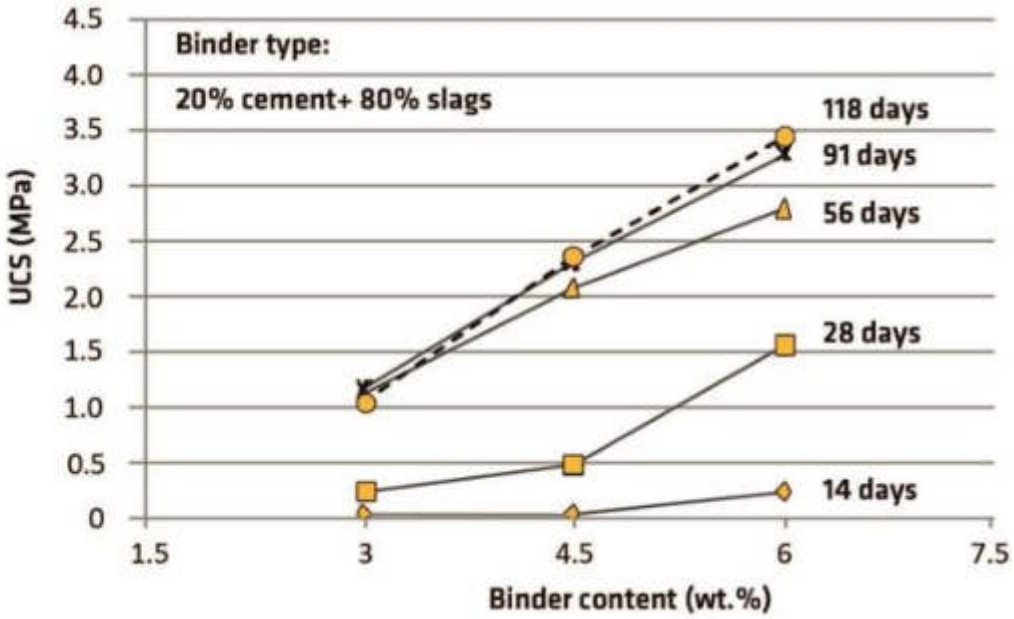
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BINDER CONTENT VS FINAL COMPRESSIVE STRENGTH



Source: Sika Services



Source: Belem T., Benzaazoua M. 2004

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PRODUCTION OF BACKFILL PASTE

TAILINGS DIFFERENCES

Tailings can look completely different and have different properties



→ Different water requirements

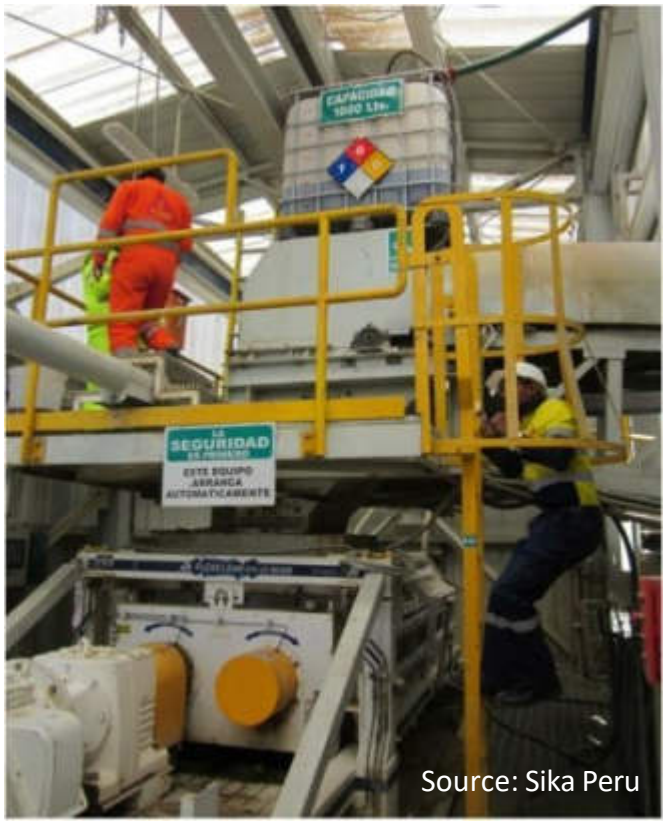
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DOSING OF ADMIXTURES

DIRECTLY INTO WETTED-OUT PASTE MIX – AT MODERATE DOSAGES



Source: Sika Peru



Source: Sika Services

BACKFILL MIX OPTIMIZATION



- By the use of Sika® Stabilizer MBF a reduction of 20-50% of cement is possible by maintaining the 28 days compressive strength.
- By optimising the mix, bleeding can be reduced significantly.

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SIKA ADMIXTURES FOR BACKFILL PASTES

Case Study VMS deposit in Europe:
1740g tailings - 130g Water - 120g Binder

No Admixture



1.5% Sika®Stabilizer-302 MBF (bwoc)



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SIKA ADMIXTURES FOR BACKFILL PASTES

Case Study, Epithermal Deposit, Latin America:
1839g tailings - 140g binder (CEM 42.5) - 365g water

No Admixture	3 % Sika®Stabilizer-305 MBF (bwoc)



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SIKA ADMIXTURES FOR BACKFILL PASTES

Case Study, Orogenic Gold Deposit, West Africa:
1153g tailings - 434g aggregates - 122g binder (CEM 42.5) - 190g water

No Admixture	2.6% Sika®Stabilizer-305 MBF (bwoc))





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WHAT CAN BE EXPECTED

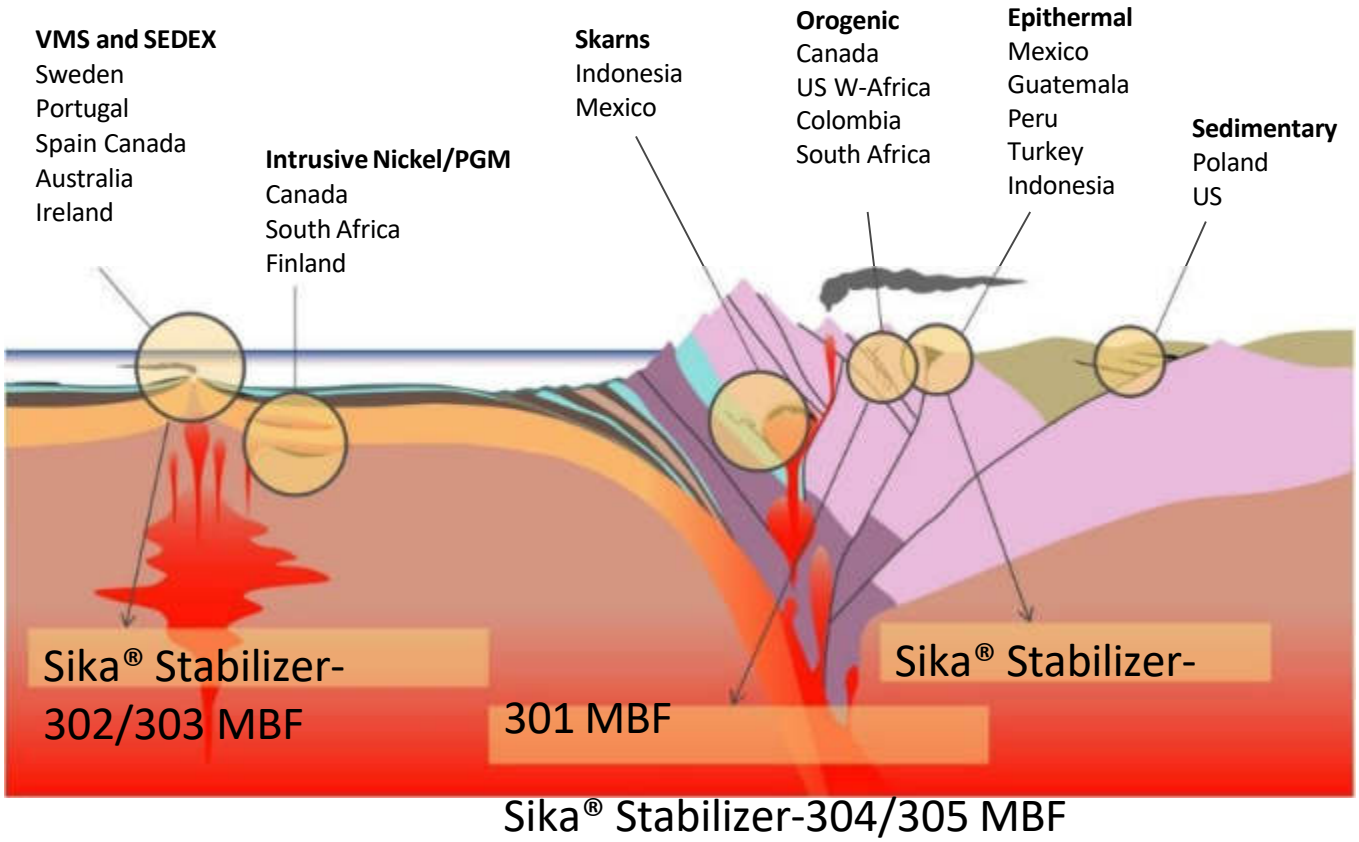
- Reduction of binder (cement) consumption by 20-50%
- A much more stable paste mix with less bleeding and less segregation
- Reduced pumping pressure and reduced wear and tear of machinery and equipment
- Higher solid content in the fill
- Possibility to improve the performance of the backfill plant that allows for higher backfill capacity and shorter mine and fill cycle time

15

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WHY DIFFERENT PASTE FILL ADMIXTURES?



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TEST PROCEDURE

- Standard paste mix is tested based on actual mix design from the mine
- Standard Hobart mixing equipment is used for the trials
- Slump is measured with a concrete or mortar cone
- Rheology is measured using a rheometer
- Sika is usually working with 7 day and 28 day compressive strengths. Compressive strength testing takes place using standard mortar bars or casted cylinders

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LAB TEST PROCEDURE



- Homogenize the paste mix containing filter dried tailings, binder and water in a standard Hobart mixer, then add admixture into the mix and homogenise again



- Use a standard concrete or mortar cone to measure the slump of the mix



- Use a calibrated transparent cylinder to measure waterbledding of paste mix over time

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LAB TEST PROCEDURE



- Use the Brookfield viscometer to measure the rheology of th paste



- Cast standard mortar bars or larger round cylinders to perform compressive strength testing after 7 days and 28 days.

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QUALITY CONTROL USING REGULAR SLUMP CONE AND CYLINDERS FOR COMPRESSIVE STRENGTH TESTING





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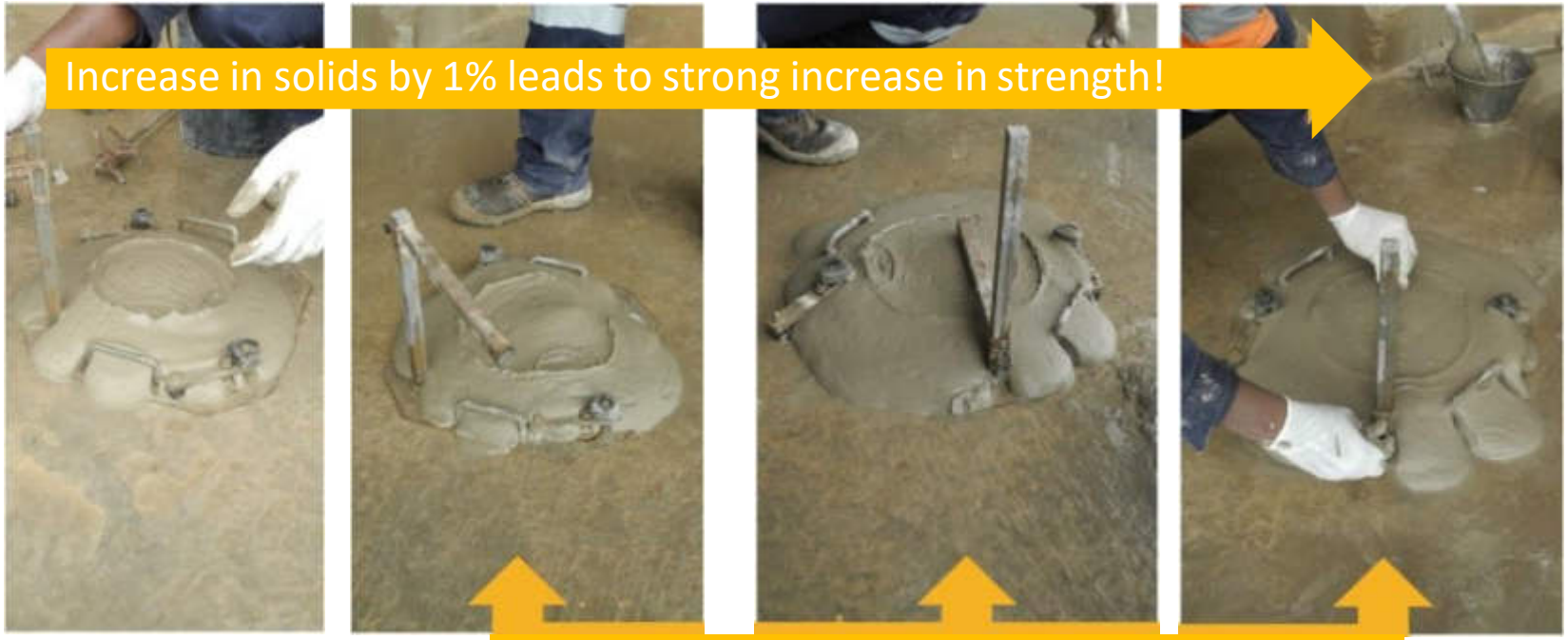
FIELD-RESULTS

0.0% Sika SK5
Solids: 78.5%
Slump: 22
UCS (28 day): 3.16 MPa

1.0% Sika SK5
Solids: 78.19%
Slump: 23-24
UCS (28 day): 2.29 MPa

2.0% Sika SK5
Solids: 78.88%
Slump: 24-25
UCS (28 day): 3.14 MPa

3.0% Sika SK5
Solids: 79.15%
Slump: 25
UCS (28 day): 4.6 MPa



Increase in solids by 1% leads to strong increase in strength!

Further water reduction possible due to high slump

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SIKA MINE PASTE BACKFILL ADMIXTURES

- **Sika® Stabilizer-301 MBF:**
  - Mainly for typical "LATAM style" epithermal, quartz-calcite hosted Ag-Zn-Pb tailings
- **Sika® Stabilizer-302 MBF:**
  - Mainly Polymetallic Zn-Pb-Cu-Ag (VMS-style- tailings)
- **Sika® Stabilizer-303 MBF:**
  - Mainly Polymetallic Zn-Pb-Cu-Ag (SEDEX-style- tailings)
- **Sika® Stabilizer-305 MBF:**
  - Mainly gold mines (orogenic-style Au tailings)
- **Sika® Stabilizer-401 MBF:**
  - Admixture to prevent high water bleeding

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BACK TO PRESENTATION

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DUST CONTROL - DUST SUPPRESSANT TECHNOLOGY

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338

DUST CONTROL

WHY DUST CONTROL?









Problems which can be caused by dust:

- Health & Safety issues
  - Dust can cause serious respiratory diseases as dust can also contain metals that are hazardous to health
  - Fine particle matter PM 10 & PM 2.5 (particulate matter) is **CRITICAL** for humans
  - reduced visibility on traffic areas (trucks, cars, planes) and endanger of operations (mines, airports, …)
- Impact on equipment
  - Dust can affect the performance of mechanical equipment and electronics (maintenance intervals, permanent damages, etc.)
- Impact on infrastructure / villages / social / fauna and flora

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DUST CONTROL

WHY DUST CONTROL: LIMITS ON EMISSIONS

European Union:

	PM10	PM2.5
Yearly average	40 µg/m^3	25 µg/m^3
Daily average (24-hour)	50 µg/m^3	None
Allowed number of exceedences per year	35	None

Australia:

	PM10	PM2.5
Yearly average	None	8 µg/m^3
Daily average (24-hour)	50 µg/m^3	25 µg/m^3
Allowed number of exceedences per year	None	None

USA:

	PM10	PM2.5
Yearly average	None	12 µg/m^3
Daily average (24-hour)	150 µg/m^3	35 µg/m^3
Allowed number of exceedences per year	1	Not applicable (3 year average of annual 98th percentile)

Japan:

	PM10	PM2.5
Yearly average	None	15 µg/m^3
Daily average (24-hour)	100 µg/m^3 200 µg/m^3 1h	35 µg/m^3
Allowed number of exceedences per year	None	None

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DUST CONTROL

MEASURES FOR DUST CONTROL



Technologies for dust control:

- Very short term
  - Water
- Medium to long terms
  - Chemical treatments
- Very long term to definitive
  - Soil cement
  - Bitumen / asphalt / concrete

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DUST CONTROL

MEASURES FOR DUST CONTROL: SHORT TERM



Characteristics:

- Depending on ambient conditions (mainly sun exposure, temperatures and wind frequency and speed), water can evaporate in a short period of time (ca. 30 min – 3h)
- Frequent repetition necessary (every day or even more than once a day)
- Generally high availability **BUT**
- Main problem: in most of the areas around the world water is a precious (and some times very expensive) resource

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DUST CONTROL

MEASURES FOR DUST CONTROL: MEDIUM TERM







Characteristics:

- Based on several chemical bases for soil treatment and stabilization
- Several action principles
- Allowing to spread on large areas
- More effective use of water

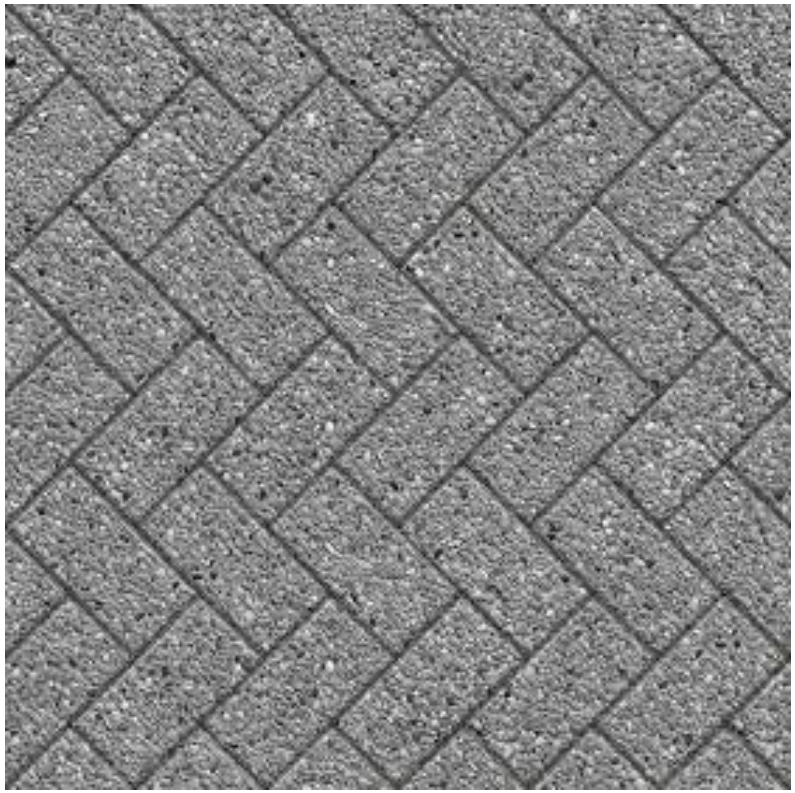
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DUST CONTROL

MEASURES FOR DUST CONTROL: LONG TERM



Characteristics:

- Based on e.g. asphalt, oils derived from coal or petroleum, concrete slabs or paving stones
- Form a coherent surface layer
- Environmental impact on plants can be critical
- Usually very expensive
- Some restriction in terms of drainage in case of rain storms

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DUST CONTROL

SIKA STABILIZER-100 (FORMERLY SIKA DUST SEAL 100)



Characteristics:

- Forms a hard, firmly bound surface
- Improvement of traction, safety and comfort for vehicle
- Decreases the rate of water penetration into treated area
- Increases run-off and reduces mud conditions
- Area may be used immediately after treatment

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DUST CONTROL

SIKA STABILIZER-100 (FORMERLY SIKA DUST SEAL 100)



Characteristics:

- Based on biodegradable polymer
- Forms layer on surface
- Highly economical
  - Dosage: 1-2 l per m²
  - Dilution: 3-10%
- Good availability (can, IBC, bulk)
- Very good cost / performance ratio
- Easy and homogeneous film coating formation using standard equipment
- Strong resistance of the film against wind erosion
- Eco assessment available
  - Not dangerous / not hazardous for environment

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DUST CONTROL

SIKA DUST SUPPRESSANT: WHAT CAN BE ACHIEVED





Results – inside the mine

- Before application

- After application
  (about the same distance)

DUST CONTROL

SIKA DUST SUPPRESSANT:

- New Application truck
  - Improved spraying area



- Dust measuring tool for permanent control







BACK TO PRESENTATION







CONCRETE REPAIR SYSTEM USING READY TO USE REPAIR MORTAR

CONCRETE REPAIR: TYPE OF DAMAGES

- **Concrete damage:**
  - Mechanical (impact, wear, overloading, vibration)
  - Chemical (Aggregate, exposure, bacterial)
  - Physical (Thermal, erosion)
  - Wrong doing when placing concrete

- **Reinforcement corrosion → Concrete damage**
  - Carbonation
  - Electrical current (Electropotential gradient varies)
  - Corrosive Contaminants (Marine)

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CONCRETE REPAIR: MANAGEMENT

- **Survey, investigations**
  → *Current conditions, history review.*
- **Diagnosis of the cause of the damage**
  → *Root causes*
- **Repair objectives**
  → *Downgrade, demolition,* ***Repair***
- **Repair products systems selection(s)**
  → *As per objectives (durability, life design, cost)*
- **Implementation, action plan, methodology**
  → *Surface preparation, equipments, application*

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CONCRETE REPAIR: PRODUCTS SYSTEM SELECTION

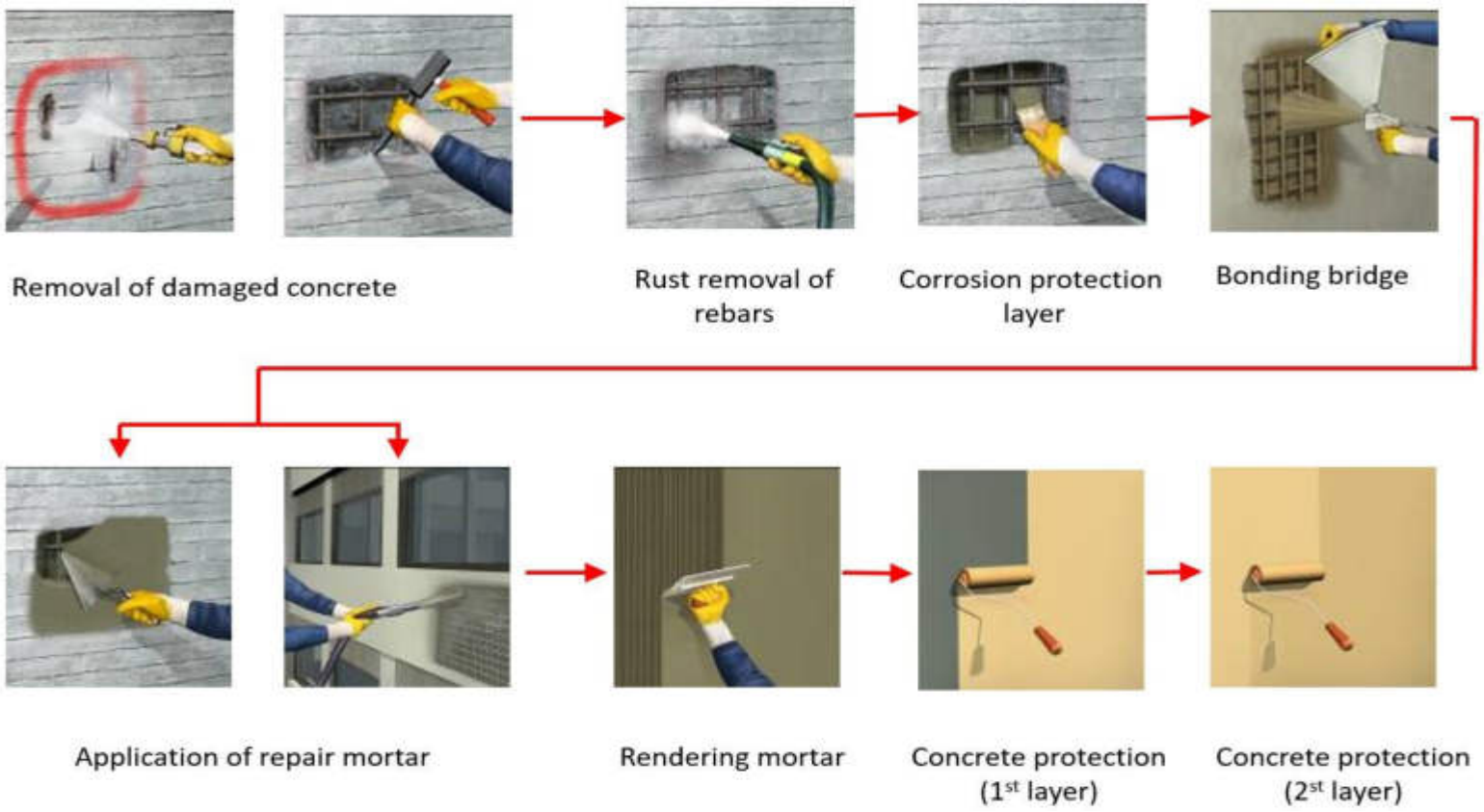
- Nature and condition of the existing structure
- Exposure, Stresses
- Durability - Maintenance
- Application duration, curing, environmental
- Appearance requirement
- Over all cost in place

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CONCRETE REPAIR: WORKING STEPS



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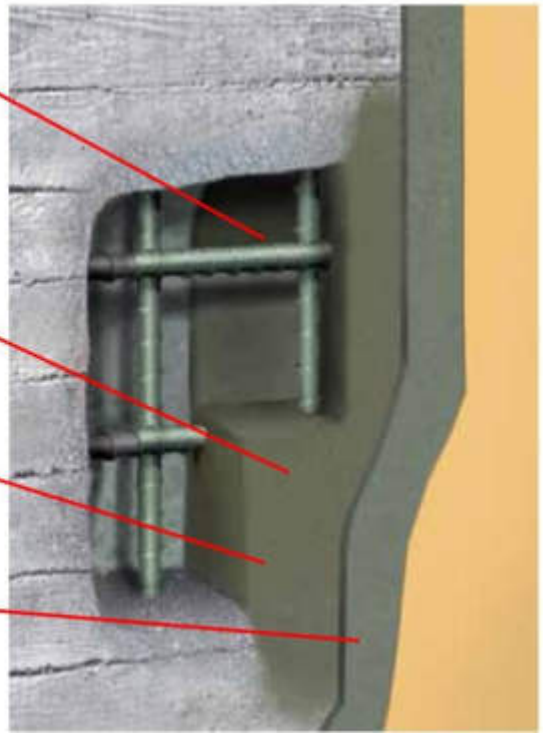
CONCRETE REPAIR SYSTEMS - USING READY TO USE MORTARS
OVER VIEW

Sika® MonoTop®, SikaTop® Armatec®, Sikadur® 32 serie
- Bonding bridge, reinforcement corrosion

Sika® MonoTop®, **Sikacrete®-5xx**
- Patching mortar — Dry spray repair

SikaGrout® **Sikadur®-4x, SikaFlow®-6xx**
- Form pour repair — Form pour repair epoxy base

Sika® Skim Coat, Lanko Skim Coat, Sika® Refit xxxxx
- Pore sealer and smoothing mortar



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CONCRETE REPAIR SYSTEMS - USING READY TO USE MORTARS

BONDING AGENT & REINFORMCEMENT CORROSION PROTECTION

Sika® MonoTop®, SikaTop® Armatec®, Sikadur® 32 serie

- Bonding bridge, reinforcement corrosion

Sika®MonoTop®,

- Patching mortar

Sikacrete®-5xx

Dry spray repair

SikaGrout®

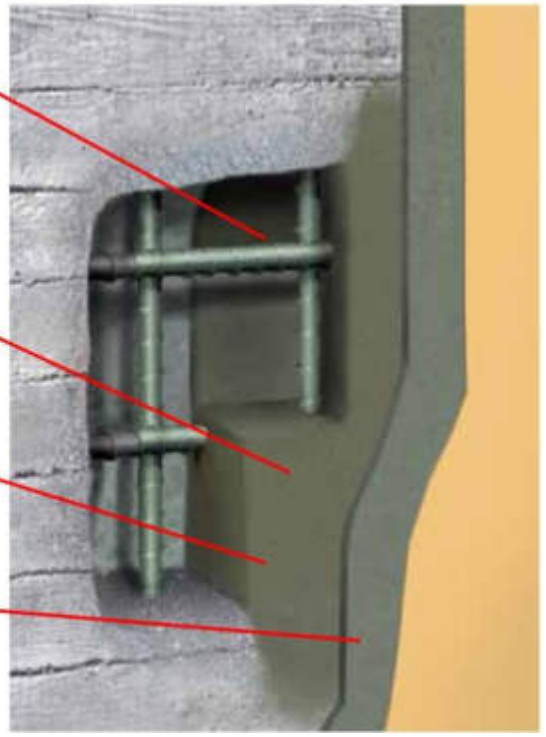
Sikadur®-4x,

SikaFlow®-6xx

- Form pour repair Form pour repair epoxy base

Sika® Skim Coat, Lanko Skim Coat, Sika® Refit xxxxx

- Pore sealer and smoothing mortar



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CONCRETE REPAIR SYSTEMS - USING READY TO USE MORTARS

BONDING AGENT & REINFORMCEMENT CORROSION PROTECTION



Sikadur®-32 serie

- Bond of far greater strength than the tensile strength of the concrete itself. Therefore, it is suitable for use wherever structural bonding of new concrete to existing one is carried out.
- Be used as a bonding agent on : concrete, mortar, steel, iron, etc.

Bond Strength: > 2.5 Mpa.
Open Times: > 2 hrs

Consumption: 0.3-0.8 kg/m2,

Sikalatex®

- Synthetic rubber emulsion that is used as admixture for cementitious bonding bridges and high quality site-mix mortars

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CONCRETE REPAIR SYSTEMS - USING READY TO USE MORTARS

REPAIR AND REPROFILLING MORTARS

Sika® MonoTop®, SikaTop® Armatec®, Sikadur® 32 serie

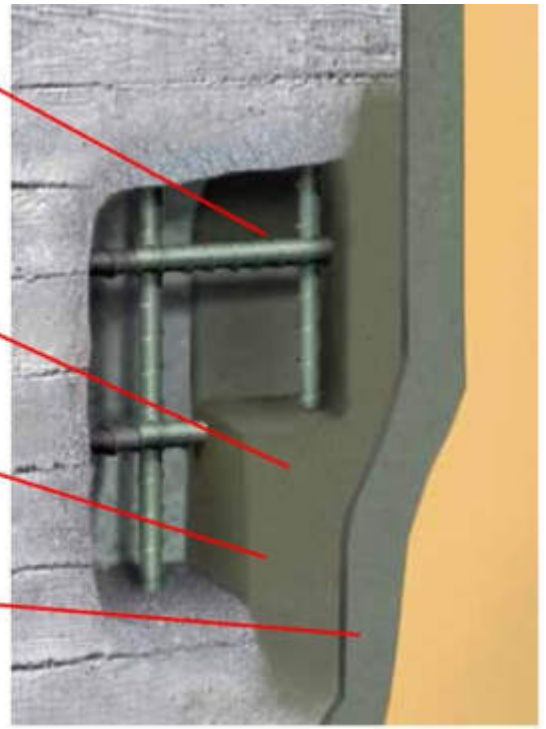
Sika®MonoTop®,
- Patching mortar

Sikacrete®-5xx
Dry spray repair

SikaGrout®
SikaFlow®-6xx
- Form pour repair

Sikadur®-4x,
Form pour repair epoxy

Sika® Skim Coat, Lanko Skim Coat, Sika® Refit xxxxx
- Pore sealer and smoothing mortar



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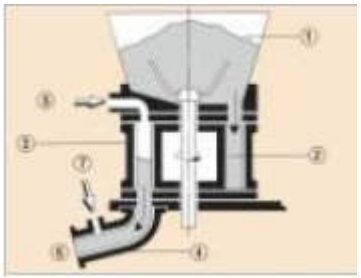
CONCRETE REPAIR SYSTEMS - USING READY TO USE MORTARS

REPAIR AND REPROFILLING MORTARS (SPRAY TYPE)



Use:





Sikacrete®-503

1-component, unaccelerated cementitious and silicafume containing dry spray mortar.

Use: For concrete structural repairs, especially for large and overhead applications.

- Suitable for dry spraying
- Sulphate resistant, Low chloride permeability
- Very low shrinkage behaviour

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359

CONCRETE REPAIR SYSTEMS - USING READY TO USE MORTARS

REPAIR AND REPROFILLING MORTARS

Sika® MonoTop®, SikaTop® Armatec®, Sikadur® 32 serie

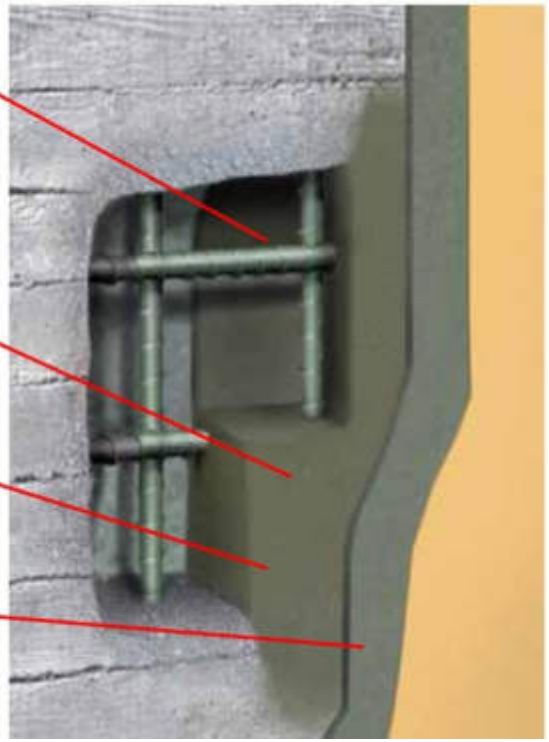
Sika® MonoTop®,
- Patching mortar

Sikacrete®-5xx
Dry spray repair

SikaGrout®

Sikadur®-4x, SikaFlow®-6xx
- Form pour repair Form pour repair epoxy base

Sika® Skim Coat, Lanko Skim Coat, Sika® Refit xxxxx
- Pore sealer and smoothing mortar



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CONCRETE REPAIR SYSTEMS - USING READY TO USE MORTARS

REPAIR AND REPROFILLING MORTARS (FORM POUR TYPES)



SikaGrout®

A non-shrink, self-leveling, cementitious grouting mortar with extended working time to suit local ambient temperatures

Use: For concrete structural repairs, especially for pouring and pumping application.

- Excellent flowability
- Good dimensional stability
- High strengths. Adjustable consistency
- No Bleeding, Non toxic non corrosive
- Impact and vibration resistant

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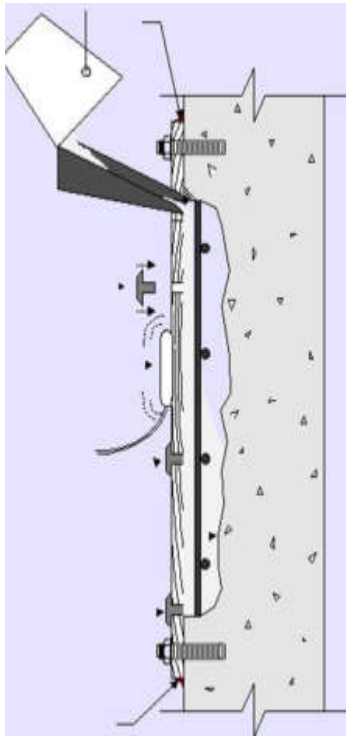
361

CONCRETE REPAIR SYSTEMS - USING READY TO USE MORTARS

REPAIR AND REPROFILLING MORTARS (FORM POUR TYPES)







SikaCrete®xx SCC / Micro SCC

Sika® Micro SCC is a self-compacting, cementitious micro concrete, with incorporated corrosion inhibitor and extended working time to suit local ambient temperatures

- Larger Repairs to Reinforced Concrete Structures exposed to industrial or marine environs.
- Method of placement: Form & Pour or Form & Pump, partial or full member replacement.
- Re-surfacing of reinforced concrete slabs
- Any concreting works where access for placement and

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362



SIKA INJECTION

CRACKS REPAIRS SOLUTIONS BY INJECTIONS

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WHAT ARE TYPICAL **ORIGIN** OF CRACKS

Physical

- Dry Shrinkage
- Crazing

Thermal

- Freeze/ Thaw cycle
- Early Thermal Contraction

Structural

- Service Load
- Blast / Impact
- Creep

Chemical

- ASR
- Steel Corrosion
- Carbonation

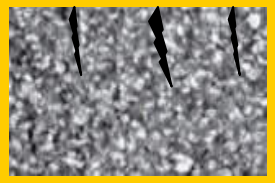
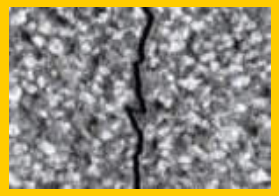
Plastic

- Plastic Shrinkage
- Plastic settlement

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WHAT TYPES OF **PATTERN** OF CRACKS MAY OCCUR

	**Surface cracks** ➔ Negative effect for durability
	**Separation cracks** ➔ Negative effect for watertightness ➔ Negative effect for durability ➔ Negative effect for strength

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WHY WE **NEED TO REPAIR CRACKS**?

	**Durability of the construction** Closing a crack to avoid the ingress of damaging substances (e.g. corrosion due to chlorides)
	**Watertightness of the construction** Closing a crack to refit the sealing function of the concrete structure.
	**Strengthening of the construction** Force transmitting injections to rebuilt the strength of a structure (e.g. after a unique overload)

CONCRETE REPAIR SYSTEMS
CONCRETE CRACK REPAIR

QUESTIONS TO BE ANSWERED:

Purpose or target for repairing the crack?

- Water sealing / stopping?
- Restore bonding?
- Both

PRE EXISITING CONDITIONS &

Purpose or target for repairing the crack?

- Material selection
- Application method

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CRACKS IN CONCRETE STRUCTURES

Objectives	State of moisture/ dampness		
	Dry & damp	wet cracks, flowing water	
		no pressure	under pressure
Closing Waterproofing	SikaInject®-216 SikaInject®- 52	SikaInject®-201 CE SikaInject®- 3**	SikaInject®-102 + SikaInject®-201 CE
Structural repair	SikaInject®- 52 SikaInject®-216	SikaInject®-216	Sika Injection-102 + Sika Injection-216
Flexible repairs	SikaInject®-102 Sika Injection-3**	Sika Injection-102 Sika Injection-3**	Sika Injection-102 + SikaInject®-201 CE

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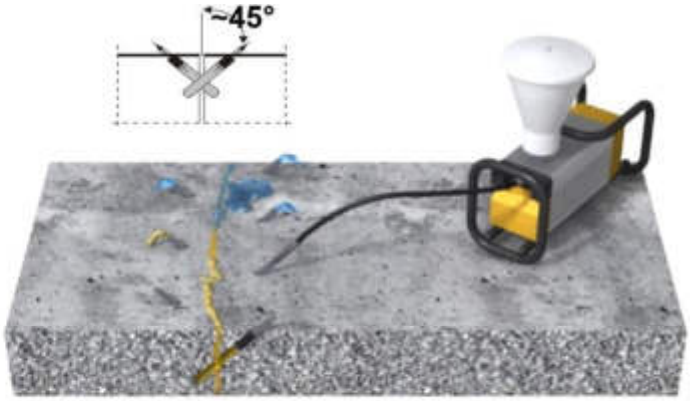
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CONCRETE REPAIR SYSTEMS

CONCRETE CRACK REPAIR – INJECTION APPLICATION CONCEPT



Surface Packers:
Repair from the surface into the crack, for small concrete layers, or when drilling is not allowed or not possible.



Mechanical drill hole packers:
Deep penetration, repair from the center into the entire crack. Higher pressure and waterstopping injections possible.

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BACK TO PRESENTATION

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